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Physics-Informed Graph Neural Networks for Fast Electromagnetic-Thermal Simulation of Induction Heating

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Abstract

Induction heating needs fast EM–thermal simulations for control and optimisation. The Finite Element Method is accurate but costly. In this thesis, we propose a physics-informed Graph Neural Network for transient heat modelling on unstructured meshes and a physics-informed EM surrogate that predicts magnetic vector potential A in the frequency domain. Each model is validated against FEM on representative geometries and material ranges. The thermal model predicts multi-step rollouts. The results indicate that a coupled surrogate model can be used for industrial induction heating problems. There is potential for real-time modelling and rapid inverse problem solving in design optimisation.

Keywords: Induction heating; Physics-informed graph neural networks; Maxwell's equations; Joule heating; Temperature-dependent material properties; Surrogate modelling

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Introduction

Controlled heat treatment is essential in many industrial processes, where the spatial and temporal evolution of temperature governs material behaviour, product quality, and process efficiency [1, 2]. Among modern heating technologies, induction heating is attractive because it enables rapid, contactless, and localised energy transfer to electrically conductive workpieces [1, 3]. In these systems, the primary focus is the transient temperature field, as it directly affects the material's thermal history.

However, accurately predicting temperature evolution is challenging because the heat source is not independently defined, but arises from Joule heating generated by an underlying electromagnetic field, resulting in a coupled multiphysics problem [4, 5]. As a result, induction heating poses a difficult challenge for computational methods that aim to balance physical accuracy, numerical efficiency, and robustness.

Modern computational methods, such as the Finite Element Method (FEM), are established academic and industrial standards for induction heating problems. Repeated multiphysics simulations are computationally demanding. Alternative methods can be a promising compromise between accuracy and speed while remaining physically credible and well verified. Therefore, fast physics-respecting surrogate approaches are worth investigating.

Classical FEM can accurately model induction heating [6]. Additionally, surrogate models such as physics-informed neural networks, neural operators, reduced-order models, and mesh-based graph neural simulators have shown potential to accelerate physical simulations [7–10].

Among these methods, graph neural networks (GNNs) are especially effective for mesh-based multiphysics problems. Unlike PINNs, which encode physics into a loss function over a fixed collocation domain, or neural operators, which learn mappings between function spaces but often rely on regular discretisations, GNNs operate directly on the graph structure of an unstructured mesh. In this graph-based method, each node acts as a spatial degree of freedom, and edges ensure local connectivity, similar to how FEM defines finite elements [11]. As a GNN passes messages through the network, it gathers information from nearby nodes, creating representations that reflect the mesh layout and the local nature of differential operators. This built-in geometric bias makes GNNs a good fit for problems where solutions change smoothly within elements but become more complex at the boundaries or interfaces. This is often the case in induction heating, where eddy currents are concentrated in thin layers near the boundary. The same goes for the temperature gradients near the coil. GNN surrogates can potentially adapt to different mesh resolutions or shapes because the learned computation depends on the graph structure rather than a fixed coordinate grid. This flexibility is useful when design parameters or workpiece shapes change between simulation runs, especially for the skin-depth-dependent mesh near the workpiece-coil interface.

However, creating reliable surrogate models for induction heating remains challenging

due to factors such as multiphysics coupling, geometric dependence, field complexity, and the need for trustworthy verification. Many studies often focus on a single subsystem [12], use simplified benchmarks [13], or lack sufficient validation [10]. Therefore, there is a need for a verified surrogate framework that effectively addresses both the electromagnetic and thermal subproblems to support the simulation of induction heating.

Accurate thermal prediction is crucial for induction heating applications. Faster computational models can facilitate design iterations, parameter studies, digital twins and future workflows focused on control [14–16]. However, speed without physical validity is ineffective in computational physics. Therefore, a surrogate approach must be both computationally efficient and rigorously verified.

This thesis aims to develop and assess a physics-informed surrogate modelling workflow for induction heating, in which electromagnetic and thermal behaviour are learned separately and coupled via Joule heating. The research investigates whether physics-informed, mesh-based surrogate models can preserve sufficient physical fidelity to be meaningful for multiphysics induction-heating workflows.

The objectives are to:

- formulate the governing EM and thermal problems for the workpiece/coil/air induction heating setup;
- implement a thermal surrogate model and assess its predictive performance;
- implement an EM surrogate model for the axisymmetric problem;
- verify the EM model on controlled benchmarks, including a magnetostatic case and a time-harmonic case;
- connect EM and thermal surrogates through the coupling interface;
- evaluate the strengths and limitations of the resulting workflow.

The problem involves two physical components – thermal and EM. However, creating a monolithic surrogate model would be challenging to train, validate, and interpret. Therefore, this thesis adopts a modular strategy: the electromagnetic and thermal surrogates are trained independently and connected through the heat source term.

The EM model predicts the time-harmonic magnetic vector potential A induced by a prescribed current through a stranded coil. The thermal model forecasts the temperature distribution on the same geometry. The EM model supplies the Joule heating, which the thermal model utilises as a source. Both models, when considered independently and in coupled form, are verified against analytical and manufactured solutions, as well as FEM simulations and benchmark cases.

A key methodological commitment of this thesis is a systematic and step-by-step verification strategy. Instead of checking the surrogate framework only at the end, the verification follows a clear hierarchy of benchmarks for both thermal and EM surrogates. It starts with simple linear diffusion, moves on to nonlinear thermal behaviour, verifies magnetostatic and eddy-current behaviour, and finishes with the full electromagnetic-thermal system. By building confidence in the models through these steps, it becomes easier to find and fix errors at each stage. This method follows best practices in computational physics and establishes a clear chain of evidence supporting the reliability of the final workflow.

Chapter 1 introduces the governing physics, centred on the heat equation and the EM-generated heat source, the FEM reference formulations and the physics-informed

machine learning background. Chapter 2 describes the thermal and EM surrogate methodology, the nondimensionalisation of the EM model, the coupling workflow, and the verification strategy. Chapter 3 presents verification results and coupled case studies. Chapter 4 discusses limitations, implications, and future work.

Chapter 1

Theoretical Background

1.1 Governing physics

1.1.1 Heat equation

The heat equation, or diffusion equation, is a second-order parabolic partial differential equation that governs the spatial and temporal distribution of temperature within a given domain. Any heat-transfer problem must satisfy the law of conservation of energy [17, 18]. According to the first law of thermodynamics, the rate of change of total internal energy in the domain must equal the net heat flux through the boundary. The mathematical bridge between temperature gradients and energy transport is provided by Fourier's Law of Heat Conduction [19], which states that the heat flux vector q is directly proportional to the negative gradient of the temperature

$$\mathbf{q} = -k\nabla u, \tag{1.1}$$

where k denotes the thermal conductivity. Substituting Fourier's Law into the energy conservation equation and applying the divergence theorem yields the general form of the heat equation

$$\rho\, c_p \frac{\partial u}{\partial t} = \nabla \cdot (k\nabla u) + Q. \tag{1.2}$$

For the equation (1.2) to make physical sense, that is, to establish a well-posed problem, we must introduce boundary conditions that define the system's interaction with its surroundings [19]. Dirichlet boundary conditions specify an exact temperature $u = g(x,t)$ at the boundary Γ_D, which physically corresponds to a surface in perfect contact with a thermal reservoir. Neumann boundary conditions prescribe the heat flux across the boundary Γ_N with normal n as $-k\,(\partial u/\partial n) = q_N$, for example, a thin electrical resistance heater applied to a surface. Homogeneous Neumann condition $(\partial u/\partial n) = 0$ which represents a perfectly insulated or adiabatic surface. Robin boundary conditions typically model convective heat transfer $-k(\partial u/\partial n) = h(u - T_\infty)$, where h is the convective heat transfer coefficient.

Combining the heat equation itself with boundary conditions, we get the following

strong form in some domain $\Omega \subset R^d$ with boundary $\Gamma = \partial\Omega$ split into $\Gamma_D \cup \Gamma_N \cup \Gamma_R$

$$\begin{cases} \rho\, c_p \frac{\partial u}{\partial t} = \nabla \cdot (k \nabla u) + Q, & \text{in } \Omega \times (0, t_f], \\ T = \bar{T} & \text{on } \Gamma_D \times (0, t_f], \\ k\, \nabla T \cdot \mathbf{n} = \bar{q}_N & \text{on } \Gamma_N \times (0, t_f], \\ k\, \nabla T \cdot \mathbf{n} = h\,(T - T_\infty) & \text{on } \Gamma_R \times (0, t_f], \\ T(\mathbf{x}, 0) = T_0(\mathbf{x}) & \text{in } \Omega. \end{cases} \tag{1.3}$$

There are multiple methods for numerically solving the heat equation. The most popular ones are the Finite Difference Method with explicit or implicit schemes [20], the Method of Lines [21], Spectral Methods [22], the Finite Element Method (covered in Section 1.2), and recently, Physics-Informed Neural Networks [7]. All methods have applications in different scenarios.

1.1.2 Electromagnetics

Maxwell's equations in differential form are given by

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}, \tag{1.4}$$

$$\nabla \times \mathbf{H} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t}, \tag{1.5}$$

$$\nabla \cdot \mathbf{B} = 0, \tag{1.6}$$

$$\nabla \cdot \mathbf{D} = \rho_f, \tag{1.7}$$

where $\mathbf{E}$ is the electric field, $\mathbf{H}$ the magnetic field intensity, $\mathbf{B}$ the magnetic flux density, $\mathbf{D}$ the electric displacement field, $\mathbf{J}$ the free current density, and ρ_f the free charge density.

Charge accumulation effects are usually negligible in the conductive materials used for induction heating (the relaxation time $\tau = \epsilon/\sigma$ is extremely short for metals, so any free charge dissipates almost instantly [23]), so (1.7) is satisfied with $\rho_f \approx 0$. However, it is included here to show the complete Maxwell system. With this assumption $\nabla \times \mathbf{H} \approx \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t}$.

In induction heating, the displacement current is typically negligible at industrial frequencies and length scales, which leads to the magnetoquasistatic (MQS) approximation of Maxwell's equations [24, 25]:

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}, \tag{1.8}$$

$$\nabla \times \mathbf{H} = \mathbf{J}, \tag{1.9}$$

$$\nabla \cdot \mathbf{B} = 0, \qquad \nabla \cdot \mathbf{J} = 0. \tag{1.10}$$

The constitutive relations are

$$\mathbf{B} = \mu_0 \mu\, \mathbf{H}, \qquad \mathbf{J} = \sigma \mathbf{E} + \mathbf{J}_{\text{str}}, \tag{1.11}$$

where μ is the relative permeability, σ the electrical conductivity, and $\mathbf{J}_{\text{str}}$ a prescribed source (stranded) current density in the coil.

A way to solve these equations is to introduce the magnetic vector potential $\mathbf{A}$ and electric scalar potential ϕ [26],

$$\mathbf{B} = \nabla \times \mathbf{A}, \qquad \mathbf{E} = -\frac{\partial \mathbf{A}}{\partial t} - \nabla\phi. \tag{1.12}$$

The magnetic reluctivity is defined as

$$\nu := \frac{1}{\mu_0 \mu}. \tag{1.13}$$

Using (1.9)– (1.12), the MQS strong form in the conducting region reads

$$\nabla \times (\nu \nabla \times \mathbf{A}) + \sigma \left(\frac{\partial \mathbf{A}}{\partial t} + \nabla\phi \right) = \mathbf{J}_{\text{str}}. \tag{1.14}$$

Charge conservation $\nabla \cdot \mathbf{J} = 0$ yields the auxiliary equation for ϕ,

$$\nabla \cdot \left[\sigma \left(\frac{\partial \mathbf{A}}{\partial t} + \nabla\phi \right) \right] = 0 \qquad \text{in conducting domains.} \tag{1.15}$$

Assuming sinusoidal steady state with angular frequency ω and using the complex representation $\mathbf{u}(\mathbf{x}, t) = \text{Re}\{\hat{\mathbf{u}}(\mathbf{x}) e^{j\omega t}\}$. Then $\partial/\partial t \mapsto j\omega$, and (1.14)– (1.15) become

$$\nabla \times \left(\nu \nabla \times \hat{\mathbf{A}} \right) + j\omega\, \sigma \left(\hat{\mathbf{A}} + \frac{1}{j\omega} \nabla \hat{\phi} \right) = \hat{\mathbf{J}}_{\text{str}}, \tag{1.16}$$

$$\nabla \cdot \left[\sigma \left(j\omega \hat{\mathbf{A}} + \nabla \hat{\phi} \right) \right] = 0. \tag{1.17}$$

For axisymmetric geometries with currents purely azimuthal, the only nonzero component is $\hat{\mathbf{A}} = \hat{A}_\varphi(r, z)\, \mathbf{e}_\varphi$. The magnetic flux density components are

$$\hat{B}_r = -\frac{\partial \hat{A}_\varphi}{\partial z}, \qquad \hat{B}_z = \frac{1}{r} \frac{\partial (r \hat{A}_\varphi)}{\partial r}. \tag{1.18}$$

Moreover, $\nabla\hat{\phi}$ has only r–z components and therefore does not contribute to the azimuthal electric field. Hence,

$$\hat{E}_\varphi = -j\omega \hat{A}_\varphi, \qquad \hat{J}_\varphi = \sigma \hat{E}_\varphi + \hat{J}_{\text{str},\varphi} = -j\omega\sigma \hat{A}_\varphi + \hat{J}_{\text{str},\varphi}. \tag{1.19}$$

A commonly used strong form for $\hat{A}_\varphi$ in (r, z) is

$$-\frac{1}{r}\frac{\partial}{\partial r}\left(r\nu \frac{\partial \hat{A}_\varphi}{\partial r} \right) - \frac{\partial}{\partial z}\left(\nu \frac{\partial \hat{A}_\varphi}{\partial z} \right) + \nu \frac{\hat{A}_\varphi}{r^2} + j\omega\sigma\, \hat{A}_\varphi = \hat{J}_{\text{str},\varphi}, \tag{1.20}$$

with regularity on the symmetry axis

$$\hat{A}_\varphi(0, z) = 0. \tag{1.21}$$

Together, the problem reads

$$\begin{cases} -\frac{1}{r}\frac{\partial}{\partial r}\left(r\nu \frac{\partial \hat{A}_\varphi}{\partial r} \right) - \frac{\partial}{\partial z}\left(\nu \frac{\partial \hat{A}_\varphi}{\partial z} \right) + \nu \frac{\hat{A}_\varphi}{r^2} + j\omega\sigma\, \hat{A}_\varphi = \hat{J}_{\text{str},\varphi}, & \text{in } \Omega \\ \hat{A}_\varphi(0, z) = 0 & \text{on } \Gamma_D. \end{cases} \tag{1.22}$$

1.1.3 Induction heating as a coupled electromagnetic-thermal process

Together, equations (1.3) and (1.22) form a coupled system that simulates an induction heating process [1, 27]. The equations are coupled through the source term Q and the temperature dependence of materials. It is acquired by solving the EM system and calculating the time-averaged Joule Heating term [25]

$$Q(x) = \frac{1}{2}\,\sigma(T)\,|E(x)|^2 \tag{1.23}$$

which is passed to the heat equation.

1.2 Finite Element Method

The study of complex physical systems relies heavily on solving partial differential equations. While analytical solutions are ideal, they are usually unfeasible for systems with complex geometries or non-linear dynamics. As a result, numerical methods such as the Finite Element Method have become essential for translating continuous physical laws into discrete systems [28, 29].

FEM involves discretising a spatial domain Ω into non-overlapping elements, such as triangles or tetrahedra. Unlike the Finite Difference Method, FEM uses a variational formulation, applying integration by parts to derive integral representations from strong-form PDEs. This approach naturally incorporates Neumann boundary conditions and is adaptable to irregular geometries.

In this variational framework, an unknown continuous solution is approximated using a trial function u_h from a finite-dimensional subspace V_h, typically made up of continuous piecewise polynomial basis functions. A Galerkin approach sets the test space equal to the trial space, ensuring that the discretised residual vanishes for any test function. The accuracy of this approximation depends on the mesh density and polynomial degree of the basis functions.

The assembly of element contributions results in a global system of coupled linear equations, characterised by a sparse, often symmetric stiffness matrix. Solving this system yields nodal coefficients that interpolate to provide a continuous physical representation. This section lays the mathematical foundation for these processes, detailing variational principles and discretisation techniques that support both traditional FEM and modern graph-based physics-informed simulations.

1.2.1 Variational formulation and weak forms

The strong forms derived in Section 1.1 require the solution to be twice continuously differentiable, a regularity that is difficult to guarantee on irregular geometries or across material interfaces. The variational, or weak, formulation relaxes this requirement and forms the mathematical foundation of the Finite Element Method [29].

The general idea is to multiply the strong-form PDE by a smooth test function v taken from a suitable function space, integrate over the domain Ω, and apply Green's first identity to transfer one order of differentiation from the unknown field onto v. For a second-order elliptic operator, Green's identity reads

$$\int_\Omega \nabla \cdot (k\nabla u)\, v\, d\Omega = -\int_\Omega k\nabla u \cdot \nabla v\, d\Omega + \oint_{\partial\Omega} k\,\frac{\partial u}{\partial n}\, v\, d\Gamma, \tag{1.24}$$

where n is the outward unit normal on $\partial\Omega$. The boundary integral in (1.24) naturally absorbs Neumann and Robin boundary conditions as natural conditions, meaning they enter the formulation without any additional constraint enforcement. Dirichlet conditions are essential conditions that are imposed directly on the trial and test spaces.

The appropriate function space for problems involving first-order derivatives is the Sobolev space

$$H^1(\Omega) := \left\{ v \in L^2(\Omega) \ : \ \nabla v \in L^2(\Omega) \right\}, \tag{1.25}$$

which only requires the solution and its gradient to be square-integrable, not classically differentiable. For problems with homogeneous Dirichlet conditions on Γ_D, the test space is restricted to

$$V := \left\{ v \in H^1(\Omega) \ : \ v = 0 \text{ on } \Gamma_D \right\}. \tag{1.26}$$

A weak solution is then a function u in the corresponding trial space that satisfies the integrated form of the PDE for all test functions $v \in V$. This is the sense in which the equation holds weakly: rather than pointwise equality, one requires equality of all inner products against the test space.

Galerkin discretisation. To convert the infinite-dimensional weak problem into a finite-dimensional linear system, we introduce a mesh that partitions Ω into non-overlapping elements–typically triangles or tetrahedra in two and three dimensions, respectively. On this mesh, we then construct a finite-dimensional subspace $V_h \subset V$ spanned by piecewise polynomial basis functions $\{\phi_j\}_{j=1}^{N}$, often chosen to be continuous and of low polynomial degree. The unknown field is then approximated as

$$u_h(\mathbf{x}) = \sum_{j=1}^{N} u_j \, \phi_j(\mathbf{x}), \tag{1.27}$$

where u_j are the unknown nodal coefficients. The Galerkin condition sets the test space equal to the trial space, V_h = test space, so that the discretised residual vanishes for every basis function ϕ_i. This choice leads to a square system of N equations in N unknowns [29].

Assembly and global system. In practice, the integrals appearing in the weak form are computed element-by-element and accumulated into a global sparse matrix. For a steady-state diffusion problem, this assembly process yields

$$\mathbf{K}\,\mathbf{u} = \mathbf{f}, \tag{1.28}$$

where the stiffness matrix $\mathbf{K}$ and the load vector $\mathbf{f}$ have entries

$$K_{ij} = \int_\Omega k \, \nabla\phi_j \cdot \nabla\phi_i \, d\Omega, \qquad f_i = \int_\Omega Q \, \phi_i \, d\Omega + \int_{\Gamma_N} q_N \phi_i \, d\Gamma. \tag{1.29}$$

Because each basis function ϕ_i has support only on the elements sharing node i, the matrix $\mathbf{K}$ is sparse: non-zero entries exist only between nodes connected by an edge in the mesh. This sparsity is fundamental both to the efficiency of FEM solvers and, as discussed in Section 1.3.2, to the connection between FEM assembly and graph-based message passing.

For transient problems, spatial discretisation introduces an additional mass matrix $\mathbf{M}$, whose entries are

$$M_{ij} = \int_\Omega \rho \, c_p \, \phi_j \, \phi_i \, d\Omega, \tag{1.30}$$

converting the original PDE into a semi-discrete system of ordinary differential equations in time. Time integration of this system is discussed in the following subsection.

1.2.2 Time discretisation for transient problems

After spatial discretisation by the Galerkin method described above, the heat equation reduces to the semi-discrete system of ordinary differential equations

$$\mathbf{M}\,\dot{\mathbf{T}}(t) + \mathbf{K}\,\mathbf{T}(t) = \mathbf{f}(t), \tag{1.31}$$

where the over-dot denotes differentiation with respect to time. Solving (1.31) requires a further discretisation in time. The most common single-step schemes can be written in the unified θ-method form

$$\mathbf{M}\,\frac{\mathbf{T}^{n+1} - \mathbf{T}^{n}}{\Delta t} + \mathbf{K}\left[\theta\,\mathbf{T}^{n+1} + (1-\theta)\,\mathbf{T}^{n}\right] = \theta\,\mathbf{f}^{n+1} + (1-\theta)\,\mathbf{f}^{n}, \tag{1.32}$$

where $\Delta t = t^{n+1} - t^{n}$ is the time step and the parameter $\theta \in [0,1]$ selects the specific scheme. The choice $\theta = 0$ recovers the forward Euler (explicit) method, $\theta = 1$ gives the backward Euler (implicit) method, and $\theta = \frac{1}{2}$ gives the Crank–Nicolson method [30].

Stability considerations. For the forward Euler scheme, the time step must satisfy the condition $\Delta t \lesssim \mathcal{O}(h^2/\alpha)$ [31], where h is the characteristic mesh size and $\alpha = k/(\rho c_p)$ is the thermal diffusivity. On fine meshes or for materials with high thermal conductivity, this constraint can force prohibitively small time steps. Implicit schemes ($\theta > 0$) are unconditionally stable: no restriction on Δt is imposed by stability, and the only constraint is accuracy. For induction heating, the thermal response evolves over timescales that are long relative to the electromagnetic period, so a moderately large time step is both computationally attractive and physically appropriate. For this reason, the backward Euler method ($\theta = 1$) is adopted throughout this work.

Backward Euler scheme. Setting $\theta = 1$ in (1.32) gives the fully implicit update

$$\left(\frac{1}{\Delta t}\mathbf{M} + \mathbf{K}\right)\mathbf{T}^{n+1} = \frac{1}{\Delta t}\mathbf{M}\,\mathbf{T}^{n} + \mathbf{f}^{n+1}, \tag{1.33}$$

which requires solving a linear system at every time step. The system matrix $\frac{1}{\Delta t}\mathbf{M}+\mathbf{K}$ is symmetric positive definite and can be factored once and reused across time steps when material properties are temperature-independent. When the source term $\mathbf{f}$ depends on temperature, as is the case in the coupled induction heating problem, the system becomes nonlinear and must be resolved iteratively at each step, as discussed in Section 2.

Connection to the surrogate model. A central design choice of the thermal surrogate developed in this work is that it is trained to replicate the backward Euler update (1.33) rather than the continuous PDE. This means the network is a learned time-stepper: given the temperature field at time t^n and the volumetric heat source $\mathbf{f}^{n+1}$, it predicts $\mathbf{T}^{n+1}$. The physics-informed loss function in Section 2 is constructed by evaluating the residual of (1.33) at the network's prediction, which penalises violations of the discrete time-stepping scheme rather than the continuous equation. This choice anchors the surrogate to the same discretisation as the reference FEM solver, making the two directly comparable.

1.2.3 Coupling strategies for induction heating simulations

The electromagnetic and thermal subproblems in induction heating are coupled through the Joule heating term Q: the electromagnetic field drives heat generation, while the resulting temperature field alters the material properties that govern the electromagnetic problem. How this feedback is handled in practice leads to a family of coupling strategies of increasing fidelity and computational cost.

One-way coupling. In a one-way-coupled strategy, the electromagnetic field is solved first, and the resulting loss distribution Q_{em} is passed to the thermal model as a prescribed heat source. The thermal problem is then solved without feeding the updated temperature field back into the electromagnetic model [32, 33]. In its simplest form, the electromagnetic problem is solved only once, using reference or initial material properties, and the resulting loss field is held fixed throughout the thermal simulation. A slightly more general variant allows the excitation itself to vary in time, but still neglects the influence of temperature on the electromagnetic coefficients. This strategy is computationally attractive and is often adequate when temperature changes are modest, when the heating interval is short, or when the temperature dependence of σ and μ is weak.

From a modelling perspective, one-way coupling assumes that

$$\mathbf{K}_{\text{em}}(\mathbf{p}(T)) \approx \mathbf{K}_{\text{em}}(\mathbf{p}(T_{\text{ref}}))\,, \tag{1.34}$$

so that the power density Q_{em} is effectively independent of the evolving temperature field. For induction heating, this approximation can be reasonable for non-ferromagnetic materials over a limited temperature range, but it becomes questionable once the electrical resistivity, magnetic permeability, or skin depth vary significantly with temperature.

Two-way coupling. In two-way coupling, the feedback from the thermal field to the electromagnetic field is retained. After each thermal update, the temperature-dependent material properties are recomputed and the electromagnetic problem is re-solved with the updated coefficients. The new electromagnetic solution produces an updated Joule loss distribution, which is then used in the next thermal step. In induction heating, this is the physically consistent approach whenever the material's properties are temperature-dependent, since the temperature field alters current penetration, magnetic flux distribution, and therefore the spatial localisation of the heat source itself. This feedback is especially important for ferromagnetic steels, where the magnetic response changes strongly near the Curie temperature [4, 6].

Choice of coupling in this work. The surrogate model developed in Chapter 2 adopts one-way coupling: the electromagnetic surrogate predicts the volumetric heating source Q, which is subsequently passed as a fixed input to the thermal surrogate. This choice is motivated by the modular decomposition described in Section 2.3 and is appropriate for the operating conditions considered here. Extension to two-way coupling is straightforward within the modular framework and is discussed as future work in Chapter 4.

The structural parallel between FEM and graph-based neural networks is worth noting explicitly. In a finite element mesh, each node carries a set of degrees of freedom, elements define the connectivity between nodes, and the assembly process

accumulates local element contributions into a global sparse matrix via nearest-neighbour interactions. This is precisely the structure of a graph: nodes correspond to mesh vertices, edges encode element connectivity, and message passing aggregates local information in the same pattern as element assembly. Consequently, Graph Neural Networks are a natural architectural choice for learning operators defined on FEM meshes, and the GNN surrogate developed in this work inherits the geometric flexibility and local interaction structure of the underlying FEM discretisation. This connection is developed further in Section 1.3.2.

1.3 Machine learning and Neural Networks

Machine learning is an exciting field that focuses on developing and studying statistical algorithms that learn from data without explicit instructions. These algorithms can interpret data behaviour and generalise to new, unseen data. As Arthur Samuel noted in 1959, the central question is: how can computers learn to solve problems without being explicitly programmed [34]?

Machine learning models can be classified into supervised, unsupervised, and reinforcement learning categories [35, 36].

Supervised learning models use labelled datasets to train algorithms for making predictions or classifications on new data. Unsupervised learning models, on the other hand, analyse unlabelled data to identify patterns or structures through clustering or association analysis. Reinforcement learning models involve an agent that learns to make decisions by interacting with an environment to maximise cumulative rewards over time.

1.3.1 Physics-Informed Neural Networks

Steve Brunton discussed in his Physics-Informed Machine Learning course [37] that physics can be integrated at any of the five stages of the machine learning process: problem definition, data curation, model selection, loss function creation, and optimisation.

The first stage involves determining the problem to be modelled. The second stage focuses on curating the data, deciding what information will inform the model. Next, you will choose the model architecture, which may include options such as Recurrent Neural Networks [38], autoencoders [39], Dynamic Mode Decomposition [40], or Sparse Identification of Nonlinear Dynamical Systems [41]. After selecting the model, you will define a loss function that quantifies good model performance. Finally, you will employ optimisation strategies to determine which algorithms to use for training the model.

Each of these stages offers an opportunity to incorporate principles of physics into the machine learning process or to discover new aspects of physics relevant to machine learning. Further research and academic papers explore various methods for integrating physics throughout these stages.

Physics-Informed Machine Learning has evolved into an extension of classical numerical techniques for solving both forward and inverse problems related to partial differential equations [10].

In a PINN, the solution $u(t, \mathbf{x})$ is approximated by a neural network $u_\theta(t, \mathbf{x})$ with parameters θ. The key idea is to train u_θ, such that it satisfies the partial differential equation and conditions as closely as possible [7] [42].

How does PINNs incorporate the PDE? For illustration, let's consider the constant-coefficient heat equation. We need to define a physics-informed loss function that

penalises the PDE's residual. First, sample a set of collocation points (also called training points) in the domain where one evaluates the PDE. These can be randomly chosen on a grid of $\mathbf{x}$ values in the domain (excluding the boundary for the PDE residual term). At each collocation point $\mathbf{x}_i$, one computes the network's prediction $u_\theta(\mathbf{x}_i)$ and then uses automatic differentiation to compute the needed derivatives u_{xx}, u_{yy}, u_{zz}. Let $\mathcal{S}_f \subset (0,T]\times\Omega$, $\mathcal{S}_D \subset (0,T]\times\Gamma_D$, $\mathcal{S}_N \subset (0,T]\times\Gamma_N$, $\mathcal{S}_R \subset (0,T]\times\Gamma_R$, and $\mathcal{S}_0 \subset \Omega$ denote collocation sets with sizes N_f, N_D, N_N, N_R, N_0, respectively. Then the PDE residual is defined as

$$R_\Omega(t,\mathbf{x};\theta) = \partial_t u_\theta(t,\mathbf{x}) \;-\; \kappa\,\Delta u_\theta(t,\mathbf{x}) \;-\; f(t,\mathbf{x}), \tag{1.35}$$

which should equal zero for an exact solution of the heat equation. The PINN loss will include a term that is the mean-squared residual over all collocation points. For example, the PDE loss term can be written as:

$$L_{\text{PDE}} = \frac{1}{N_f} \sum_{(t_i,\mathbf{x}_i)\in\mathcal{S}_f} \Big(\partial_t u_\theta(t_i,\mathbf{x}_i) - \kappa\,\Delta u_\theta(t_i,\mathbf{x}_i) - f(t_i,\mathbf{x}_i)\Big)^2 \tag{1.36}$$

which drives the neural network to satisfy the equation at those points. All derivatives are obtained via automatic differentiation through the neural network's computation graph, a built-in capability of modern deep learning frameworks. This means we do not need to derive finite difference formulas. The network can be differentiated analytically, providing a mesh-free method for evaluating PDE terms at arbitrary points.

How are boundary and initial conditions enforced? PINNs incorporate these by adding additional loss terms. We sample points on the domain boundaries and penalise the deviation from the known conditions. For instance, suppose we have Dirichlet boundary conditions specifying $u(t,x,y,z)$ on the domain boundaries. We can define boundary loss

$$L_{\text{BC}} = \frac{1}{N_D} \sum_{(t_j,\mathbf{x}_j)\in\mathcal{S}_D} \Big(u_\theta(t_j,\mathbf{x}_j) - g_D(t_j,\mathbf{x}_j)\Big)^2, \tag{1.37}$$

where g_D is the given boundary value, this term ensures the network outputs the correct temperature on the boundaries.

Time-dependent problems have an initial condition that must also be satisfied. The initial condition loss is

$$L_{\text{IC}} = \frac{1}{N_0} \sum_{\mathbf{x}_k\in\mathcal{S}_0} \Big(u_\theta(0,\mathbf{x}_k) - u_0(\mathbf{x}_k)\Big)^2. \tag{1.38}$$

If the boundary conditions are Neumann or Robin, a similar approach is taken, except we penalise the derivative (computed by automatic differentiation [43]) minus the specified value. The loss terms are formulated accordingly. For example, for a Neumann boundary, one would include $\big(\partial_n u_\theta(t,\mathbf{x}) - g_N(t,\mathbf{x})\big)^2$ in L_{BC} for points on that boundary. Finally, we combine all these contributions into a single total loss that the network will minimise:

$$L_{\text{Total}} = w_f L_{\text{PDE}} + w_D L_D + w_N L_N + w_R L_R + w_0 L_{\text{IC}} \;+\; w_{\text{data}} L_{\text{Data}} \;\;\text{(optional)}, \tag{1.39}$$

where $w_f, w_D, w_N, w_R, w_0, w_{\text{data}} > 0$ are tunable weights, where we have also included L_{data} to represent any reference solution data or observational data from the domain. The L_{data} term would be a mean squared error between the network prediction and known solution values at a set of data points in the domain. In many cases where no additional interior data is available, L_{data} is omitted and the PINN is entirely self-supervised by $L_{\text{PDE}} + L_{\text{BC}} + L_{\text{IC}}$ – making it an unsupervised learning approach guided by physics.

During training, the PINN adjusts its weights θ to minimise L_{total}. This is done using standard optimisation algorithms, such as gradient descent or Adam [44], and the gradients of the loss with respect to θ are computed by backpropagation. Importantly, backpropagation naturally incorporates the PDE constraint because the PDE residual term uses gradients of the network, relying on automatic differentiation.

Mesh-free collocation provides geometric flexibility but can struggle with high-frequency oscillations, a phenomenon known as spectral bias [45]. This issue can negatively impact the convergence of the training process and the model's accuracy. A viable strategy to mitigate this problem is to employ input encoding techniques. This method requires transforming the input data into a higher-dimensional feature space using high-frequency functions. Such an approach is detailed in the study by Tancik et al. [46], in which Fourier features are used to enhance neural network capabilities.

While vanilla Physics-Informed Neural Networks have shown promise as mesh-free solvers, they face significant roadblocks in realistic engineering applications [47]. Most point-wise PINN formulations rely on fully-connected networks that approximate continuous functions, leading to poor scalability and high training costs due to the need to compute automatic differentiation over vast collocation points [48]. Furthermore, PINNs often struggle with "soft" boundary enforcement, requiring the tedious tuning of penalty parameters to balance the PDE residuals with initial and boundary conditions in non-convex optimisation spaces [49].

To address these limitations, researchers have developed integrated frameworks called variational PINNs (V-PINNs) [50]. V-PINNs utilise the variational form of the system. By shifting the differential burden to the test functions, V-PINNs can effectively reduce the required order of differentiation for the neural network, thereby mitigating the computational complexity and non-convexity of the optimisation landscape. This approach allows the massive number of collocation points required by strong-form PINNs to be replaced by a relatively small number of quadrature points, which significantly reduces the high-dimensional automatic differentiation costs associated with high-order derivatives [50].

An advancement in the field of partial differential equations with neural networks is the Physics-Informed Graph Neural Galerkin Network, which marries the variational structure of PDEs with the non-Euclidean flexibility of Graph Convolutional Networks [47].

1.3.2 Graph neural networks for mesh-based simulation

Traditional grid-based learning approaches can be difficult to apply to high-fidelity physical simulations on complex geometries. In particular, Convolutional Neural Networks (CNNs) require the computational domain to be represented on a regular lattice of pixels or voxels, whereas many physics solvers, such as the Finite Element Method, operate on unstructured meshes. This mismatch is especially problematic for domains with curved boundaries, heterogeneous materials, and locally refined discretisations. Graph Neural Networks provide a natural alternative, since they operate directly on non-Euclidean data and can therefore represent irregular computational meshes without rasterisation.

A graph is denoted by $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \theta)$, where $\mathcal{V}$ is the set of nodes, $\mathcal{E}$ is the set of edges and θ is a vector representing properties that apply to the entire system also called global features [51]. In a mesh-based setting, nodes typically correspond to mesh vertices, while edges encode the connectivity of neighbouring nodes. Global vectors can describe material constants or time steps. Node attributes may contain local

state variables and material properties, whereas edge attributes may encode geometric relations such as relative position, distance, or interface information. This gives GNNs a strong relational inductive bias: the model learns how information is exchanged locally between neighbouring entities, rather than assuming a regular Cartesian structure.

Most GNNs are built on a message-passing scheme, in which each node iteratively aggregates information from its neighbourhood [52]. A common formulation is

$$\mathbf{m}_i^{(\ell+1)} = \sum_{j \in \mathcal{N}(i)} \phi_m\left(\mathbf{h}_i^{(\ell)}, \mathbf{h}_j^{(\ell)}, \mathbf{e}_{ij}\right), \tag{1.40}$$

$$\mathbf{h}_i^{(\ell+1)} = \phi_u\left(\mathbf{h}_i^{(\ell)}, \mathbf{m}_i^{(\ell+1)}\right), \tag{1.41}$$

where $\mathbf{h}_i^{(\ell)}$ is the latent representation of node i at message-passing step ℓ, $\mathbf{e}_{ij}$ denotes the features of edge (i, j), $\mathcal{N}(i)$ is the neighbourhood of node i, ϕ_m is a learnable message function, and ϕ_u is a learnable node-update function. After several message-passing steps, information from increasingly distant parts of the graph is incorporated into each node representation. If a graph-level output is required, a readout operation aggregates the final node embeddings into a global quantity.

For physical simulation, this locality is highly attractive because it mirrors the structure of numerical discretisations: information is exchanged primarily between neighbouring nodes or elements, and global behaviour emerges from repeated local interactions. In this sense, message passing can be interpreted as a learned analogue of stencil operations or local element assembly. For mesh-based PDE problems, GNNs therefore offer a representation that is much better aligned with the underlying discretisation than image-based neural architectures.

This idea is central to MeshGraphNets [9, 53], where an unstructured mesh is interpreted as a graph and processed by an encoder–processor–decoder architecture. The processor typically comprises several message-passing blocks, enabling the network to propagate information across the mesh and learn the evolution of physical fields. Because the learned dynamics are defined on graph connectivity rather than on a fixed pixel grid, such models can transfer more naturally across geometries and mesh resolutions than standard grid-based networks.

In the context of induction heating, this is particularly relevant. The computational domain typically includes irregular workpiece geometries, surrounding air regions, and excitation coils, often with strongly heterogeneous, temperature-dependent material parameters. A graph representation can incorporate these directly as node and edge features, such as spatial coordinates, boundary markers, electrical conductivity, magnetic permeability, thermal properties, and source-related quantities. Message passing then allows the model to learn how electromagnetic and thermal information propagates across the discretised domain while remaining compatible with the underlying FEM mesh.

Physics-informed variants extend this idea further by constraining the network with the governing equations, rather than relying purely on supervised field data. In a physics-informed MeshGraphNet [13], the graph architecture retains the geometric flexibility of mesh-based message passing, while the loss function additionally penalises violations of the PDE system and boundary conditions. For induction heating, this is a particularly promising direction, since generating high-quality coupled electromagnetic–thermal simulation data is computationally expensive, and embedding physical structure into the learning process can improve both data efficiency and physical consistency.

Chapter 2

Methods

This chapter presents the methods used in this work. All the code written throughout this thesis is available here [54]. The repository allows the experiments reported to be fully reproduced.

2.1 Why physics-informed mesh-based learning for induction heating?

Induction heating involves the iterative coupling of electromagnetic and thermal fields. While high-fidelity Finite Element Method simulations are accurate, they can be expensive for tasks such as repeated simulations, parameter studies, optimisation, or future control settings.

On the other hand, purely data-driven surrogate models typically require a large number of FEM-generated labels, which also come with high computational costs. Physics-informed training addresses this issue by utilising the governing equations and the FEM structure directly, thereby reducing the dependence on expensive labelled data. Additionally, since the geometry involved is mesh-based and can vary, graph-based models are often more suitable than regular-grid models for this application.

The overarching goal is therefore to learn a surrogate $\mathcal{S}_\theta$ that replicates the input–output behaviour of the FEM solver $\mathcal{S}_{\text{FEM}}$, but at a fraction of its computational cost. Concretely, the FEM solver defines a mapping

$$\mathcal{S}_{\text{FEM}} : (\Omega, \mathbf{p}, \mathbf{J}_{\text{src}}) \mapsto (\mathbf{A}, T), \tag{2.1}$$

where Ω is the problem domain (discretised as a mesh), $\mathbf{p}$ collects material and geometric parameters, $\mathbf{J}_{\text{src}}$ is the applied source current density, and $(\mathbf{A}, T)$ are the resulting magnetic vector potential and temperature fields, respectively. The learned surrogate approximates this mapping,

$$\mathcal{S}_\theta \approx \mathcal{S}_{\text{FEM}}, \tag{2.2}$$

while satisfying the constraint

$$\text{cost}(\mathcal{S}_\theta) \ll \text{cost}(\mathcal{S}_{\text{FEM}}). \tag{2.3}$$

Achieving this favourable trade-off motivates combining physics-informed training–which encodes the coupled electromagnetic and thermal equations as soft constraints on $\mathcal{S}_\theta$–with a graph-based architecture that naturally respects the unstructured mesh topology of Ω.

2.2 Graph-based representation of the computational mesh

Both the electromagnetic and thermal surrogates operate on a graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ constructed directly from the finite-element mesh. This section describes how the mesh produced by `Netgen` [55] and a `Netgen` wrapper of Open Cascade Technology `netgen.occ` is converted into that graph, and motivates the connectivity choices made for the two element types encountered in practice.

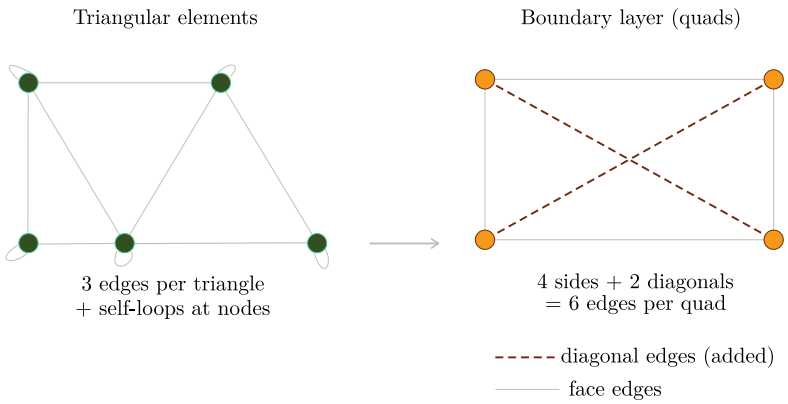


Figure 2.1: Graph construction from the finite-element mesh. *Left:* a patch of triangular bulk elements; each pair of vertices sharing a triangle face is connected by a directed edge (both directions), and a self-loop is added at every node. *Right:* a quadrilateral boundary-layer element; in addition to the four face edges, both diagonals are included so that every vertex pair within the element lies within one message-passing hop of each other.

2.2.1 Mesh generation

Geometry is defined using the `occ` constructive solid geometry kernel included in `Netgen` [55]. `Netgen` then discretises the resulting domain into an unstructured two-dimensional mesh. The bulk of the domain–the coil, air region, and workpiece interior–is meshed with triangular elements, while a structured boundary layer of quadrilateral elements is generated near the workpiece surface to resolve the steep field gradients associated with the electromagnetic skin effect and the thermal boundary layer.

2.2.2 Node set

Each mesh vertex becomes a graph node $v_i \in \mathcal{V}$. Node attributes encode the local geometric position, boundary-condition type, material properties, and physical state, as detailed in the surrogate-specific subsections below. Nodes carrying prescribed Dirichlet boundary conditions are retained in the full graph but removed from the prediction graph before message passing; the network therefore never needs to learn a fixed boundary value.

2.2.3 Edge construction

Directed edges are created between every pair of vertices that share a mesh element, following the finite-element connectivity. For a triangular element with vertices $\{i, j, k\}$ (see left panel of Figure 2.1), this adds the six directed edges $(i \to j)$, $(j \to i)$, $(i \to k)$, $(k \to i)$, $(j \to k)$, $(k \to j)$. Self-loops $(i \to i)$ are added at every node so that a node's own state is included in the aggregation step during message passing.

Boundary-layer elements are quadrilateral. Connecting only the four sides of a quad would leave the two diagonal node pairs without a direct message-passing path, producing an asymmetric local receptive field relative to the triangular region. To restore full connectivity within each element, edges are also added along both diagonals of every quadrilateral. A boundary-layer quad with vertices $\{i, j, k, l\}$ (see right panel of Figure 2.1), therefore, contributes the same directed edges as two triangles sharing the diagonal–exactly the connectivity that a diagonal split of the quad into triangles would yield. This ensures that every node pair co-located in the same element is always within one message-passing hop of each other, regardless of element shape.

Boundary-segment edges between vertices on shared element faces are included in both the triangular and quadrilateral regions, so the graph connectivity mirrors the full finite-element adjacency structure.

2.2.4 Summary of graph properties

The resulting graph is directed, with self-loops, and respects the unstructured topology of the `Netgen` mesh. Edge features encode the relative spatial displacement and Euclidean distance between the connected nodes in physical coordinates, normalised by the characteristic length scale $r_\star$ (see Section 2.3) in the EM case. The graph is batched across multiple problems using `PyTorch Geometric` [56], which treats each problem's graph as a disconnected component of a single large graph, enabling a single GPU forward pass over a batch of geometrically distinct meshes.

2.3 Modular Decomposition into EM and Thermal Surrogate Models

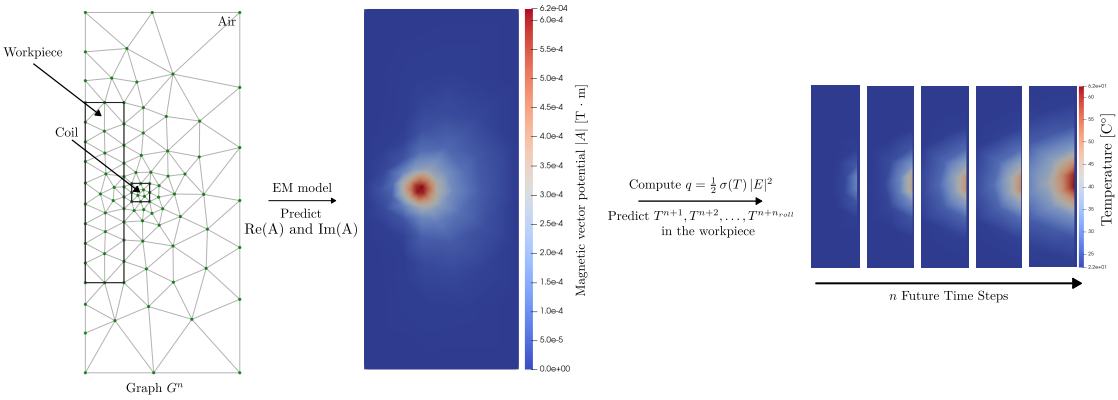


Figure 2.2: Overview of the coupled surrogate method for the induction heating simulation. First, the graph is assembled. Then the EM model predicts the real and imaginary parts of the magnetic vector potential A. The electric field $E = -j\omega A$ and Joule heating term $Q_J = \frac{1}{2}\,\sigma(T)\,|E|^2$ are computed. The Joule heating term is passed to the thermal surrogate, which predicts the n future temperature states.

In this section, we examine the rationale for adopting a modular decomposition approach rather than training a single, monolithic multiphysics model. This choice is sensible for several reasons related to the distinct characteristics of the electromagnetic and thermal subproblems.

The EM and thermal models operate under different mathematical frameworks and use distinct physical time descriptions. They also yield different outputs, scales, and verification requirements. By utilising a modular strategy, we simplify the processes of training, verification, and interpretation.

The thermal problem is transient and governed by the heat equation, which is naturally suited to temporal analysis. In contrast, the EM problem is time-harmonic, expressed in terms of complex fields and solved under specific operating conditions rather than via a transient simulation. The thermal model's main objective is to predict the scalar temperature, whereas the EM model targets a complex magnetic vector potential system that ultimately generates the Joule heating term Q_J.

The evaluation of the thermal model primarily focuses on tracking temperature changes. In contrast, the EM model is assessed by the behaviour of fields and, more importantly, by derived quantities such as Q_J and the total power. It is essential to verify the EM model independently, including aspects such as magnetostatics and eddy-current behaviour. Differences in training stability, normalisation, and residual construction further separate the two subproblems. Using a monolithic model would complicate identifying which component failed if the resulting coupled temperature is incorrect.

In the context of the current work, the coupling between the two models is one-way: the EM model predicts the volumetric heating source, and the thermal model utilises this source to forecast the evolution of temperature.

Formally, the coupled problem is decomposed into a sequence of two surrogate evaluations per time step (see Figure 2.2). At each step n, the EM surrogate $\mathcal{M}^{\mathrm{EM}}$ maps the geometry $\mathcal{G}$, the source current density $\mathbf{J}_{\mathrm{src}}$, and the temperature-dependent material properties $\sigma(T^n)$ and $\mu(T^n)$ to the magnetic vector potential $\mathbf{A}^n$. The Joule heating Q^n_{Joule} is then computed as a pointwise post-processing step from $\mathbf{A}^n$, before being passed as a source term to the thermal surrogate $\mathcal{M}^{\mathrm{thermal}}$, which advances the temperature field to T^{n+1}. The coupled problem can be decomposed as:

$$\mathbf{A}^n = \mathcal{M}^{\mathrm{EM}}\left(\mathcal{G}, \mathbf{J}_{\mathrm{src}}, \sigma(T^n), \mu(T^n)\right), \tag{2.4}$$

$$Q^n_{\mathrm{Joule}} = \frac{1}{2}\,\sigma(T^n)\,\omega^2\,|\mathbf{A}^n|^2, \tag{2.5}$$

$$T^{n+1} = \mathcal{M}^{\mathrm{thermal}}\left(\mathcal{G}, T^n, Q^n_{\mathrm{Joule}}, k(T^n), \rho, c_p\right). \tag{2.6}$$

2.4 Thermal model

The thermal surrogate models the transient heat conduction problem on a fixed finite-element mesh. This section derives the weak formulation and its backward Euler time discretisation, which together define the discrete residual used as the physics-informed training loss. The surrogate takes the current nodal temperature field and a volumetric heat source as inputs and predicts a bundle of future temperature states in a single forward pass. The following subsections describe the residual construction, the graph representation of mesh and boundary data, and the training strategy, respectively.

2.4.1 Weak form and time discretisation used in this work

To apply the Finite Element Method, we first derive the weak formulation of the strong form of the heat equation. Let the boundary be decomposed as $\partial\Omega = \Gamma_D \cup \Gamma_N$ with

$\Gamma_D \cap \Gamma_N = \varnothing$, where Dirichlet boundary conditions are prescribed on Γ_D and Neumann boundary conditions on Γ_N. We define the test space

$$V := \left\{ v \in H^1(\Omega) \ : \ v = 0 \text{ on } \Gamma_D \right\}. \tag{2.7}$$

Multiplying Equation (1.2) by a test function $v \in V$, integrating over Ω, applying Green's identity to the diffusion term, and using the Neumann boundary condition yields the weak formulation:

$$\int_\Omega \rho c_p \frac{\partial T}{\partial t} \, v \, d\Omega \ + \ \int_\Omega k \nabla T \cdot \nabla v \, d\Omega \ = \ \int_\Omega q(T) \, v \, d\Omega \ + \ \int_{\Gamma_N} h^N \, v \, d\Gamma. \tag{2.8}$$

Here, $q(T)$ denotes the volumetric heat source. In the linear case, $q(T) = Q$ is independent of temperature.

We now introduce the finite element approximation

$$T_h(\mathbf{x}, t) = \sum_{j=1}^{N_\phi} T_j(t) \, \phi_j(\mathbf{x}), \tag{2.9}$$

where $\{\phi_j\}_{j=1}^{N_\phi}$ are the finite element basis functions and $\mathbf{T}(t) = [T_1(t), \ldots, T_{N_\phi}(t)]^\top$ is the vector of nodal degrees of freedom. Choosing the test functions $v_h = \phi_i$ gives the semi-discrete system

$$\mathbf{M} \, \dot{\mathbf{T}} + \mathbf{K} \, \mathbf{T} - \mathbf{f}(\mathbf{T}) - \mathbf{g} = \mathbf{0}, \tag{2.10}$$

with

$$M_{ij} := \int_\Omega \rho c_p \, \phi_j \phi_i \, d\Omega, \tag{2.11}$$

$$K_{ij} := \int_\Omega k \nabla \phi_j \cdot \nabla \phi_i \, d\Omega, \tag{2.12}$$

$$f_i(\mathbf{T}) := \int_\Omega q(T_h) \, \phi_i \, d\Omega, \tag{2.13}$$

$$g_i := \int_{\Gamma_N} h^N \phi_i \, d\Gamma. \tag{2.14}$$

Hence, $\mathbf{M}$ is the mass matrix, $\mathbf{K}$ is the conductivity matrix, $\mathbf{f}(\mathbf{T})$ is the nonlinear source vector, and $\mathbf{g}$ is the Neumann load vector.

Using the backward Euler method from time level t^n to $t^{n+1} = t^n + \Delta t$ gives

$$\mathbf{M} \, \frac{\mathbf{T}^{n+1} - \mathbf{T}^n}{\Delta t} + \mathbf{K} \, \mathbf{T}^{n+1} - \mathbf{f}(\mathbf{T}^{n+1}) - \mathbf{g}^{n+1} = \mathbf{0}. \tag{2.15}$$

Equivalently, at each time step we seek $\mathbf{T}^{n+1}$, such that

$$\mathbf{R}(\mathbf{T}^{n+1}) := \mathbf{M} \, \frac{\mathbf{T}^{n+1} - \mathbf{T}^n}{\Delta t} + \mathbf{K} \, \mathbf{T}^{n+1} - \mathbf{f}(\mathbf{T}^{n+1}) - \mathbf{g}^{n+1} = \mathbf{0}. \tag{2.16}$$

Because the source term depends on temperature, the Equation (2.16) is, in general, nonlinear and must be solved iteratively.

2.4.2 Physics-informed loss

The thermal surrogate predicts a bundle of future temperatures $\hat{\mathbf{T}}^{n+1}, \hat{\mathbf{T}}^{n+2}, \ldots, \hat{\mathbf{T}}^{n+w}$ using the current state $\mathbf{T}^n$ and the volumetric heating input $\mathbf{f}^n$. Instead of training on labelled FEM outputs, the model learns by penalising any violations of the backward Euler residual described in Section 1.2, Equation (2.15). Here is a reminder of that residual

$$\mathbf{R}(\mathbf{T}^{n+1}) := \mathbf{M}\,\frac{\mathbf{T}^{n+1}-\mathbf{T}^n}{\Delta t} + \mathbf{K}\,\mathbf{T}^{n+1} - \mathbf{f}^{n+1} - \mathbf{g}^{n+1} = \mathbf{0}, \tag{2.17}$$

which can be rewritten in an equivalent implicit form:

$$\left(\frac{1}{\Delta t}\mathbf{M} + \mathbf{K}\right)\mathbf{T}^{n+1} - \frac{1}{\Delta t}\mathbf{M}\,\mathbf{T}^n - \mathbf{f}^{n+1} - \mathbf{g}^{n+1} = \mathbf{0}. \tag{2.18}$$

If we use the surrogate prediction $\hat{\mathbf{T}}^{n+1}$ instead of the exact solution $\mathbf{T}^{n+1}$, the equation is usually not satisfied exactly. The difference defines the physics residual at the first predicted step

$$\mathbf{R}^{n+1} = \left(\frac{1}{\Delta t}\mathbf{M} + \mathbf{K}\right)\hat{\mathbf{T}}^{n+1} - \frac{1}{\Delta t}\mathbf{M}\,\mathbf{T}^n - \mathbf{f}^{n+1} - \mathbf{g}^{n+1}, \tag{2.19}$$

where $\mathbf{M}$ and $\mathbf{K}$ are the FEM mass and conductivity matrices, $\mathbf{f}^{n+1}$ is the volumetric heat source vector, and $\mathbf{g}^{n+1}$ is the Neumann boundary load.

For a multi-step rollout over w time steps, the residual is calculated recursively. The surrogate prediction $\hat{\mathbf{T}}^{n+k-1}$ from the previous step replaces the known state $\mathbf{T}^n$ in the mass-matrix term. For $k = 1, \ldots, w$

$$\mathbf{R}^{n+k} = \left(\frac{1}{\Delta t}\mathbf{M} + \mathbf{K}\right)\hat{\mathbf{T}}^{n+k} - \frac{1}{\Delta t}\mathbf{M}\,\hat{\mathbf{T}}^{n+k-1} - \mathbf{f}^{n+k} - \mathbf{g}^{n+k}, \tag{2.20}$$

where, $\hat{\mathbf{T}}^n \equiv \mathbf{T}^n$ is set as the initial state. Because of this recursive structure, errors from earlier steps carry forward, which encourages the model to make predictions that are consistent over time.

Before calculating the loss, the residual at each node is normalised by the corresponding diagonal entry of the implicit system matrix

$$D_i = \left[\operatorname{diag}\left(\frac{\mathbf{M}}{\Delta t} + \mathbf{K}\right)\right]_i, \tag{2.21}$$

giving the Jacobi-scaled residual

$$\widetilde{R}_i^{n+k} = \frac{R_i^{n+k}}{D_i}. \tag{2.22}$$

This scaling is important because the raw residual values can vary a lot between interior nodes, boundary nodes, and areas with different materials or heat sources. Dividing by D_i makes the residual dimensionless and puts contributions from different parts of the mesh on a similar scale. If radiation is present, the relevant radiation Jacobian terms are included in D_i.

The scalar physics-informed training loss is defined as the mean squared normalised residual over all N_{active} free nodes and all w predicted time steps in the bundle

$$\mathcal{L}_{\text{phys}} = \frac{1}{w\,N_{\text{active}}}\sum_{s=1}^{w}\sum_{i=1}^{N_{\text{active}}}\left(\widetilde{R}_i^{n+s}\right)^2. \tag{2.23}$$

By minimising $\mathcal{L}_{\text{phys}}$, the surrogate learns to produce temperature predictions that match the backward Euler FEM discretisation at each step of the rollout, without needing any labelled solution data.

2.4.3 Graph inputs and outputs

The thermal surrogate model operates on a graph representation of the finite-element mesh used for the transient heat-conduction problem as described in Section 2.2.

Before the neural network is evaluated, the full graph is reduced to the active prediction domain. For standard thermal problems, the active graph contains only the free degrees of freedom, i.e., all non-Dirichlet nodes. Dirichlet boundary values are therefore not learned directly. The network predicts temperatures only at free nodes, after which the full temperature vector is reconstructed by inserting the prescribed boundary temperatures at Dirichlet nodes. For workpiece-only induction-heating problems (experiments 3.3.1, 3.3.2 and thermal generalisation and ablation study), the active graph is instead restricted to nodes belonging to the workpiece region. The original full-mesh indexing is stored in a node-mapping array, so that predictions on the active graph can be expanded back to the full finite-element vector before evaluating the residual.

Each node is described by boundary, state, material, and source information. For constant-conductivity thermal problems, the current implementation uses the node feature vector

$$\mathbf{x}_i = \left[\mathbf{b}_i, T_i^n, m_i, q_i^N, T_i^D, h_i, T_i^\infty, \widehat{Q}_i, \log_{10}(Q_\star + 1)\right], \tag{2.24}$$

where $\mathbf{b}_i$ is a one-hot encoding of the node type, T_i^n is the current nodal temperature, m_i is the material or conductivity field, q_i^N is the prescribed Neumann flux value, T_i^D is the prescribed Dirichlet temperature, h_i is the Robin or convection heat-transfer coefficient, T_i^∞ is the corresponding ambient temperature, $\widehat{Q}_i$ is the normalised nodal heat-source pattern, and $Q_\star$ is the heat-source magnitude scale. The node-type encoding is given in table 2.1.

Node type	Encoding component
Dirichlet	0
Neumann	1
Interior	2
Robin/convection/radiation	3

Table 2.1: One-hot node-type categories used in the thermal graph representation. Exactly one component is active for each node.

The constant-conductivity feature vector has 13 scalar entries: four node-type indicators and nine scalar thermal features. This encoding gives the network access to the current thermal state, the local boundary condition, the local material field, and the spatial heat-source distribution. In induction-heating problems, the volumetric heat source is obtained from an electromagnetic eddy-current solve and sampled onto the thermal graph as a nodal source field.

For temperature-dependent conductivity problems, the material part of the feature vector is extended. Instead of using only a scalar material field, the graph includes

$$[k(T_i^n), k_{\mathrm{ref},1}, \ldots, k_{\mathrm{ref},N_{\mathrm{ref}}}], \tag{2.25}$$

where $k(T_i^n)$ is the conductivity evaluated at the current nodal temperature and $k_{\mathrm{ref},j}$ are fixed samples from the one-dimensional conductivity lookup table. The node feature dimension, therefore, becomes

$$13 + N_{\mathrm{ref}}. \tag{2.26}$$

This is an important implementation detail: the network is not merely told that the material is nonlinear. It is given both the current local conductivity and a sampled representation of the conductivity law.

Each directed edge carries geometric and state-difference information between a sender node i and a receiver node j,

$$\mathbf{e}_{ij} = \left[x_j - x_i, y_j - y_i, \sqrt{(x_j - x_i)^2 + (y_j - y_i)^2}, T_j^n - T_i^n \right]. \tag{2.27}$$

For axisymmetric meshes, the spatial coordinates correspond to the two-dimensional computational domain, with the radial coordinate entering the finite-element weak form separately through the axisymmetric Jacobian. The relative positions and distances are static and can be cached, while the temperature-difference feature is rebuilt whenever a graph is constructed. This gives the message-passing network information about both local mesh geometry and the current thermal gradient proxy across graph edges.

The graph creator also stores a small global feature vector,

$$\mathbf{g} = [t^n, N], \tag{2.28}$$

where N is the number of graph nodes.

The thermal network predicts a temporal bundle of future temperatures on the active graph. For each active node, the output is

$$\mathbf{y}_i = \left[T_i^{n+1}, T_i^{n+2}, \ldots, T_i^{n+w} \right], \tag{2.29}$$

where w is the bundle length, called `time_window` in the implementation. The default value used in the thermal experiments is $w = 20$.

Thus, the network output has shape $N_{\text{active}} \times w$. The predicted active-node temperatures are expanded back to the full finite-element vector before residual evaluation. For ordinary thermal problems, this reconstruction is

$$T_{\text{full}}^{n+s} = \begin{cases} T_{\text{pred}}^{n+s}, & \text{on free nodes,} \\ T^{D,n+s}, & \text{on Dirichlet nodes,} \end{cases} \qquad s = 1, \ldots, w. \tag{2.30}$$

For workpiece-only induction-heating problems, the prediction is inserted only on workpiece nodes, while non-workpiece nodes are filled with their initial temperature when reconstructing the full mesh state.

The MeshGraphNet architecture used for the thermal surrogate consists of a two-layer node encoder, a two-layer edge encoder, a processor with 12 message-passing layers, and a decoder that maps the final node latent state to the temporal temperature bundle. The latent dimension is 128. Each processor layer updates edge features from sender-node, receiver-node, edge, and graph latents; updates node features from the current node latent and aggregated incoming edge messages; and updates the graph latent from mean node and edge latents. For the default bundle length $w = 20$, the decoder uses a small one-dimensional convolutional output head over the node latent representation. For other supported bundle lengths, the output head is adjusted or replaced by a linear projection.

2.4.4 Training strategy

The thermal surrogate is trained with a FEM based residual as described in section 2.4.2. Dirichlet values are enforced before evaluating the residual. The residual is then

filtered back to the free degrees of freedom. For workpiece-only problems, it is further restricted to workpiece nodes.

For nonlinear thermal problems, the residual assembly is updated with the predicted state. If the conductivity is temperature-dependent, the stiffness matrix is reassembled using $k(T_{\text{pred}})$. The temporal bundle is trained autoregressively.

During training, Gaussian noise is injected into the temperature state used to construct the graph features,

$$T^n_{\text{graph}} = T^n + \epsilon, \qquad \epsilon \sim \mathcal{N}(0, \sigma^2_{\text{noise}}). \tag{2.31}$$

The finite-element residual, however, is still evaluated using the noiseless physical state. This separates robustness training from the physics target: the model learns to predict physically consistent future states from slightly perturbed graph inputs, which should reduce sensitivity to accumulated rollout error [9].

For a single thermal problem, each epoch starts from the initial condition and rolls through the full time interval in temporal windows. Each window performs one optimiser step: a graph is built from the current state, the model predicts a temporal bundle, the finite-element residual loss is computed, gradients are backpropagated, and the rollout advances using the final predicted state. The default optimiser is Adam with a starting learning rate 10^{-3} to 10^{-5}, followed by an exponential learning-rate scheduler with decay factor 0.9995. Gradients are clipped to a norm of 1.0.

The trainer also supports multi-problem training. In this setting, each training problem has its own mesh, graph creator, boundary data, source data, and finite-element residual assembly. At each epoch, the problem order is shuffled. If the batch size is greater than one, active-node graphs from multiple problems are batched using PyTorch Geometric [56] and passed through the neural network in a single forward pass. The finite-element residual is then evaluated separately for each problem, because the matrices, boundary conditions, sources, and active-node masks are problem-specific. The batch loss is the average of the individual problem losses. This is the training mode used for induction-heating generalisation studies over different current and frequency settings (see Section 3.4).

Validation and evaluation are performed by autoregressive rollout over the full time grid. A reference solution is generated to compute relative L^2 errors over all nodes, or only over workpiece nodes for workpiece-only induction-heating cases. Predicted and reference fields are exported to `VTK` [57] for visualisation, and `matplotlib` [58] is used for creating figures.

2.5 Electromagnetic model

The electromagnetic surrogate targets the time-harmonic magnetoquasistatic problem in an axisymmetric domain. Unlike the thermal model, the problem is steady-state: given a fixed operating point defined by coil current and frequency, the surrogate predicts the complex magnetic vector potential in a single forward pass, from which the volumetric Joule heating source fed to the thermal model is derived. The formulation is nondimensionalised with respect to field variables and material coefficients while retaining the physical mesh coordinates, a non-standard choice motivated by the goal of sharing a single mesh between the EM and thermal solvers. The magnetostatic case is introduced first as a simpler, real-valued special case before the full complex eddy-current formulation is developed.

2.5.1 Weak form of MQS

Using $A \equiv \hat{A}_\varphi(r,z)$ and test function $v \equiv v_\varphi(r,z)$, we obtain the axisymmetric bilinear form

$$a(A,v) = \int_\Omega \nu \left(r\, A_{,z}\, v_{,z} + \frac{1}{r} \frac{\partial(rA)}{\partial r} \frac{\partial(rv)}{\partial r} \right) d\Omega \; + \int_{\Omega_{\text{cond}}} j\, \omega\, \sigma\, r\, A\, v\, d\Omega, \tag{2.32}$$

and the linear functional for a prescribed stranded current density in the coil,

$$f(v) = \int_{\Omega_{\text{coil}}} r\, J_\varphi\, v\, d\Omega, \qquad J_\varphi = \frac{N_{\text{turns}}\, I_{\text{coil}}}{S_{\text{coil}}}. \tag{2.33}$$

The variational problem is: find $A \in V$, such that

$$a(A,v) = f(v) \quad \forall v \in V. \tag{2.34}$$

2.5.2 Nondimensionalisation

In this work, the MQS equations (see Equation (2.32)) are nondimensionalised, but the finite element mesh is kept in physical coordinates (r,z). That idea is motivated by the goal of using a single mesh for both the EM and thermal parts. Therefore, the weak form is not assembled on a dimensionless domain $\tilde{\Omega}$. Instead, the dimensionless weak form is pulled back to the physical domain Ω. This distinction is important for the implementation.

We introduce the characteristic scales for length r_*, magnetic vector potential A_* and magnetic permeability μ_* and define

$$\nu_* = \frac{1}{\mu_*}, \qquad J_* = \frac{A_*}{\mu_* r_*^2}, \qquad \sigma_* = \frac{J_*}{\omega A_*} = \frac{1}{\mu_* \omega r_*^2}. \tag{2.35}$$

The dimensionless material coefficients and field variables are then

$$\tilde{A} = \frac{A}{A_*}, \qquad \tilde{v} = \frac{v}{A_*}, \qquad \tilde{\nu} = \frac{\nu}{\nu_*}, \qquad \tilde{\sigma} = \frac{\sigma}{\sigma_*}, \qquad \tilde{J}_\varphi = \frac{J_\varphi}{J_*}. \tag{2.36}$$

If we scale the domain, the nondimensional weak form would read:

$$\int_{\tilde{\Omega}} \tilde{\nu} \left(\tilde{r}\, \tilde{A}_{,\tilde{z}}\, \tilde{v}_{,\tilde{z}} + \frac{1}{\tilde{r}} \frac{\partial(\tilde{r}\tilde{A})}{\partial \tilde{r}} \frac{\partial(\tilde{r}\tilde{v})}{\partial \tilde{r}} \right) d\tilde{\Omega} + \int_{\tilde{\Omega}_{\text{cond}}} j\, \tilde{\sigma}\, \tilde{r}\, \tilde{A}\, \tilde{v}\, d\tilde{\Omega} = \int_{\tilde{\Omega}_{\text{coil}}} \tilde{r}\, \tilde{J}_\varphi\, \tilde{v}\, d\tilde{\Omega}. \tag{2.37}$$

Since the numerical discretisation is performed on the physical mesh, the dimensionless weak form (Equation (2.37)) must be rewritten in terms of the physical coordinates (r,z). Using

$$d\tilde{\Omega} = \frac{1}{r_*^2}\, d\Omega, \qquad \frac{\partial}{\partial \tilde{z}} = r_* \frac{\partial}{\partial z}, \qquad \frac{\partial}{\partial \tilde{r}} = r_* \frac{\partial}{\partial r}, \tag{2.38}$$

we obtain

$$\frac{\partial(\tilde{r}\tilde{A})}{\partial \tilde{r}} = \frac{\partial}{\partial \tilde{r}} \left(\frac{r}{r_*} \tilde{A} \right) = \frac{\partial(r\tilde{A})}{\partial r}, \tag{2.39}$$

and analogously for the test function $\tilde{v}$. Therefore, the pulled-back dimensionless weak form on the physical domain becomes

$$\int_\Omega \tilde{\nu} \left(\frac{r}{r_*}\, \tilde{A}_{,z}\, \tilde{v}_{,z} + \frac{1}{r\, r_*} \frac{\partial(r\tilde{A})}{\partial r} \frac{\partial(r\tilde{v})}{\partial r} \right) d\Omega + \int_{\Omega_{\text{cond}}} j\, \tilde{\sigma}\, \frac{r}{r_*^3}\, \tilde{A}\, \tilde{v}\, d\Omega = \int_{\Omega_{\text{coil}}} \frac{r}{r_*^3}\, \tilde{J}_\varphi\, \tilde{v}\, d\Omega. \tag{2.40}$$

Equation (2.40) is the form that is assembled numerically in the present work. It is dimensionless with respect to the field variables and material coefficients, while the integration is still performed on the physical mesh.

With the choice

$$\mu_* = \mu_0, \tag{2.41}$$

the dimensionless reluctivity becomes

$$\tilde{\nu} = \frac{\nu}{\nu_*} = \frac{1/(\mu_0 \mu_r)}{1/\mu_0} = \frac{1}{\mu_r}. \tag{2.42}$$

Likewise,

$$\tilde{\sigma} = \frac{\sigma}{\sigma_*}, \qquad \tilde{J}_\varphi = \frac{J_\varphi}{J_*}, \tag{2.43}$$

with

$$J_\varphi = \frac{N_{\text{turns}} I_{\text{coil}}}{S_{\text{coil}} f}, \tag{2.44}$$

where f is the fill factor. Hence, when the weak form is assembled in the form (2.40), the factors involving ω, μ_0, and r_* are absorbed into the dimensionless coefficients $\tilde{\sigma}$, $\tilde{\nu}$, and $\tilde{J}_\varphi$, while the geometric factors r/r_*, $1/(r r_*)$, and r/r_*^3 remain explicitly in the integrals because the mesh itself is not rescaled.

2.5.3 Magnetostatic case

When $\sigma = 0$ in all regions, the imaginary part of the sesquilinear form in (2.32) vanishes, and the problem becomes real-valued. In this case, the discrete system takes the form

$$Ka = f, \tag{2.45}$$

where K is the stiffness matrix and a is the vector of unknown degrees of freedom. Assuming that Dirichlet constraints remove all null modes, the reduced matrix on the free degrees of freedom is symmetric and positive definite.

For this real coercive problem, the solution can be characterised as the minimiser of the quadratic energy functional

$$\Pi(a) = \frac{1}{2} a^\top K a - f^\top a. \tag{2.46}$$

Its gradient is

$$\nabla \Pi(a) = Ka - f, \tag{2.47}$$

so the stationary point of Π satisfies the discrete system $Ka = f$. Since K is symmetric positive definite, this stationary point is the unique minimiser:

$$a^\star = \operatorname*{argmin}_{b \in \mathbb{R}^{N_f}} \Pi(b), \tag{2.48}$$

where N_f is the number of free degrees of freedom.

At the exact discrete solution $a^\star$, where $Ka^\star = f$, the energy takes the value

$$\Pi(a^\star) = \frac{1}{2}(a^\star)^\top K a^\star - f^\top a^\star = \frac{1}{2}(a^\star)^\top f - (a^\star)^\top f = -\frac{1}{2}(a^\star)^\top f. \tag{2.49}$$

Hence, the minimum value of the energy functional is generally not zero. This is expected and does not indicate an error. Moreover, for any a, the difference to the minimum satisfies

$$\Pi(a) - \Pi(a^\star) = \frac{1}{2}(a - a^\star)^\top K(a - a^\star) \geq 0, \tag{2.50}$$

which shows that the energy gap is a natural error measure in the K-norm.

In practice, this means that an energy-based loss does not converge to zero, but rather to the problem-dependent minimum value $-\frac{1}{2}(a^\star)^\top f$.

2.5.4 Eddy current case

For the full-time-harmonic electromagnetic problem, the PI-GNN output changes from one channel to two channels, corresponding to the real and imaginary parts of the solution.

Let us consider the harmonic weak form: find $u \in V$, such that for all $v \in V$,

$$a(u, v) = \ell(v), \tag{2.51}$$

where the sesquilinear form is written as

$$a(u, v) = a_r(u, v) + i\, a_i(u, v), \tag{2.52}$$

with a_r denoting the magnetic stiffness contribution and a_i the conductivity contribution.

After finite element discretisation and the enforcement of Dirichlet boundary conditions, we get

$$(K_r + iK_i)(u_r + iu_i) = f, \qquad (f \text{ typically real}), \tag{2.53}$$

which is equivalent to the real block system

$$\begin{bmatrix} K_r & -K_i \\ K_i & K_r \end{bmatrix} \begin{bmatrix} u_r \\ u_i \end{bmatrix} = \begin{bmatrix} f \\ 0 \end{bmatrix}. \tag{2.54}$$

In general, the block matrix in (2.54) is nonsymmetric whenever $K_i \neq 0$. Since the Hessian of a twice continuously differentiable real-valued scalar functional is symmetric, the full harmonic problem cannot, in general, be obtained as the Euler–Lagrange equation of a real scalar minimisation functional. Only in the special case $K_i = 0$, i.e., magnetostatics, we recover the usual symmetric coercive structure as discussed previously in Subsection 2.5.3.

Therefore, instead of searching for a true energy functional, we minimise a least-squares-type residual functional. Define the complex residual

$$r(u) := (K_r + iK_i)u - f. \tag{2.55}$$

A plain least-squares choice would be

$$\mathcal{J}_{\mathrm{LS}}(u) = \frac{1}{2}\, r(u)^{\mathrm{H}} r(u). \tag{2.56}$$

However, this treats the residual in the Euclidean coefficient norm and ignores the finite element metric. A more finite-element-consistent choice is to measure the residual in the discrete dual norm induced by the L^2 inner product, using the lumped mass matrix M_L:

$$\mathcal{J}(u) = \frac{1}{2}\, r(u)^{\mathrm{H}} M_L^{-1} r(u). \tag{2.57}$$

Here, M_L denotes the lumped axisymmetric mass matrix. Starting from the consistent mass matrix

$$(M)_{ij} = \int_\Omega \hat{r}\, \phi_i \phi_j \, d\Omega, \tag{2.58}$$

the lumped mass entries are defined by row summation,

$$m_i := \sum_j M_{ij}, \qquad M_L = \mathrm{diag}(m_i). \tag{2.59}$$

Hence,

$$\|r\|^2_{M_L^{-1}} = r^{\mathrm{H}} M_L^{-1} r = \sum_i \frac{|r_i|^2}{m_i}, \tag{2.60}$$

which is the discrete Riesz-map weighting associated with the finite element L^2 inner product, and therefore provides a natural norm for residuals viewed as linear functionals on the test space [59].

Splitting the residual into real and imaginary parts,

$$r_r = K_r u_r - K_i u_i - f, \qquad r_i = K_i u_r + K_r u_i, \tag{2.61}$$

the functional becomes

$$\mathcal{J}(u) = \frac{1}{2}\left(r_r^\top M_L^{-1} r_r + r_i^\top M_L^{-1} r_i \right). \tag{2.62}$$

In practice, the weighting is especially simple because M_L is diagonal after lumping. Therefore, no linear solve is required:

$$\mathcal{J}(u) = \frac{1}{2} \sum_i \left(\frac{(r_r)_i^2}{m_i} + \frac{(r_i)_i^2}{m_i} \right). \tag{2.63}$$

After restricting to the free degrees of freedom, this is evaluated with the corresponding diagonal entries of $M_{L,ff}$.

2.5.5 Graph inputs and outputs

The electromagnetic surrogate model operates directly on a graph representation of the finite-element mesh. For the axisymmetric two-dimensional problems considered here, each graph node corresponds to one mesh vertex in the (r, z) domain. The default graph connectivity follows the finite-element topology: directed edges are created between vertex pairs belonging to the same element, boundary-segment edges are included, and self-loops are added.

Dirichlet boundary conditions are not learned directly. Nodes located on prescribed Dirichlet boundaries are marked during graph construction and are removed from the prediction graph before message passing. The neural network, therefore, predicts only the free degrees of freedom. Before evaluating the finite-element residual, the full solution vector is reconstructed by inserting the network prediction at free nodes and the prescribed boundary values at Dirichlet nodes. This avoids asking the network to relearn fixed boundary values and makes the physics loss consistent with the constrained FEM system.

Each node is described by geometric, boundary, material, and source information. The current two-dimensional electromagnetic model uses the following node feature vector:

$$\mathbf{x}_i = \left[\frac{r_i}{r_\star}, \frac{z_i}{r_\star}, \texttt{is_dirichlet}_i, \texttt{is_interior}_i, m_i, A_i^D, \ln(1+\sigma_i), \ln\left(1+\mu_r^i\right), J_i \right], \tag{2.64}$$

where $r_\star$ is the characteristic length scale, m_i is the material encoding given by Table 2.2, A_i^D is the prescribed Dirichlet value, σ_i is the nondimensional nodal conductivity, and J_i is the nondimensional coil current-density source term. The conductivity feature is nonzero in the conducting workpiece, while the current-density feature is nonzero in the coil region. This gives the network access to both the local material distribution and the excitation pattern.

Region	Encoding
Air	0
Coil	1
Workpiece	2

Table 2.2: Material encoding used in the electromagnetic graph representation.

Each directed edge carries relative geometric information between a sender node i and a receiver node j,

$$\mathbf{e}_{ij} = \left[\frac{r_j - r_i}{r_\star}, \frac{z_j - z_i}{r_\star}, \frac{\sqrt{(r_j - r_i)^2 + (z_j - z_i)^2}}{r_\star} \right]. \tag{2.65}$$

These edge features provide the message-passing network with local metric information without requiring the mesh to be structured. They also make the model sensitive to element size and orientation, which is important because the finite-element residual depends on spatial derivatives of the magnetic vector potential.

In addition to node and edge attributes, each graph contains global parameters describing the operating point of the electromagnetic problem. The current implementation uses two global features: a frequency-like parameter,

$$\kappa = \omega \mu_\star r_\star^2, \tag{2.66}$$

and the normalised coil current. Both are logarithmically scaled before being passed to the network. These global quantities are encoded once per graph and injected into every message-passing layer, allowing a single surrogate model to condition its prediction on changes in excitation frequency and current.

The network output depends on the electromagnetic formulation. The active implementation is the complex eddy-current formulation, where the network predicts the real and imaginary parts of the azimuthal magnetic vector potential $A = A_{\text{real}} + iA_{\text{imag}}$ at each free node. The real magnetostatic formulation predicts only the real vector potential. A mixed A–ϕ formulation is present as a scaffold but is not part of the current working training path and is left for future work. An overview is presented in a table 2.3.

Formulation	Status	Output per free node
Real magnetostatic	supported	A_{real}
Complex eddy-current	active default	$A_{\text{real}}, A_{\text{imag}}$
Mixed A–ϕ	scaffolded, not fully working	$A_{\text{real}}, A_{\text{imag}}, \phi_{\text{real}}, \phi_{\text{imag}}$

Table 2.3: Network outputs for the electromagnetic formulations implemented in the codebase.

For complex eddy-current simulations, the free-node prediction is inserted into the full vector as

$$A_{\text{full}} = \begin{cases} A_{\text{pred}}, & \text{on free nodes,} \\ A^D, & \text{on Dirichlet nodes,} \end{cases} \tag{2.67}$$

with the imaginary part of the Dirichlet value set to zero. In experiments, axis regularity is enforced by interpreting the raw neural-network output as a scaled quantity $\tilde{A}$ and applying

$$A = \frac{r}{r_\star}\tilde{A}. \tag{2.68}$$

This transformation imposes regular behaviour near the symmetry axis $r = 0$, which is physically appropriate for the azimuthal magnetic vector potential in an axisymmetric formulation.

2.5.6 Training strategy

The electromagnetic surrogate is trained by minimising the finite-element-consistent least-squares residual functional $\mathcal{J}(u)$ (see Equation (2.63)). Dirichlet values are enforced before evaluating the residual, which is then restricted to the free degrees of freedom.

Since the eddy-current problem is time-harmonic, there is no time-stepping loop and no temporal bundle: the network predicts the full steady-state solution in a single forward pass. Each training step, therefore, constructs a graph from the mesh and operating-point parameters, evaluates the model to obtain the real and imaginary parts of the magnetic vector potential at all free nodes, reconstructs the full solution vector, and computes $\mathcal{J}$ as the training loss.

The architecture is identical to the thermal surrogate - a two-layer node encoder, a two-layer edge encoder, a 12-layer message-passing processor with latent dimension 128, and a decoder - except that the output head is a linear projection to two scalar channels (A_{real}, A_{imag}) per free node rather than a temporal bundle.

The default optimiser is Adam with an initial learning rate of 10^{-3}, followed by an exponential learning-rate scheduler with decay factor 0.9995, and gradient clipping to a norm of 1.0. Multi-problem training follows the same batching strategy as the thermal case: graphs from different operating points (e.g., current, frequency, μ_r or σ settings) are batched via PyTorch Geometric for a single forward pass, while the residual $\mathcal{J}$ is evaluated separately per problem and the batch loss is taken as the average across the problems in the batch.

Evaluation is performed by a single forward pass over each validation problem. Relative L^2 errors are computed over all free nodes, and predicted fields are exported to VTK for visualisation.

2.6 Error metrics and verification

Before presenting the verification and generalisation results, we define the error metrics and scalar diagnostic quantities used throughout the results to assess surrogate accuracy and computational performance.

Pointwise relative error

For error distribution visualisation at time step n, the relative metric is used

$$\epsilon_{\text{rel}}^n = \frac{\tilde{u}_v^n - u_v^n}{\max_{v,n}(u_v^n) - \min_{v,n}(u_v^n)}, \tag{2.69}$$

where $\tilde{u}_v^n$ and u_v^n denote the surrogate prediction and reference solution at node v and time step n, respectively. The denominator normalises by the global range of the

reference field across all nodes and time steps, so that the metric remains bounded and comparable across problems with different field magnitudes. This quantity is used in the two-dimensional error field plots throughout Sections 3.1–3.3 to identify where spatial errors are concentrated.

Relative L_2 error

The relative L_2 error is used to measure the overall accuracy of the surrogate prediction against the reference solution across the full spatial field and rollout horizon:

$$\varepsilon_{L_2} = \sqrt{\frac{\sum_{j=1}^{N_t} \sum_{i=1}^{N_\phi} |\tilde{y}_i^j - y_i^j|^2}{\sum_{j=1}^{N_t} \sum_{i=1}^{N_\phi} |y_i^j|^2}}, \tag{2.70}$$

where N_t is the number of time steps in the rollout, N_ϕ is the number of mesh nodes, and y_i^j and $\tilde{y}_i^j$ are the reference and predicted field values at node i and time step j. For stationary problems such as the electromagnetic surrogate, $N_t = 1$ and the metric reduces to a standard spatial L_2 norm. This metric is used throughout the results chapter as the primary scalar accuracy measure for both field-level comparisons and generalisation studies.

Inference speed-up

The inference speed-up

$$S = \frac{t_{\text{FEM}}}{t_{\text{surrogate}}} \tag{2.71}$$

is used to quantify the computational advantage of the surrogate relative to the FEM reference solver for an equivalent problem. Values of $S > 1$ indicate that the surrogate is faster than FEM; values below unity indicate that the surrogate is slower. As discussed in Section 3.4, S is evaluated from mean runtimes over repeated inference runs after a warm-up pass, and excludes the one-time offline training cost. The surrogate runtime includes graph construction, data transfer to the compute device, and neural-network inference, but excludes model loading from disk.

Scalar diagnostic quantities

In addition to the field-level metrics above, several scalar quantities are reported to facilitate direct comparison with reference data. For the electromagnetic surrogate, these include the total deposited power $P = \int_{\Omega_{\text{wp}}} Q \, d\Omega$ and the pointwise maximum Joule heat $Q_{\max}$. For the thermal surrogate, they include the final-time maximum temperature $T_{\max}(t_f)$ and the boundary-integrated heat fluxes q_{conv} and q_{rad}. These quantities aggregate the field prediction into physically interpretable outputs and are particularly sensitive to localised errors near the workpiece surface, providing a complementary diagnostic to the global L_2 metric.

Chapter 3

Results and Discussion

This chapter explains how we verify and evaluate the physics-informed graph neural network used for induction heating simulations and discusses the results. Before combining the two surrogate models, we carry out a step-by-step verification process.

First, we independently verify the thermal model, ensuring it accurately reproduces transient heat diffusion, handles boundary conditions, responds to heating sources, and accounts for nonlinear temperature effects. The electromagnetic surrogate is tested in a similar manner, in which we compare its predictions against standard magnetostatic and harmonic eddy-current test cases involving systems with one- and two-stranded coils.

Next, we link the two surrogates via the Joule heat source to form a comprehensive model of induction heating. The coupling interface is tested and verified against the TEAM 36 problem. We then conduct generalisation tests and an ablation study, and measure the achieved computational speed-up.

Model predictions are compared with analytical solutions and finite-element reference data (via our in-house solver, powered by `NGSolve` [60]) using both visual field comparisons and numerical error metrics. Sections 3.1, 3.2, and 3.3 verify each surrogate model on the problems for which it was trained. Finally, Section 3.4 tests the generalisation capabilities of the models on unseen cases, and Section 3.5 performs a physics-driven vs data-driven ablation study.

3.1 Verification of the thermal surrogate

We begin by verifying the thermal surrogate, as it is the second step in the induction-heating process and must accurately model heat transfer before we can test the full system. The verification moves from simple to more complex cases: first, an analytical transient diffusion problem, then a boundary-condition test, followed by a source-driven heating case, and finally a nonlinear temperature-dependent problem. This order helps us identify specific issues, such as errors in transient behaviour, problems with boundary conditions, incorrect responses to heat sources, or performance drops when nonlinear materials are introduced.

3.1.1 Analytical transient diffusion test

For the first verification test, we use a standard transient diffusion problem with a known analytical solution. This test looks at basic diffusion without internal heat sources, nonlinear effects, or changing material properties. It allows us to verify that

the PI-GNN predicts the correct time-dependent decay, satisfies homogeneous Dirichlet boundary conditions, and gives stable results over time.

The problem is posed on the unit square domain $\Omega = (0,1)^2$ with constant material properties $\rho = 1, c_p = 1, k = 0.1$ and zero forcing term. The initial condition is chosen as

$$T(x, y, 0) = 100 \sin(\pi x) \sin(\pi y), \tag{3.1}$$

which is an eigenfunction of the Laplace operator with homogeneous Dirichlet boundary conditions. The corresponding initial–boundary value problem reads

$$\begin{aligned} \rho c_p \partial_t T - \nabla \cdot (k \nabla T) &= 0, \quad (x, y) \in (0,1)^2,\ t > 0, \\ T(x, y, t) &= 0, \quad (x, y) \in \partial\Omega,\ t > 0, \\ T(x, y, 0) &= 100 \sin(\pi x) \sin(\pi y). \end{aligned} \tag{3.2}$$

For this problem, the exact solution is given by

$$T(x, y, t) = 100 e^{-2k/\rho c_p \pi^2 t} \sin(\pi x) \sin(\pi y), \tag{3.3}$$

which makes it possible to compare the surrogate's predictions with the analytical solution at every time step.

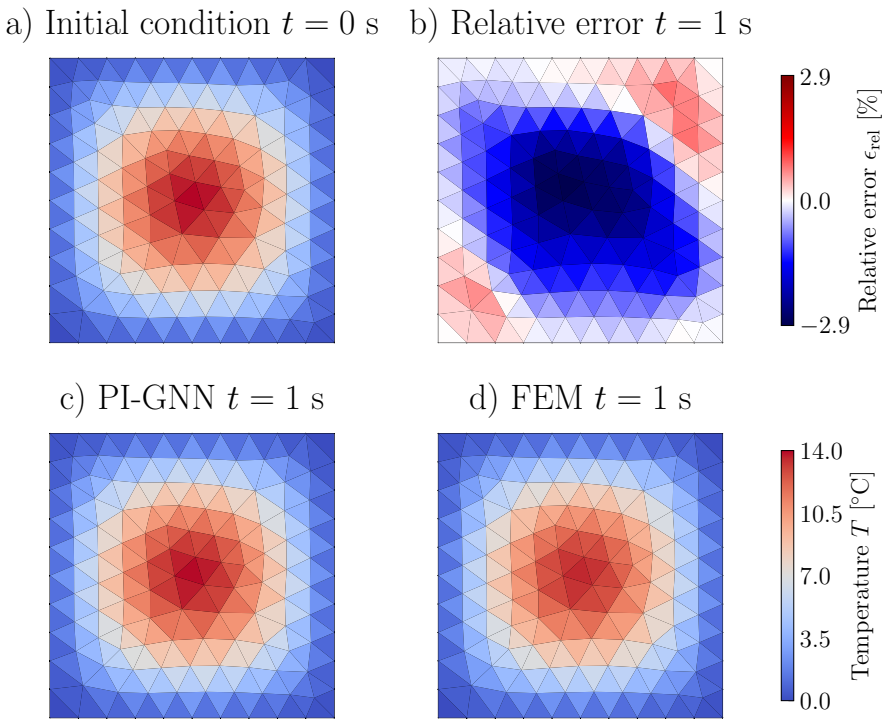


Figure 3.1: Analytical transient diffusion test on the unit square with homogeneous Dirichlet boundary conditions. Shown are a) the initial condition at $t = 0$ s, b) the relative error field, c) the reference solution given by FEM at $t = 1$ s and d) the PI-GNN prediction at $t = 1$ s.

Figure 3.1 shows the initial condition, the reference solution at 1s, the PI-GNN prediction at the same time, and the relative error field. The predicted temperature field matches the expected decayed eigenmode shape. The relative error is small, ranging in $[-2.9; 1.1]\%$, and smooth across the space, with no noticeable unwanted oscillations. Figure 3.2 shows how the middle probe at the centre of the domain changes over time

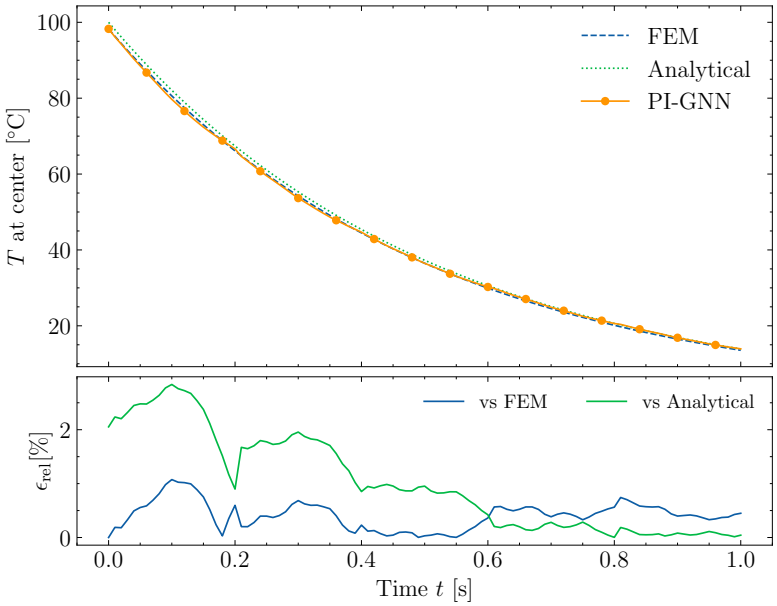


Figure 3.2: Temporal evolution of the relative L_2 error for the analytical transient diffusion test over the rollout interval $t \in [0, 1]$ s.

with FEM and analytical references. The relative error stays below 3% for the analytical solution and below 1% for FEM, using the same mesh discretisation throughout the interval $t \in [0, 1]$ s. Figure 3.3 shows how the relative L_2 error changes over time for

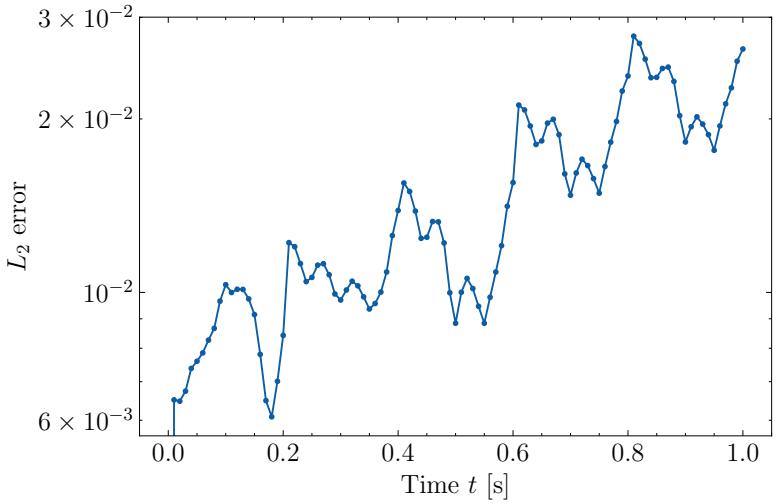


Figure 3.3: Temporal evolution of the relative L_2 error for the analytical transient diffusion test. The piecewise jumps are consistent with temporal bundling, in which the PI-GNN predicts 20 future time steps per rollout block before updating the graph state.

the analytical transient diffusion test. The error remains between about 6×10^{-3} and 3×10^{-2} throughout the simulation, indicating that the PI-GNN maintains a stable rollout without major error growth. There is a gradual increase in error over time, consistent with the expected accumulation of approximation errors in autoregressive prediction. The non-smooth behaviour and local jumps that appear at regular intervals are due to the temporal-bundling strategy: the network predicts a block of 20 future steps from a single input graph, then assembles a new graph for the next bundle. As

a result, small discontinuities at bundle boundaries are expected and do not indicate instability.

Since this benchmark uses constant coefficients, has no source term, and has an exact analytical solution, these results indicate that the surrogate can reproduce the decay of this single-mode diffusion benchmark before moving on to more complex thermal tests.

3.1.2 Boundary condition verification

This test, inspired by [61], evaluated whether the thermal surrogate can handle mixed boundary conditions in a transient heat-conduction problem. Unlike the analytical diffusion test in Section 3.1.1, which used only homogeneous Dirichlet boundaries, this case applies inhomogeneous Dirichlet, homogeneous Neumann, and Robin boundary conditions on a rectangular domain $\Omega = (0, 2) \times (0, 1)$. The goal is to check if the PI-GNN accurately captures the effects of set boundary temperatures, thermal insulation, and combined convective and radiative heat loss on the predicted temperature field.

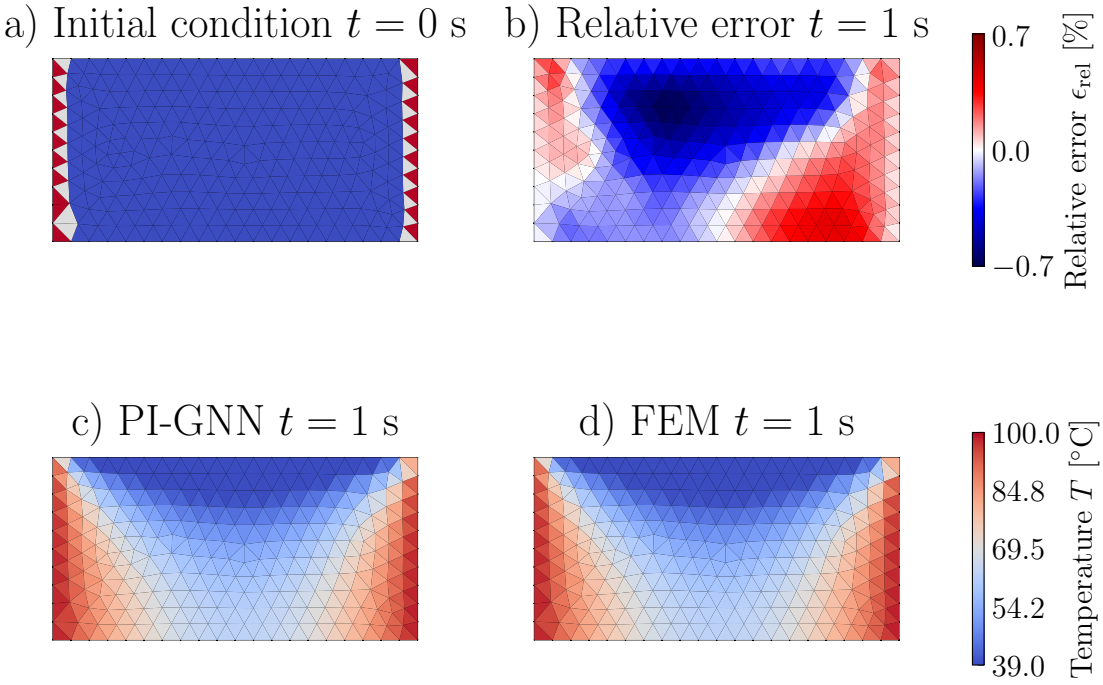


Figure 3.4: Boundary-condition verification on a rectangular domain with mixed thermal boundaries. The figure compares the FEM reference solution and the PI-GNN prediction, together with the corresponding error field, for a case with prescribed temperature boundaries on the left and right, an insulated bottom boundary, and convective–radiative heat loss at the top boundary.

The left and right boundaries are kept at $T_{\text{hot}} = 100\,^{\circ}\text{C}$, creating inhomogeneous Dirichlet conditions. The top boundary faces an ambient temperature of $T_{\text{amb}} = 20\,^{\circ}\text{C}$ and loses heat to the environment by convection with convective heat transfer coefficient $h = 10\,\text{W/m}^2\text{·K}$ and radiation with emissivity $\epsilon = 0.8$, resulting in a mixed boundary condition, modelled with a Robin boundary condition. The bottom boundary is thermally insulated, which means it has a homogeneous Neumann condition. This setup serves as a controlled benchmark to assess whether the surrogate can match both the temperature distribution and the heat fluxes induced by the boundaries. This verification is particularly relevant because, in induction-heating applications, surface heat losses can

strongly affect how the temperature changes over time. A thermal surrogate that gets the interior field right but misses the correct boundary heat exchange would not be reliable in a coupled setting. The problem reads

$$\begin{aligned} \rho c_p \partial_t T - \nabla \cdot (k \nabla T) &= 0, \quad (x, y) \in \Omega,\ t > 0, \\ T &= T_{\text{hot}}, \quad (x, y) \in \Gamma_D, \\ -k \nabla T \cdot n &= 0, \quad (x, y) \in \Gamma_N, \\ -k \nabla T \cdot n &= h_{\text{eff}} \, (T - T_{\text{amb}}), \quad (x, y) \in \Gamma_R. \end{aligned} \tag{3.4}$$

Figure 3.4 shows a comparison between the reference FEM solution and the PI-GNN prediction for the mixed-boundary-condition test case. The surrogate captures the main features of the temperature field, such as higher temperatures near the left and right heated boundaries and the cooling effect along the top boundary exposed to the ambient environment. The spatial temperature gradient between the hot-side boundaries and the cooled top boundary is also well represented, suggesting that the model correctly accounts for the combined effects of Dirichlet, insulated, and environmental heat-loss boundaries.

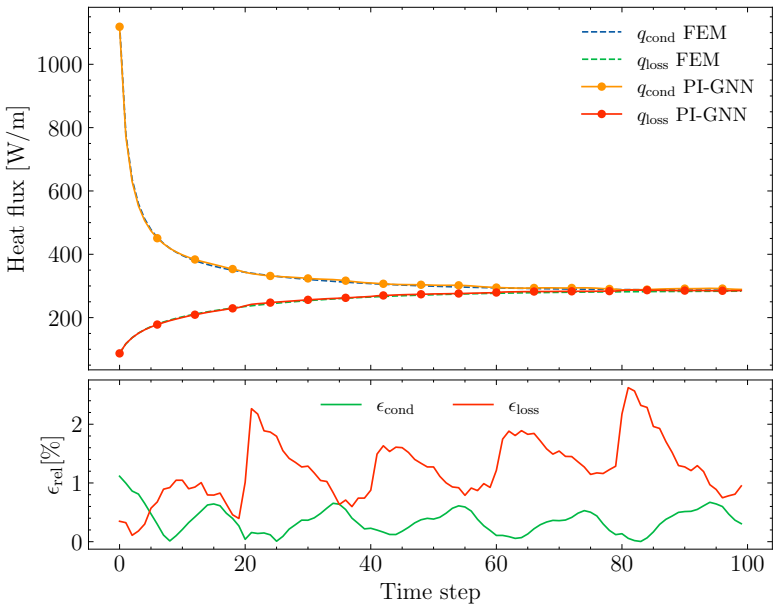


Figure 3.5: Comparison of the total heat flux through the top boundary and conductive heat flux from the sides for the FEM reference and the PI-GNN prediction in the mixed-boundary-condition test.

To assess the boundary treatment more directly, Figure 3.5 compares the heat flux lost $q_{\text{loss}} = q_{\text{conv}} + q_{\text{rad}}$ through the top boundary and gained q_{gain} from the side boundaries for both the FEM reference and the PI-GNN prediction. The convection, radiation and conductive heat flux are given by $q_{\text{conv}} = \left| \frac{1}{|\Gamma|} \int_\Gamma h \, \{T_{\text{amb}} - T(x, y)\} \, d\Gamma \right|$, $q_{\text{rad}} = \left| \frac{1}{|\Gamma|} \int_\Gamma \varepsilon \sigma \, \{T_{\text{amb}}^4 - T(x, y)^4\} \, d\Gamma \right|$ and $q_{\text{cond}} = \left| \frac{1}{|\Gamma|} \int_\Gamma -k \nabla T(x, y) \, d\Gamma \right|$ at each time step. The close agreement indicates that the surrogate reproduces the interior temperature distribution and matches the selected averaged boundary-flux quantities for this case. This is important for later induction-heating simulations, since inaccurate boundary fluxes would cause systematic errors in the thermal response over time.

3.1.3 Source-driven heating test

This test assesses whether the thermal surrogate accurately reproduces the system's response to an internal volumetric heat source.

The governing equation for this case is

$$\rho c_p \, \partial_t T - \nabla \cdot \big(k(T)\nabla T\big) = q(x, y), \tag{3.5}$$

with $\rho = 1, c_p = 1$ and $k = 3$. This step is important before moving on to induction-heating simulations, where the thermal field comes from Joule heating created by the electromagnetic problem. In this verification, the source term is set directly instead of coming from the electromagnetic surrogate, allowing the thermal model to be tested on its own. While this source does not capture all the details of electromagnetic power deposition in induction heating, it does show two key features: heating is concentrated near the surface, and the deposited power is not uniformly distributed.

The problem is set up on a rectangular domain. The left boundary is isolated, while the other boundaries have convective boundary conditions. The system follows the transient heat equation with a source term that changes with position. This source is meant to imitate the localised heating seen in induction heating, where eddy currents are strongest near the surface and decrease rapidly as you move deeper. This effect is described by the skin depth

$$\delta = \sqrt{\frac{2}{\omega \sigma \mu_0 \mu_r}}, \tag{3.6}$$

This value sets the distance over which the electromagnetic field, and the resulting Joule heating, fades with depth. The source term for this test is built to have the same spatial pattern

$$q(x, y) = q_0 \exp\left(-\frac{D - x}{\delta}\right) \exp\left(-\frac{(y - H/2)^2}{2\sigma_y^2}\right). \tag{3.7}$$

The first exponential term shows how Joule heating fades with depth, with δ acting as the skin depth from electromagnetic physics. Heating is strongest at the surface ($x = D$) and decreases over the distance δ as you move inward. The second term creates a Gaussian shape in the vertical direction, so heating is highest at the middle and drops off toward the edges. This matches the localised effect of the inductor. Together, these terms create a source that captures the two main features of induction heating: surface focus and side-to-side localization, without needing the electromagnetic model to be active. This approach links the thermal test directly to the electromagnetic physics of skin effect in Section 3.2.3. The same length scale δ that controls field penetration in the full problem is used here to set the thermal load, making sure the test matches real operating conditions.

Figure 3.6 compares the PI-GNN prediction with the FEM reference solution at $t = 1\,\mathrm{s}$. Panel (a) shows the initial condition at $t = 0\,\mathrm{s}$, where the temperature is uniform throughout the domain. Panel (d) shows the FEM reference at $t = 1\,\mathrm{s}$, where the localised volumetric source creates a clear hot spot near the centre-right, with temperatures reaching about 265 °C. The PI-GNN prediction in panel (c) closely matches this pattern, capturing both the location and size of the temperature peak. The relative error field in panel (b) supports this, showing that the maximum absolute relative error stays below 0.1% in most of the domain, with the largest errors near the heated area where temperature gradients are steepest. This indicates that the surrogate can accurately resolve a spatially non-uniform heat source.

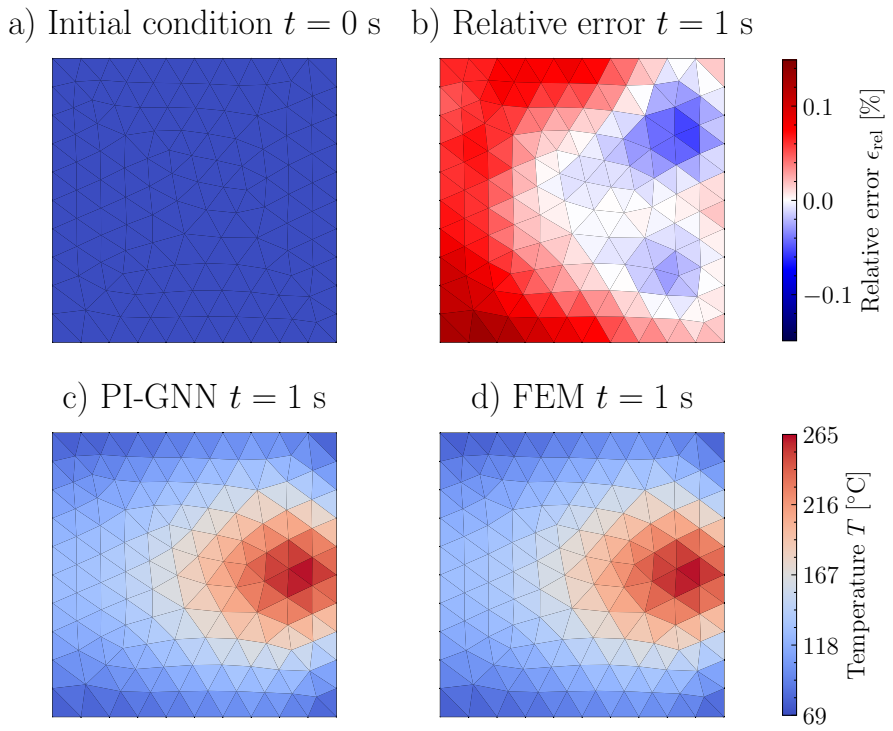
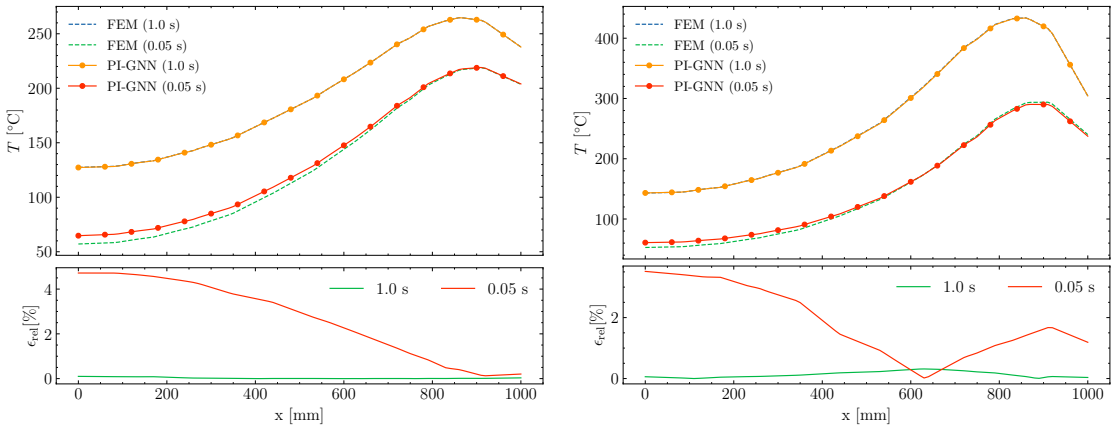


Figure 3.6: Volumetric source term verification test. The figure shows b) the relative error of c) predicted temperature and d) FEM reference solution.



(a) Constant $k = 3$

(b) Temperature varying thermal conductivity k given by a piecewise linear function (Equation (3.8)).

Figure 3.7: Temperature distribution over the middle line $y = 0.5\,\text{m}$ given by thermal surrogate and FEM for two time intervals $t = 0.05\,\text{s}$ and $t = 1\,\text{s}$ respectively. The figure also shows the relative error ϵ_{rel} for both times.

Figure 3.7a shows a one-dimensional view along the centreline $y = 0.5$m for the constant conductivity case $k = 3$. PI-GNN predictions are compared with the FEM reference at two simulation times: $t = 0.05\,\mathrm{s}$ and $t = 1\,\mathrm{s}$. At $t = 1\,\mathrm{s}$, the surrogate closely matches the FEM curve across the domain, and the relative error in the lower panel stays below 0.1% everywhere. At $t = 0.05\,\mathrm{s}$, when the thermal field is changing quickly and gradients are steep, the difference is larger. The relative error is about 5% near the left boundary and drops toward the right end, where the temperature is smoother. These results indicate that the surrogate's accuracy depends on how quickly the solution changes, with larger errors during rapid temperature changes and better performance as the temperature becomes smoother over time.

3.1.4 Nonlinear temperature-dependent thermal test

This test examines whether the thermal surrogate remains accurate when the heat equation is nonlinear, given that material properties depend on temperature. We use the same source-driven heating problem as in Section 3.1.3, but now the thermal conductivity varies with temperature rather than being constant. This setup allows us to focus on nonlinear heat conduction while keeping the geometry, boundary conditions, and heating pattern fixed.

The temperature dependence of thermal conductivity is given by the values listed in the table

$$k(T) = \begin{cases} 3.0, & T = 0\,^\circ\mathrm{C}, \\ 2.0, & T = 100\,^\circ\mathrm{C}, \\ 1.0, & T = 300\,^\circ\mathrm{C}, \end{cases} \tag{3.8}$$

with piecewise linear interpolation between the tabulated points used in the numerical model.

Since the thermal conductivity decreases with temperature, heat transport becomes less effective in the hotter regions of the domain. Compared with the constant-conductivity source-driven test, this is expected to produce a more localised thermal response, steeper gradients near the heated region, and a modified temperature evolution over time.

This test introduces nonlinearity only through the temperature dependence of thermal conductivity. It does not yet verify the full case in which k, c_p, and ρ vary with temperature simultaneously. However, it provides a controlled first assessment of the surrogate's robustness to state-dependent material coefficients.

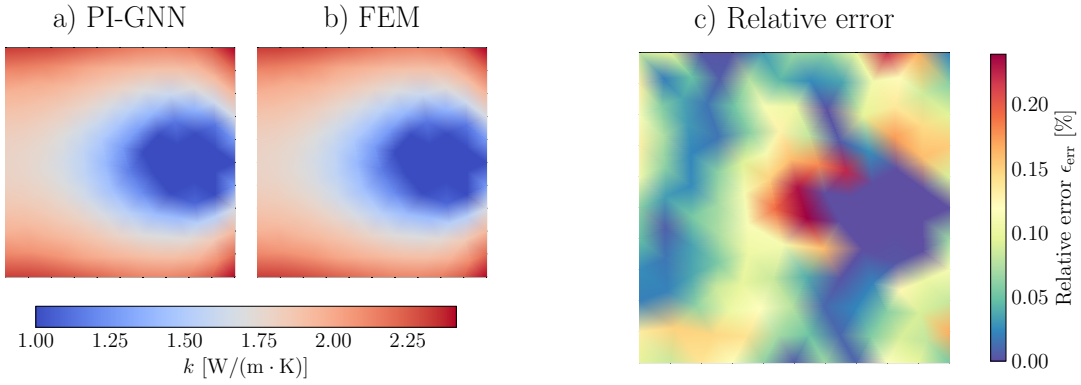


Figure 3.8: The figure shows k field: a) and b) PI-GNN vs FEM, c) relative error.

In the constant $k = 3$ case shown in Figure 3.7a, heat spreads more efficiently across the domain. In contrast, in the varying $k(T)$ case shown in 3.7b, the hotter region has lower conductivity, so heat moves away less effectively. This leads to a steeper temperature profile and a more pronounced hotspot. This difference is most pronounced at later times: the varying-$k(T)$ case attains a higher maximum temperature and exhibits a sharper profile in the hot region than the constant-k case. The PI-GNN results are close to the FEM profile in both cases, but the nonlinear case is a bit more challenging, especially at earlier times.

The larger errors seen early on are due to the link between prediction accuracy and how the stiffness matrix is assembled. As explained in Section 2.4.4, the backward Euler residual used in training evaluates k at the predicted state T^{n+1}, so the stiffness matrix $\mathbf{K}$ is rebuilt using $k(T_{\text{pred}})$ at each step. When the temperature field changes quickly at the start, even small errors in predicted temperature lead to an incorrectly assembled $\mathbf{K}$, since the constitutive law $k(T)$ directly connects temperature changes to conductivity changes. This makes the residual an imperfect training signal, so the surrogate has to adjust for a stiffness matrix that does not fully match the real physical state. As the solution gets closer to steady state and the temperature changes more slowly, these errors decrease, which matches the near-zero relative error seen at $t = 1$s. The same process, where early prediction errors affect later steps through a state-dependent material law, is also seen in the electromagnetics case in Section 3.3.2 for the TEAM 36 benchmark. This result shows it happens in nonlinear thermal problems as well.

To see if the surrogate captures the material's nonlinear behaviour, we compared the conductivity field from the PI-GNN temperature prediction with the FEM-derived conductivity field in Figure 3.8. The results show that the surrogate matches both the temperature field and the state-dependent conductivity distribution described by the constitutive law $k(T)$.

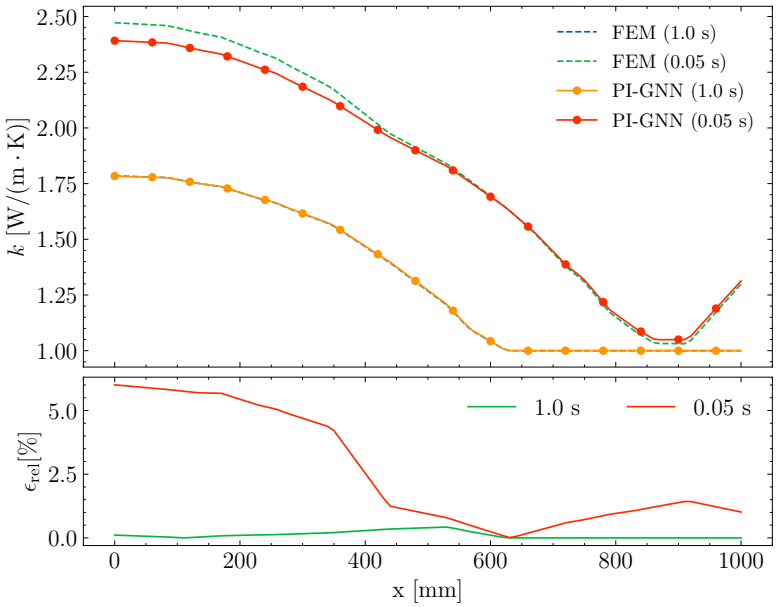


Figure 3.9: Computed k-field from temperature for FEM and PI-GNN surrogate for two time intervals $t = 0.05$s and $t = 1$s respectively. The relative error for $t = 1$s shows almost no deviation from the FEM.

The line plot along the midline in Figure 3.9 also indicates that the surrogate accurately reproduces the temperature-dependent conductivity field derived from the

predicted temperature for this controlled $k(T)$ test. The relative error at $t = 0.05$s starts at 5% for the beginning of the slice, and slowly reaches around 1% near the end of the line. The final error at $t = 1$s is near zero. This decrease in error over time matches the earlier explanation: as the temperature field becomes steady, the difference between the surrogate and true stiffness matrices becomes very small, and the physics-informed residual gives a more accurate training signal. Overall, these results show that even small temperature errors can cause noticeable changes in effective conductivity when the constitutive law depends on state, so accurate temperature prediction is essential for reliable material-field recovery.

3.2 Verification of the electromagnetic surrogate

The electromagnetic surrogate is verified in a similar step-by-step manner as the thermal model. We begin with a magnetostatic test case to confirm that the surrogate correctly resolves the real-valued magnetic vector potential operator in the absence of eddy currents, before moving on to the full harmonic eddy-current problem. Both tests compare the PI-GNN predictions against FEM reference solutions using visual field comparisons and numerical error metrics, and are carried out for systems with one and two stranded coils to assess whether the surrogate generalises to multi-source configurations.

The EM problem is posed in terms of the magnetic vector potential A. The surrogate must learn to solve both the real-valued magnetostatic operator and the complex-valued harmonic operator that governs eddy-current induction at a given excitation frequency. Unlike the thermal surrogate, the electromagnetic problem is stationary: for a given coil current and frequency, the field solution does not evolve in time but is solved once per time step of the coupled induction-heating simulation to provide the Joule heating source term to the thermal model.

For both cases, we compare the magnetic vector potential A directly, as it is the primary output of the surrogate.

3.2.1 Magnetostatic test

Two configurations are considered: a system with one stranded coil and a system with two stranded coils placed in the domain. Both cases use homogeneous Dirichlet boundary conditions on the outer boundary of the computational domain, enforcing that the magnetic vector potential vanishes sufficiently far from the source region. The coil is prescribed with a current of 3000A and $\mu_r = 100$ in the workpiece domain and is 1 in the air and coil domains.

Figure 3.10 shows the predicted magnetic vector potential field alongside the FEM reference and the relative error for the single-coil configuration. The PI-GNN reproduces the characteristic field distribution of an isolated coil, including the peak concentration of flux at the coil location and the smooth radial decay towards the outer boundary. The relative error remains below 0.1% across the domain and is largest near the left boundary of the workpiece, where the field gradients are steepest, while the error is negligibly small in the far-field.

Figure 3.11 shows the same comparison for the two-coil system, where the peak vector potential reaches approximately $5.0\,\mathrm{T}\cdot\mathrm{m}$, roughly $1.4\times$ higher than in the single-coil case due to the superposition of both sources. This configuration is more demanding because the combined flux pattern is more elongated and exhibits a flatter maximum region

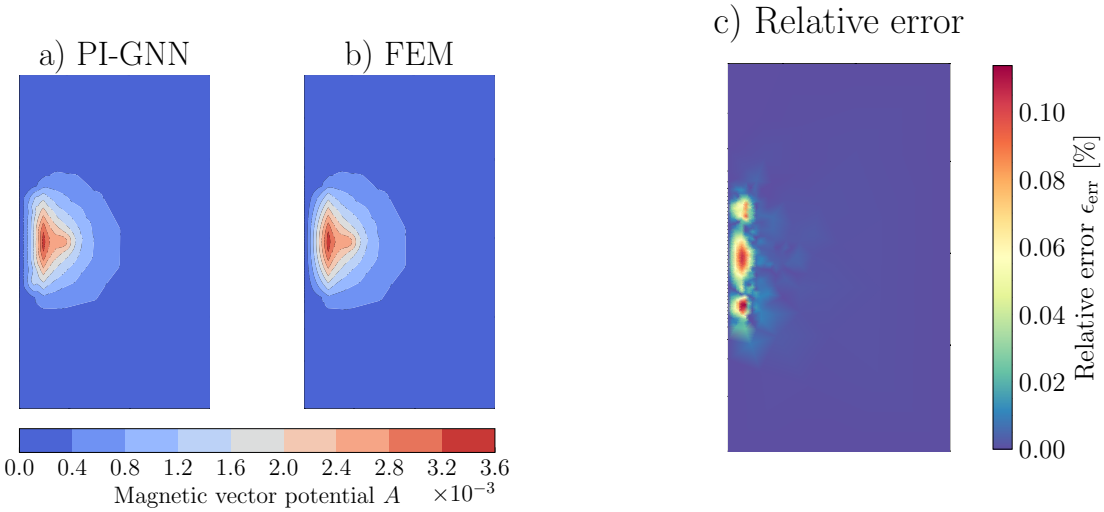


Figure 3.10: One coil system. a) and b) PI-GNN vs FEM, c) Relative error

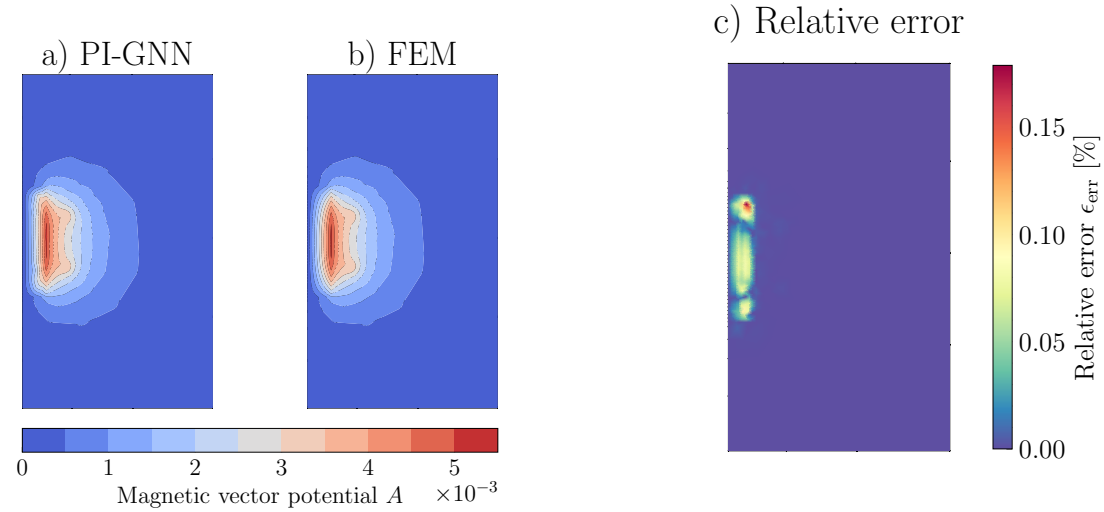


Figure 3.11: Two-coil system. a) and b) PI-GNN vs FEM, c) Relative error

between the two coils. The surrogate reproduces this more complex field distribution accurately, although the relative error is higher than in the single-coil case, reaching up to approximately 0.15% near the left top boundaries of the workpiece, where the field gradients are sharpest.

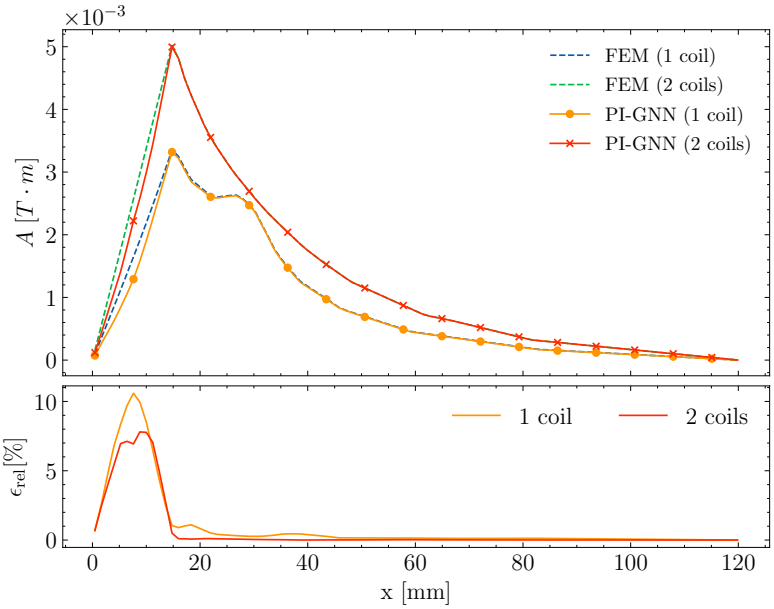


Figure 3.12: The figure shows the magnetic vector potential A over the middle of the workpiece line for one and two-stranded coil cases.

Figure 3.12 compares the magnetic vector potential predictions from the PI-GNN and FEM, as well as their relative errors, using the range-normalised metric from Equation X. The largest errors appear near the axis of symmetry, reaching almost 10% for the one-coil case and 7% for the two-coil case. Section 2.5.5 explains that the axis-regularity condition is enforced by substituting $A = (r/r_\star)\tilde{A}$ (Equation (2.68)), which removes the coordinate singularity and keeps the transformed field $\tilde{A}$ smooth and bounded at $r = 0$. Since the error metric in Equation (2.69) uses the global field range for normalisation, the higher error near the axis shows a real absolute difference in the predicted field, not just an effect of a small denominator. Two main factors likely cause this. First, while the axis-regularity transformation ensures the correct behaviour as r approaches zero, it does not guarantee that the GNN will be accurate at every point near $r = 0$. The network still needs to learn how $\tilde{A}$ changes smoothly near the axis from the training data. Second, if the mesh does not have more nodes near $r = 0$, the spatial gradients there are sampled less often, so the physics-informed loss has fewer local constraints. We will see later in the eddy-current study that the magnetic vector potential A is represented more accurately near the axis of symmetry due to the boundary layer on the boundaries. Even so, the axis error stays limited and does not spread into the main part of the domain, where most engineering quantities are measured.

3.2.2 Harmonic eddy-current test

Having confirmed that the surrogate correctly resolves the static magnetic operator, we now verify its performance on the full harmonic eddy-current problem. This is the physically relevant case for induction heating: a time-harmonic coil current at a prescribed frequency f induces eddy currents in the electrically conducting workpiece,

and the resulting Joule heating drives the thermal response. The real and imaginary parts of A are coupled through the conductivity term, making this problem strictly more demanding than the magnetostatic case. To isolate the effect of eddy-current induction from the skin-layer concentration, the eddy-current verification uses $\mu_r = 1$ throughout. The influence of relative permeability on the skin depth is then examined separately in Section 3.2.3.

As in the magnetostatic test, both one- and two-coil configurations are considered. The excitation current of coils is 3000 A (amplitude) and frequency is 3000 Hz. The PI-GNN predictions are compared against the FEM reference solutions using field plots and numerical error metrics.

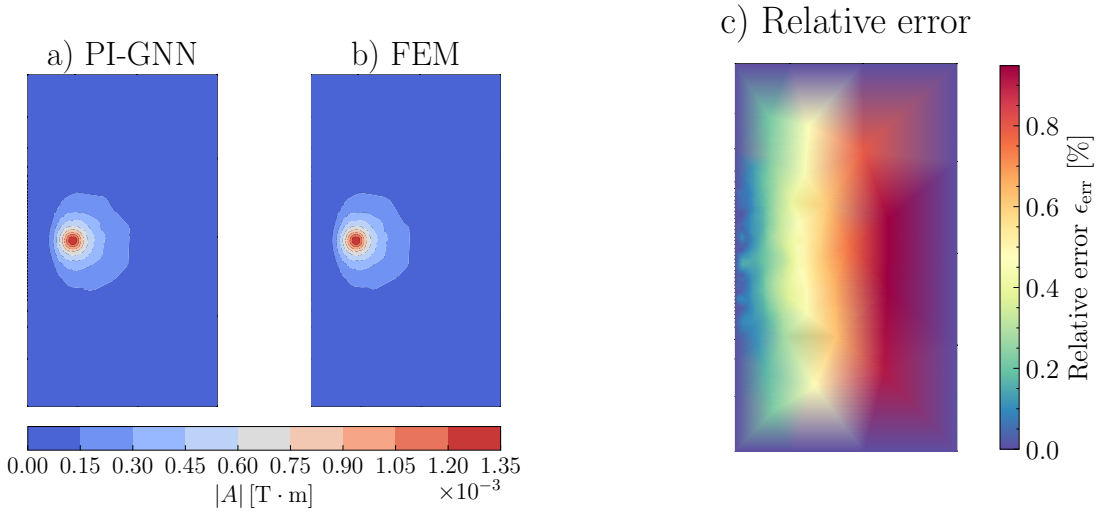


Figure 3.13: One-coil harmonic eddy-current test. a) and b) PI-GNN vs FEM, c) Relative error

Figure 3.13 shows the two-dimensional field plots of the absolute value of magnetic vector potential $|A|$ for the single-coil configuration. The PI-GNN prediction reproduces the characteristic concentric contour structure of the induced field, centred on the coil position, with a peak magnitude of approximately 1.35 mT · m at the coil centre decaying radially outward. The spatial distribution is in close visual agreement with the FEM reference. The relative error map (panel c) shows that ε_{rel} remains below 0.9% across the domain. The largest error appears in the far-field, away from both the coil and the workpiece, which is the opposite of what might be expected because surrogate errors are usually highest where field gradients are steep. This likely happens because far-field nodes have very small $|A|$ values, so relative error amplification becomes the main effect. This may also relate to the low-conductivity artefact discussed in Section 3.1.4. Importantly, the error stays below 1% everywhere, and the higher relative error in the air domain does not affect the Joule heating source.

The real and imaginary parts of A show different error patterns, as seen in Figures 4.1 and 4.2. The real part follows the pattern of $|A|$, with relative errors staying below 1%. In contrast, the imaginary part reaches relative errors of up to about 6% in certain areas. This difference is expected for physical reasons: the imaginary part of A is influenced by the eddy-current term $\sigma\mathbf{E} = -j\omega\sigma A$. Since $\sigma = 1$ is in the air domain, the imaginary component is much smaller than the real part. As a result, even small absolute errors lead to higher relative errors, which is the same effect that causes the far-field behaviour of $|A|$ described earlier. Importantly, the imaginary part of A affects the induced eddy currents and the Joule heating source only in the conducting workpiece, where the errors

are well below 6%.

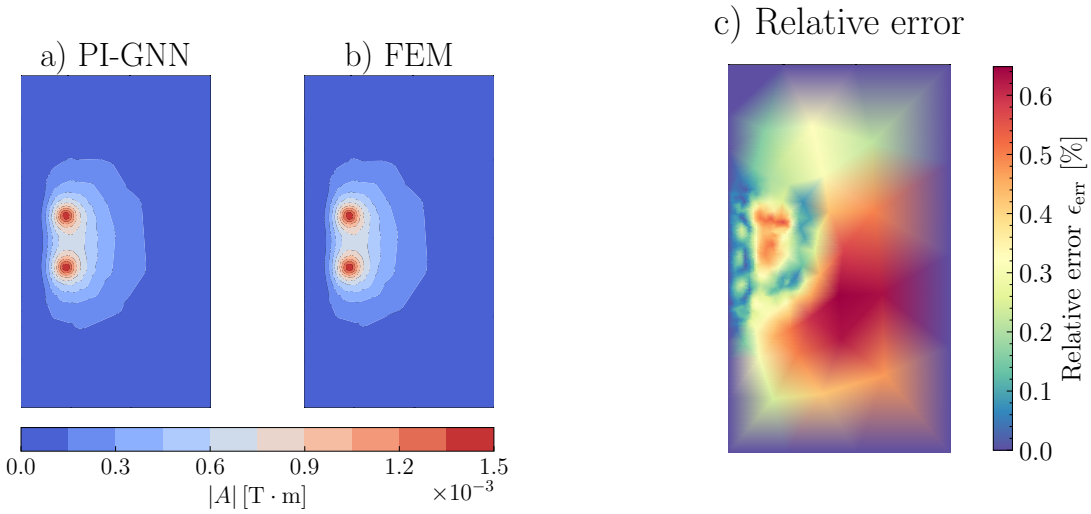


Figure 3.14: Two-coil system. a) and b) PI-GNN vs FEM, c) Relative error

Figure 3.14 presents the corresponding results for the two-coil system. The induced field now exhibits two overlapping lobes centred on the respective coil positions, with peak magnitudes of approximately 1.5 mT · m. The surrogate again captures the overall topology of the field distribution, including the elongated shape of the combined source pattern and the saddle-like structure in the inter-coil region. However, the relative error (panel c) is smaller than in the single-coil case: ε_{rel} reaches up to approximately 0.6–0.7% in localised regions in the air domain and between the two coils, where the coupling between the two eddy-current sources produces sharper spatial variations.

The real and imaginary parts show different error patterns, as seen in Figures 4.3 and 4.4. The real part behaves similarly to $|A|$, with relative errors staying below 0.6%. The imaginary part, on the other hand, has relative errors above 3% for the same reasons as in the one-coil case: its value is smaller in the low-conductivity air, which makes relative errors larger. However, the location of the highest error differs from that in the single-coil case. Instead of being in the far field, the largest error is now found between the coils, where the two eddy-current sources combine and create sharper changes in the imaginary part. This shows the more complex field shape in the two-coil setup and matches the slightly higher coupling effects already seen for $|A|$ in that area.

Figure 3.15 presents the induced current density magnitude $|J|$ extracted along two orthogonal cross-sections of the workpiece: one along its length (on the right boundary of the workpiece, $x \in [70, 140]$ mm). The PI-GNN predictions are compared against the FEM reference for the one- and two-coil configurations. Along the length of the workpiece, the one-coil case produces a bell-shaped current density profile with a single peak of approximately 2.6×10^7 A/m^2 located near the centre of the coil footprint at $x \approx 90$ mm, decaying smoothly towards both ends of the workpiece. The two-coil configuration yields a bimodal distribution with two distinct peaks of equal magnitude ($\approx 3.0 \times 10^7$ A/m^2) at approximately $x \approx 90$ mm and $x \approx 120$ mm, respectively, with a local minimum in the inter-coil region that descends to approximately 2.3×10^7 A/m^2.

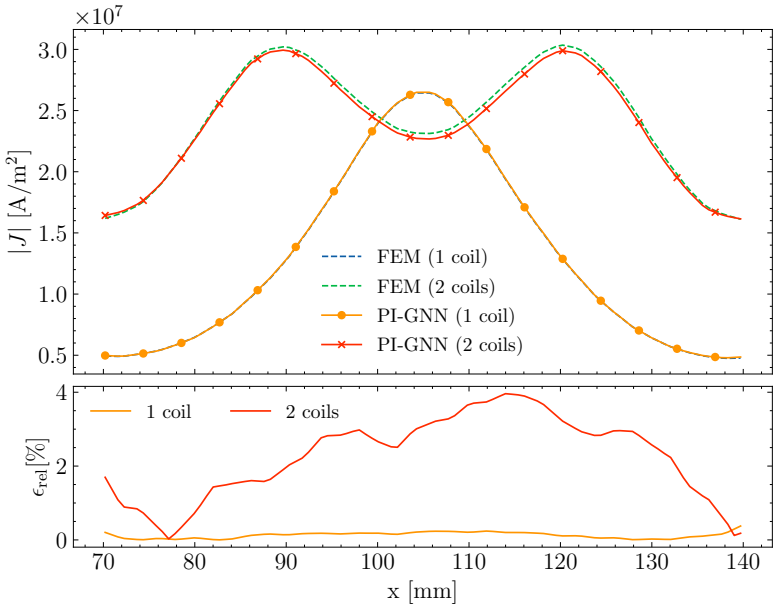


Figure 3.15: Current density across the length of the workpiece

3.2.3 Skin-layer test

Industrial meshes for induction heating problems strongly depend on a phenomenon known as the skin effect. At high frequencies or high relative permeabilities, eddy currents concentrate near the surface of the workpiece within a characteristic depth known as the skin depth,

$$\delta = \sqrt{\frac{2}{\omega \sigma \mu_0 \mu_r}}, \tag{3.9}$$

where ω is the angular frequency, σ the electrical conductivity, and μ_0 is the permeability of free space. As μ_r increases, δ decreases, and the current becomes increasingly concentrated near the surface. Resolving this behaviour accurately requires the surrogate to capture steep gradients across a narrow boundary layer, which is one of the more demanding aspects of the electromagnetic problem.

This test verifies that the surrogate correctly handles skin-layer physics across different relative permeabilities μ_r of the workpiece. A separate surrogate is trained for each value of $\mu_r \in \{1, 10, 50, 100\}$, keeping all other parameters fixed. To make the skin-layer effect more visible, we use a 3-coil system, which produces a stronger and more spatially extended eddy-current response than the 1- and 2-coil configurations used in the previous tests.

Figure 3.16 shows the normalised current density across the thickness of the workpiece for all four cases. For $\mu_r = 1$, the current is distributed broadly across the workpiece, consistent with a large skin depth. As μ_r increases to 10, 50, and 100, the current becomes progressively confined to a thin surface layer, and the interior of the workpiece carries negligible induced current. The PI-GNN predictions follow the FEM reference profiles closely in all four cases, capturing both the shape of the current distribution and the rate of decay into the workpiece interior. The relative error remains below approximately 2.5% across all μ_r values, with the largest deviations occurring near the surface where the field gradients are steepest. The consistent error level across all tested permeabilities shows that separately trained surrogates can reproduce normalised

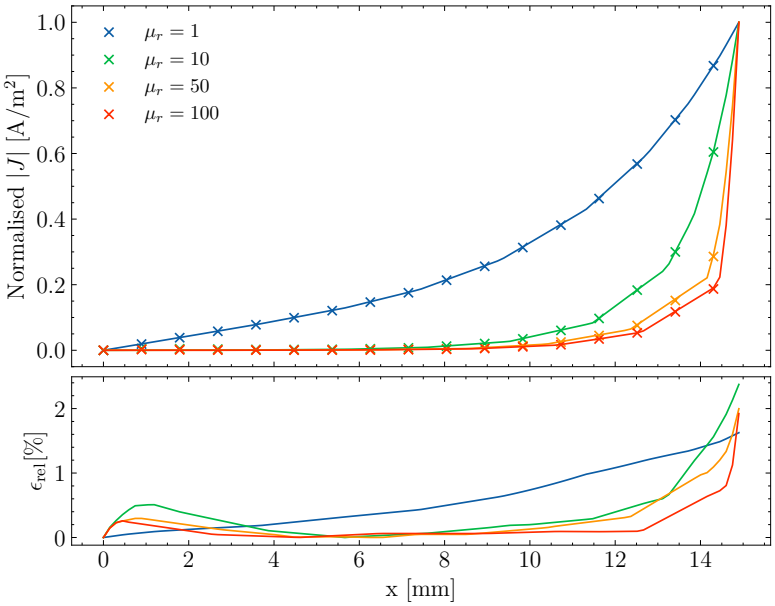


Figure 3.16: Normalised current density across the thickness of the workpiece for different relative permeabilities μ_r. Lines show FEM reference solutions. Markers show PI-GNN predictions. The lower panel shows the relative error ϵ_{rel} for each case.

current-density profiles for the tested μ_r values.

3.3 Verification of the coupled induction-heating framework

This section verifies the accuracy of the coupled electromagnetic and thermal surrogate chain for induction heating simulations. It presents two test cases, each more complex than the last. The first case, explained in Section 3.3.1, tests the one-way coupled surrogate on a simple aluminium workpiece with fixed electromagnetic properties. It also breaks down the error to show the separate effects of the electromagnetic and thermal surrogates. The second case, described in Section 3.3.2, uses a simplified version of the TEAM 36 benchmark on a ferromagnetic workpiece with high relative permeability and temperature-dependent thermal conductivity. This is the most challenging scenario in the thesis. Together, these cases cover a range of material nonlinearity and rollout lengths, showing how accurate the PI-GNN surrogate chain is in different situations.

3.3.1 One-way coupled induction heating

This test verifies the one-way coupled surrogate chain in which the electromagnetic surrogate predicts the Joule heat source and the thermal surrogate advances the temperature field using this source. The test is not a two-way coupled validation, since the electromagnetic material properties are not updated from the evolving temperature field. Its purpose is therefore limited to verifying the EM–thermal source transfer and the resulting thermal rollout under fixed electromagnetic properties.

This test uses the material and source parameters listed in Table 3.1. The workpiece is treated as non-magnetic aluminium with $\mu_r = 1$, giving a skin depth of $\delta \approx 1.5\,\text{mm}$ at $f = 3\,\text{kHz}$. The geometry is the same as in the previous skin-layer test (see Subsection 3.2.3). Natural convection is applied on top, right and bottom, while the left is treated as insulated. The EM surrogate first computes the magnetic vector potential

field, from which the Joule-heating source is derived and then passes it directly to the thermal surrogate without feedback. The coupled rollout is performed for 100 time steps with $\Delta t = 0.1\,\mathrm{s}$, giving a final time of $t = 10\,\mathrm{s}$.

Property	Symbol	Value	Unit
Thermal Material properties (workpiece)			
Density	ρ	2700	$\mathrm{kg/m^3}$
Specific heat capacity	c_p	933.3	J/kgK
Thermal conductivity	k	211	W/mK
EM Material properties			
Workpiece electrical conductivity	σ	3.7×10^7	S/m
Air electrical conductivity	σ	0	S/m
Coil electrical conductivity	σ	0	S/m
Workpiece relative permeability	μ_r	1	–
Air relative permeability	μ_r	1	–
Coil relative permeability	μ_r	1	–
EM Source properties			
Current	I	3000	A
Frequency	f	3000	Hz
Natural Convection BC			
Convective heat transfer coefficient	h	10	$\mathrm{W/m^2K}$
Ambient temperature	T_∞	20	°C

Table 3.1: Overview of material properties for induction heating verification test.

Figure 3.17 shows the predicted and reference temperature fields at the final time $t = 10\,\mathrm{s}$, together with the relative error field. The PI-GNN reproduces the characteristic heating pattern of induction heating: temperatures are highest near the surface of the workpiece, where Joule heating is concentrated due to the skin effect, and decay towards the interior and the cooler ends. The relative error remains within $\pm 1.6\%$ across the domain, with the largest deviations occurring near the top and bottom ends of the workpiece, where the temperature gradients are steepest. The interior of the domain is predicted with high accuracy, consistent with the smooth thermal diffusion that dominates the response away from the surface.

Table 3.3 summarises the key scalar quantities for the full one-way coupled surrogate (Case D) against the FEM reference (Case A). The total deposited power is underpredicted by 2.62%, consistent with the slight underestimation of the magnetic vector potential field observed in the electromagnetic verification. The maximum Joule heat error of 5.52% is larger, as expected: the maximum is a pointwise quantity sensitive to localised field errors near the surface, whereas the total deposited power reflects a volume-averaged quantity that is less affected by such local deviations. Despite this, the final maximum temperature is reproduced with an error of 1.22%, smaller than the source-field error of 4.83% in the L_2 sense. This reduction reflects the smoothing effect of thermal diffusion: spatial errors in the Joule heat source are attenuated as heat spreads through the workpiece over time, so that the temperature field is inherently less sensitive to localised EM inaccuracies than the source itself.

To separate the individual contributions of the electromagnetic and thermal surrogate errors, four one-way coupled variants are compared in Figure 3.18. An overview of cases is given by table 3.2. Case A uses the FEM Joule heat source together with the FEM thermal solver and is treated as the reference. Case B replaces only the Joule heat source

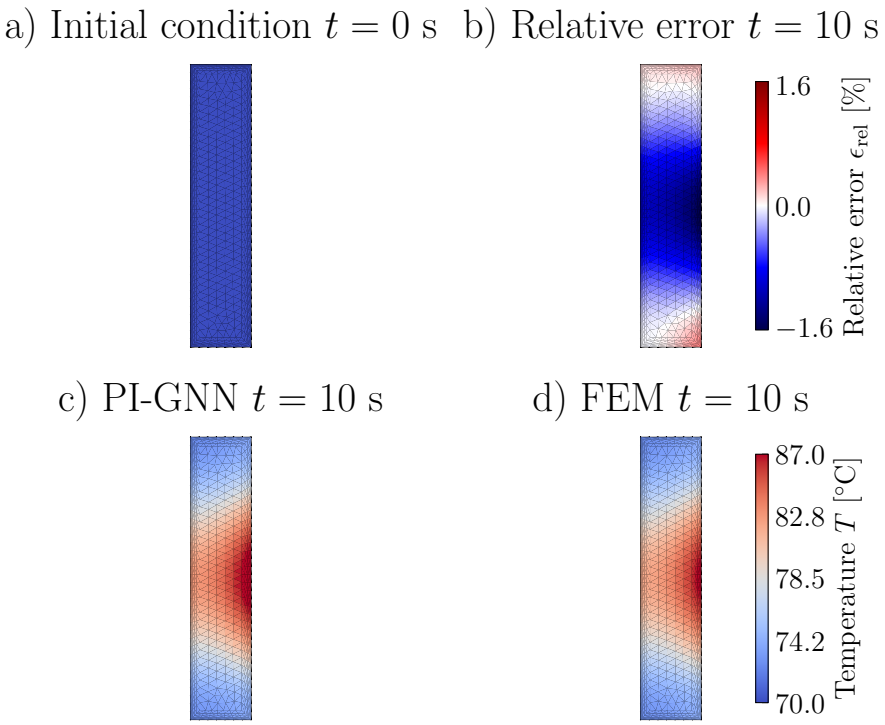


Figure 3.17: PI-GNN vs FEM and relative error

Case	EM source	Thermal solver	Purpose
A	Q_{FEM}	FEM	Reference
B	$Q_{\text{PI-GNN}}$	FEM	Physical effect of EM source error
C	Q_{FEM}	PI-GNN	Thermal surrogate error
D	$Q_{\text{PI-GNN}}$	PI-GNN	Full one-way coupled surrogate

Table 3.2: Four cases for error investigation for the one-way coupled induction-heating verification.

with the EM-surrogate prediction, while retaining the FEM thermal solver. Case C uses the FEM Joule heat source but replaces the thermal solver with the PI-GNN thermal surrogate. Case D is the full one-way coupled surrogate chain.

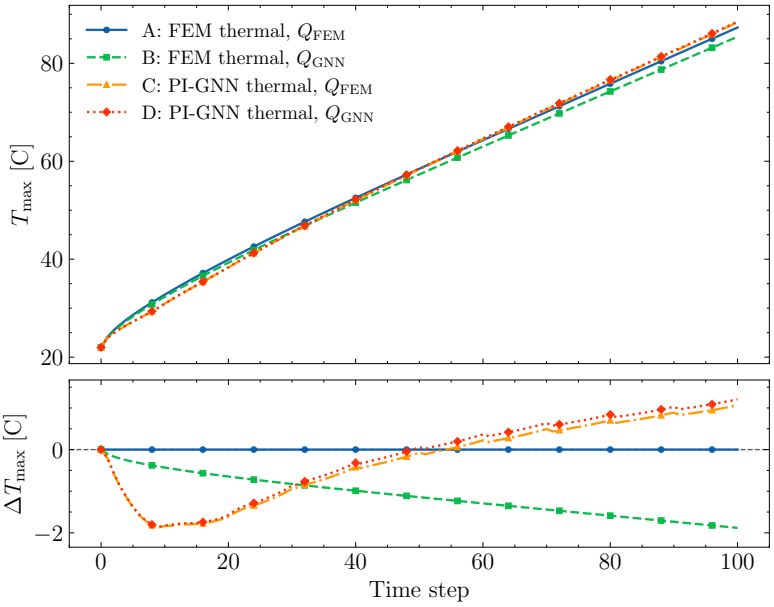


Figure 3.18: Error decomposition for the one-way coupled induction-heating verification. Case A uses the FEM Joule heat source with the FEM thermal solver and serves as the reference. Case B uses the PI-GNN Joule heat source with the FEM thermal solver, isolating the thermal consequence of the electromagnetic source error. Case C uses the FEM Joule heat source with the PI-GNN thermal surrogate, isolating the thermal surrogate error. Case D uses the PI-GNN Joule heat source with the PI-GNN thermal surrogate and represents the full one-way coupled surrogate chain. The lower panel shows $\Delta T_{\max} = T^{X}_{\max} - T^{A}_{\max}$ for cases $X \in \{B, C, D\}$.

The comparison shows that the EM-source error by itself causes a consistent underprediction of the maximum temperature throughout the rollout, as seen in case B. In case C, the thermal surrogate behaves differently: it starts by underpredicting the maximum temperature, but after time step ≈ 50, it crosses the reference and then overpredicts it. Because these two errors have opposite effects later in the rollout, their combination in case D results in a net error that is smaller than either B or C alone.

However, this partial cancellation happens by chance under these specific operating conditions and is not a planned feature of the surrogate chain. The EM-source error and the thermal-rollout error come from separate model approximations and are not linked in a way that would make them cancel each other out. In other operating regimes, such as with higher current frequencies, different material properties, or longer heating times, both errors could have the same effect and add together, leading to a much larger total error than either one alone. The reported 0.63% final error in case D should be seen in this context: it shows the accuracy of the full surrogate for this particular test, and the four-case breakdown in Figure 3.18 helps explain how this accuracy is achieved.

3.3.2 Reduced TEAM Problem 36 benchmark

The TEAM 36 benchmark [62] is a known validation problem for coupled electromagnetic–thermal solvers. In its original form, it requires a fully two-way coupled treatment in which the relative permeability $\mu_r(H, T)$ and electrical conductivity $\sigma(T)$

Quantity	PI-GNN	FEM	Relative error
Deposited power P, kW	47.04	48.30	2.62%
Maximum Joule heat $Q_{\max}, \mathrm{MW/m^3}$	0.214	0.226	5.52%
Source-field error $\varepsilon_Q^{L_2}$	–	–	4.83%
Final maximum temperature $T_{\max}(t_f), °\mathrm{C}$	88.37	87.30	1.22%
Final temperature-field relative error $\varepsilon_T^{L_2}(t_f)$	–	–	0.63 %

Table 3.3: Overview of scalar quantities for one-way coupled induction heating verification test.

both vary with the evolving magnetic field and temperature, introducing strong nonlinearities into both the electromagnetic and thermal problems simultaneously. Reproducing this benchmark in full would require iterative EM–thermal coupling at each time step, which lies outside the scope of the present one-way coupled surrogate chain. Instead, a one-way coupled reduced variant is studied here. This reduction isolates the performance of each surrogate on the TEAM 36 geometry under realistic excitation conditions, while keeping the full two-way coupled treatment for future work.

The workpiece geometry and coil configuration follow the TEAM 36 specification. The electromagnetic surrogate is driven by a coil current of $I = 4950\,\mathrm{A}$ (amplitude) at a frequency of $f = 2000\,\mathrm{Hz}$, with the workpiece assigned a relative permeability $\mu_r = 100$, and electrical conductivity $\sigma = 4761904\,\mathrm{S/m}$. This high permeability, combined with the relatively high frequency, yields a skin depth of approximately $\delta \approx 0.52\,\mathrm{mm}$, making accurate resolution of the surface current layer substantially more demanding than the $\mu_r = 1$ aluminium case examined in Section 3.3. The workpiece density and specific heat capacity are held constant at $\rho = 7870\,\mathrm{kg/m^3}$ and $c_p = 461\,\mathrm{J/kgK}$, while the thermal conductivity $k(T)$ follows the temperature-dependent tabulated values given by table 4.1 (piecewise linear interpolation). The transient simulation spans $t_f = 25\,\mathrm{s}$ with a time step $\Delta t = 0.1\,\mathrm{s}$, giving 250 rollout steps. This is the longest and most materially nonlinear case considered in this thesis, and it represents the edge of the current surrogate's design envelope.

Electromagnetic predictions

Figure 3.19 shows the predicted and reference magnetic vector potential fields $|A|$ together with the pointwise relative error. The PI-GNN reproduces the flux distribution associated with the high-permeability workpiece, with peak values of approximately $4.3\,\mathrm{mT \cdot m}$ concentrated in the coil region and decaying smoothly towards the air domain boundary. The overall field topology is well captured, although the relative error map (panel c) reveals localised deviations up to approximately 2.5% in the region next to the coil-workpiece interface, where the steep field gradient caused by the thin skin layer is hardest to resolve. The error is significantly lower in the air domain, near the outer boundary and close to the symmetry axis of the workpiece.

The scalar quantities summarised in Table 3.4 are consistent with this field-level picture. The total deposited power is underpredicted by 2.09%, and the maximum Joule heat by 3.68%, with a source-field relative L_2 error of 3.21%. These values are moderately larger than those obtained in the one-way coupled aluminium test (Section 3.3.1), which is expected. The high relative permeability $\mu_r = 100$ compresses the eddy-current layer and increases the field gradient near the surface, making the electromagnetic problem intrinsically harder to approximate for a fixed mesh resolution.

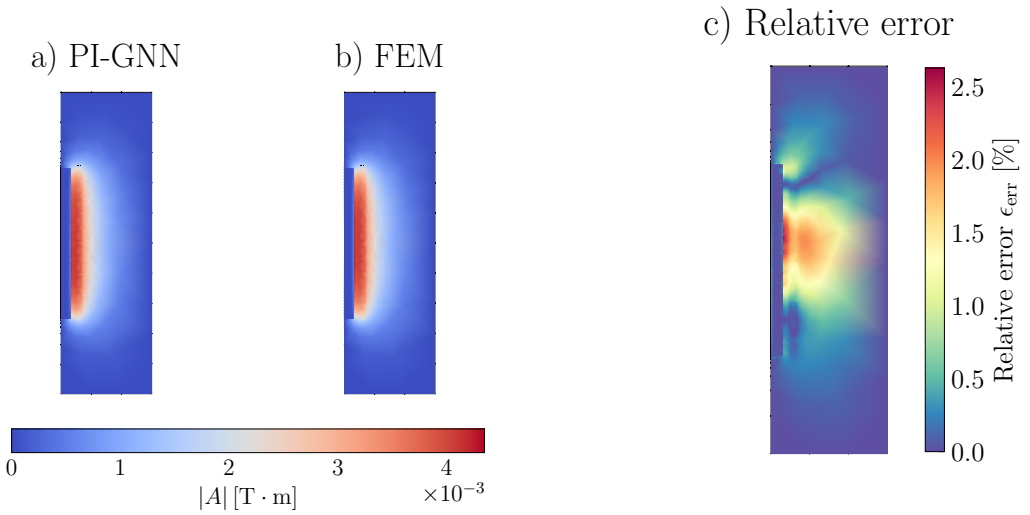


Figure 3.19: The prediction and FEM reference of the absolute value of the magnetic vector potential $|A|$ and relative error.

Quantity	PI-GNN	FEM	Relative error
Deposited power P, MW	4.42	4.52	2.09%
Maximum Joule heat $Q_{\max}$, GW/m^3	5.68	5.90	3.68%
Source-field error $\varepsilon_Q^{L_2}$	–	–	3.21%
Final maximum temperature $T_{\max}(t_f)$, °C	1160.37	1206.42	3.82%
Final temperature-field relative error $\varepsilon_T^{L_2}(t_f)$	–	–	3.88 %

Table 3.4: Overview of scalar quantities for the reduced TEAM 36 problem.

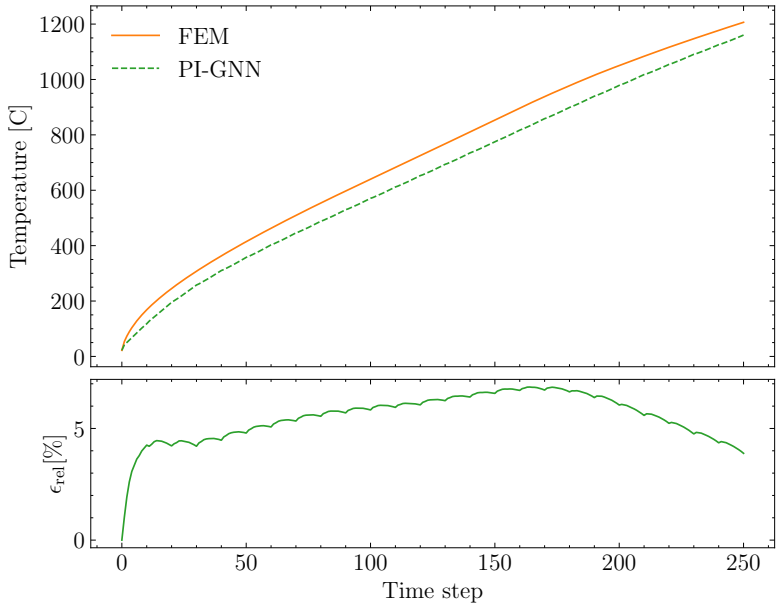


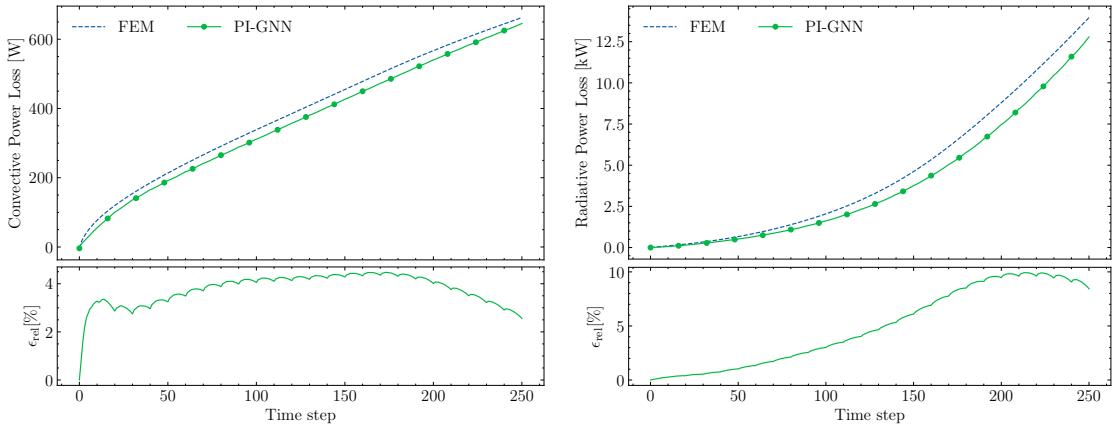
Figure 3.20: Max temperature evolution predicted by the thermal surrogate vs FEM solution. The lower panel shows the relative error field.

Thermal predictions and rollout

Figure 3.20 shows the evolution of the maximum workpiece temperature over the 250-step rollout. The PI-GNN surrogate tracks the FEM reference closely for the first half of the simulation, then accumulates a systematic underprediction that reaches approximately 3.8% at the final time $t_f = 25\,\mathrm{s}$, where the predicted maximum temperature is 1160.4 °C against the FEM reference of 1206.4 °C (Table 3.4). The final temperature-field relative L_2 error of 3.88% is consistent with the maximum-temperature error, suggesting that the deviation is distributed across the domain rather than confined to a single localised region. The growing underprediction over time is partly attributable to error accumulation in the autoregressive rollout, and partly to the temperature dependence of the thermal conductivity. As $k(T)$ decreases at higher temperatures, the effective heat transport slows, and small early-time temperature errors are amplified into larger deviations in the thermal diffusion rate over subsequent steps.

The surface heat losses shown in Figure 3.21 provide an additional diagnostic of the surrogate's boundary accuracy. Convective losses (panel a) are underpredicted by approximately $3-4\%$ throughout the rollout, broadly consistent with the temperature-field error. Radiative losses (panel b), which scale with T^4, are more sensitive to temperature errors. The relative error grows from near zero at early times to approximately $9-10\%$ by the end of the simulation, before decreasing slightly. This nonlinear amplification is expected from the Stefan-Boltzmann law. At the high temperatures encountered here (approaching 1200 °C), a 3.8% underprediction in temperature translates into a substantially larger relative error in radiated power due to the quartic dependence on absolute temperature.

The 9-10% error in radiated power results from a clear sequence of error propagation. The EM surrogate's 3.21% L_2 source error creates a heat source that is consistently too low from the start, leading to a cold bias in the temperature path similar to the underprediction seen in Case B of Section 3.3.1. This cold bias is not just added on; as the temperature drops below the reference, the temperature-dependent thermal conductivity



(a) Convective Power Loss

(b) Radiative Power Loss

Figure 3.21: Convective (panel a) and radiative (panel b) power losses over time for the reduced TEAM 36 problem.

$k(T)$ changes, which affects heat diffusion and increases the bias over the 250-step rollout. The Stefan-Boltzmann law then further increases the cold bias because of its strong dependence on temperature, causing the radiative loss error to be much larger than the original temperature error. In summary, the error propagates as follows: EM source error → systematic cold bias → increased by $k(T)$ feedback → further amplified by the T^4 dependence of radiative loss.

Overall, the reduced TEAM 36 benchmark demonstrates that the PI-GNN surrogate chain remains functional at the edge of its design envelope: a ferromagnetic workpiece with high relative permeability, a thin skin layer, a 250-step transient rollout, and a temperature-dependent thermal conductivity. The final-time errors of approximately 3.8–3.9% in both the scalar and field-level temperature quantities are larger than those obtained in the simpler aluminium test, but remain within a practically useful range for engineering applications. The error budget is shared roughly equally between the electromagnetic and thermal surrogates, with the electromagnetic source error (3.21% in L_2) slightly attenuated by thermal diffusion in the temperature field, analogously to the behaviour observed in Section 3.3.1.

3.4 Generalisation and surrogate performance

This section evaluates whether the trained surrogates remain accurate outside the specific configurations used during verification. Since the electromagnetic and thermal surrogates are trained independently, the tests are separated into electromagnetic-only, thermal-only, and coupled evaluations where appropriate. The electromagnetic surrogate is assessed through errors in the magnetic vector potential. The thermal surrogate is assessed through the transient temperature field, maximum temperature evolution, and final-time errors. Unless stated otherwise, all comparisons are made against FEM reference solutions generated with the same geometry, material properties, boundary conditions, and operating parameters.

For each generalisation test, a new model is trained in a specified range. This is done to study the model generalisation for each parameter separately.

3.4.1 Unseen geometries

This subsection looks at how well the electromagnetic surrogate can handle geometries it did not see during training. Three types of geometric changes are studied: the workpiece aspect ratio, which depends on billet height and diameter (G1); the lateral offset of the induction coil from the workpiece centerline (G2); and the separation between the coil and workpiece, set by the coil's inner diameter (G3). Table 3.5 shows the parameter ranges for each case.

Case	Change	Why it matters	Range [mm]
G1	workpiece aspect ratio	domain-size generalisation	H: $[50; 90]$, D: $[20; 40]$
G2	coil offset	source-location generalisation	$[-25; 25]$
G3	air gap	sensitivity to magnetic coupling	$[40; 60]$

Table 3.5: Geometry-variation cases used to assess generalisation of the electromagnetic surrogate.

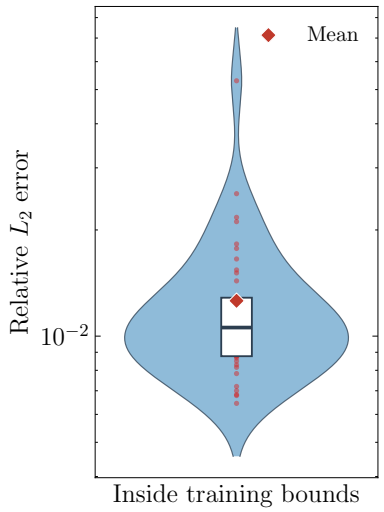
This study focuses only on the electromagnetic surrogate, and this is a deliberate choice. In the full induction-heating problem, changes in geometry affect the thermal field in two ways: directly, by changing the workpiece and its boundaries, and indirectly, by changing how electromagnetic power is deposited. To properly test how well the thermal model handles new geometries, one would need to vary the electromagnetic source and the thermal geometry separately. However, the electromagnetic surrogate calculates the magnetic vector potential across the entire domain, including air, coil, and workpiece. This makes it the part of the model most directly affected by changes in geometry.

For each geometry, we create the training set using Latin hypercube sampling within the specified training range. To test interpolation, we use another Latin hypercube sample with a different random seed but the same parameter limits. For extrapolation, we use geometries outside the training range, including the one-dimensional sweeps shown in Figures 4.5 and 4.6. We measure error using the relative discrete L_2 error in the magnitude of the magnetic vector potential.

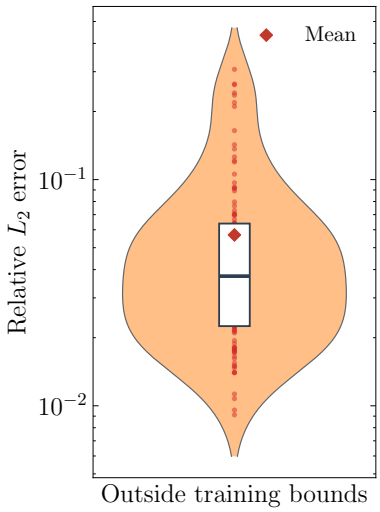
Figure 3.22 shows how the errors are distributed for both interpolation and extrapolation. Within the training range, the surrogate stays accurate, with a mean relative L_2 error of 1.26×10^{-2} and a median of 1.06×10^{-2} across 42 test geometries. The interquartile range is tight, from 8.77×10^{-3} to 1.29×10^{-2}, which means the model is not just accurate on average but also consistent throughout the sampled geometry space. The highest interpolation error is 5.30×10^{-2}, pointing to a few challenging cases rather than a general failure.

When tested outside the training range, the error distribution shifts higher and becomes much wider. In 115 extrapolation cases, the mean error rises to 5.69×10^{-2} and the median to 3.74×10^{-2}. Compared to interpolation, the median error is about 3.5 times higher and the mean is about 4.5 times higher. The interquartile range also grows from 2.25×10^{-2} to 6.39×10^{-2}, which means extrapolation is not only less accurate but also less predictable. This matters because the main limitation of the surrogate outside the training domain is not just a general drop in accuracy, but a higher chance of running into geometry-dependent failure modes.

The one-dimensional sweeps in Figures 4.5 and 4.6 help show which geometric factors cause this drop in performance. The billet-height sweep reveals that accuracy drops off smoothly as we move outside the training range. Within the training range $H \in [50, 90]$ mm, errors stay near 10^{-2}, but they rise on both sides of this interval. The



(a) Interpolation inside the training bounds.



(b) Extrapolation outside the training bounds.

Figure 3.22: This figure shows the distribution of relative L_2 errors in $|A|$ for geometries not seen during training. The interpolation set was created within the training hyper-rectangle using a separate Latin hypercube seed. The extrapolation set includes geometries outside the training range. In the box plot, the box shows the interquartile range, the horizontal line marks the median, and the diamond marker represents the mean.

increase is especially noticeable for billets that are much shorter or taller, where the electromagnetic domain and the way the induced field spreads out are quite different from the training cases. This pattern suggests that the surrogate model can interpolate reliably but struggles to predict accurately when the workpiece length is far outside what it has previously seen.

The billet-diameter sweep shows a more severe effect. When the diameter stays within the training range $D \in [20, 40]$ mm, the error stays low. But once the diameter exceeds this range, the error rises quickly and worsens further as the diameter increases. This is the most obvious failure among the geometry tests. It shows that extrapolating the diameter is not a simple scaling issue. Changing the billet diameter affects the coil to workpiece clearance, the magnetic coupling, and the area where eddy currents form. As a result, using much larger diameters changes both the geometry and how the electromagnetic coupling works. The surrogate model should not be expected to give reliable results for billet diameters far outside the training range unless those cases are included during training.

The offset sweep in Figure 4.6a shows a different trend. The error is lowest near the centre of the training range and grows roughly evenly as the coil moves beyond the training limits. Even at the largest offsets tested, the errors are much smaller than the most severe errors seen when extrapolating diameter. This suggests the surrogate model has learned how moving the electromagnetic source affects the results, as long as the offset does not create a fundamentally different field pattern.

The coil inner diameter sweep in Figure 4.6b also shows moderate extrapolation. When the coil's inner diameter goes beyond the upper training limit, the magnetic coupling between the coil and the workpiece becomes weaker, and the error increases. However, this increase is gradual, not sudden, which means this type of extrapolation is less harmful than increasing the billet diameter. This difference makes physical sense: increasing the coil size primarily reduces the coupling strength and alters how the field

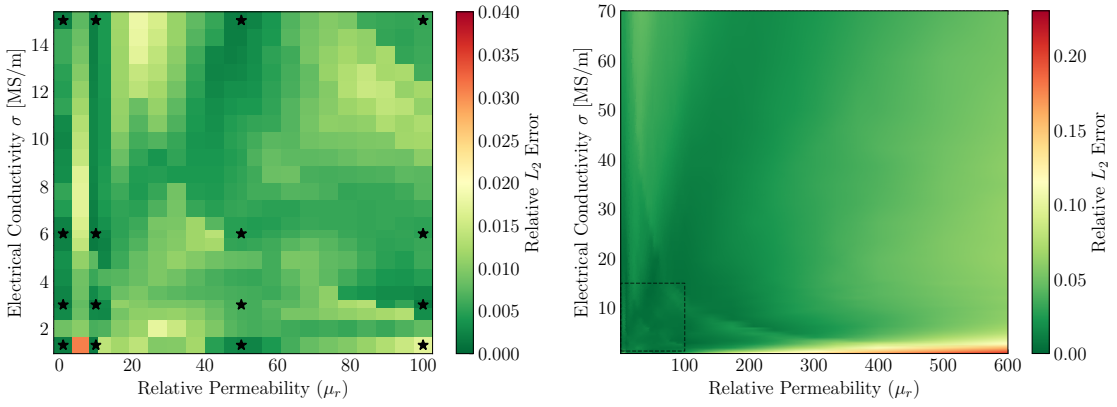
fades, while increasing the workpiece size can also shift the position of the conducting boundary relative to the coil and might move the setup into a different geometric category.

Overall, these results show that the electromagnetic surrogate performs well within the tested geometry range, but it should be used with caution outside that range. The model generalises best when the source position is changed or when the coil's inner diameter is modified moderately. It performs worst when extrapolating to much larger workpiece diameters, where the higher error suggests a true out-of-distribution failure, not just a minor drop in accuracy. For practical use, it is important to clearly define the acceptable geometry range and focus extra training data on large-diameter and extreme aspect-ratio cases.

3.4.2 Unseen operating conditions

Electromagnetic surrogate

To evaluate the generalisation capability of the electromagnetic surrogate to unseen material properties, the model is tested over a two-dimensional sweep of relative permeability μ_r and electrical conductivity σ of the workpiece, while keeping the process parameters fixed at $I = 3\,\mathrm{kA}$ and $f = 3\,\mathrm{kHz}$. The surrogate is trained on a sparse grid of 16 parameter combinations, with $\mu_r \in \{1, 10, 50, 100\}$ and $\sigma \in \{1.3, 3, 6, 15\}\,\mathrm{MS/m}$, for 10 000 epochs at approximately 2 seconds per epoch. These training points are chosen to span a representative range of ferromagnetic and non-magnetic workpiece materials, from weakly conducting non-magnetic metals to highly permeable steels. The model is then evaluated both on unseen interpolation points within the training range and on extrapolation points well beyond it, in order to assess how well the learned field representation captures the underlying physics of eddy-current penetration as material properties vary.

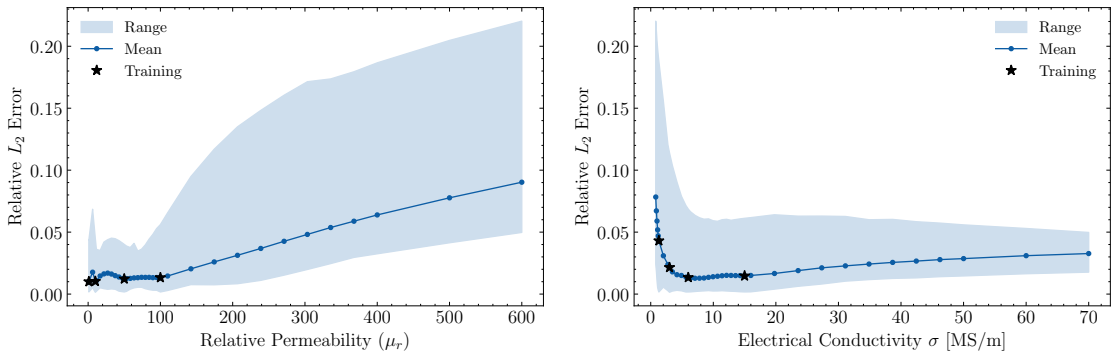


(a) Interpolation (b) Extrapolation

Figure 3.23: Generalisation of the EM surrogate model to the material properties σ and μ_r of the workpiece. Panel a) on the left shows interpolation between the 16 training points marked as black stars on the figure. Panel b) on the right shows extrapolation beyond the training range, shown in panel a), marked as a dotted line.

Figure 3.23 shows the relative L_2 error of the electromagnetic surrogate's predicted magnetic vector potential A over a two-parameter sweep in relative permeability μ_r and electrical conductivity σ. The surrogate remains accurate across most of the tested parameter space, with errors generally below a few per cent. This indicates that the

model captures the combined effect of μ_r and σ on the eddy-current field, including the associated change in skin depth. The largest errors occur at the lowest conductivities, where the induced fields and Joule heat source are weaker, and the relative error becomes more sensitive to small absolute deviations. At low conductivity, the skin depth is large relative to the workpiece, so the induced eddy currents and the resulting perturbation of A are weak. A small absolute prediction error, therefore, yields a disproportionately large relative L_2 error.



(a) μ_r (b) σ

Figure 3.24: Relative L_2 error of the electromagnetic surrogate as a function of (a) relative permeability μ_r, averaged over all tested σ values, and (b) electrical conductivity σ, averaged over all tested μ_r values. The shaded band shows the min-max range; black stars indicate training values.

Beyond the training range, errors increase gradually with μ_r, reaching approximately 9% at $\mu_r = 600$ (see Figure 3.24a). At the same time, extrapolation in σ remains accurate (see Figure 3.24b), suggesting the model has learned a representation that generalises more readily along the conductivity axis than the permeability axis.

The surrogate is further evaluated on its ability to generalise across unseen process parameters, namely coil current I and excitation frequency f, with material properties fixed at $\mu_r = 1$ and $\sigma = 4.76\,\mathrm{MS/m}$. The model is trained on a uniform 4×4 grid with $I \in \{2, 3, 4, 5\}\,\mathrm{kA}$ and $f \in \{2, 3, 4, 5\}\,\mathrm{kHz}$, and evaluated over a denser sweep extending to $I = 8\,\mathrm{kA}$ and $f = 8\,\mathrm{kHz}$.

Figure 3.25 shows that the surrogate interpolates accurately within the training range, with relative L_2 errors remaining below 2% across the interior of the parameter space. This is expected, as both I and f enter the magnetoquasistatic formulation linearly through the source term and the $j\omega$ factor, respectively, making the field response smooth and easy to interpolate. Errors increase towards the extrapolation boundary, reaching up to approximately 30% in the top-left corner ($I \approx 1.5\,\mathrm{kA}$, $f \approx 8\,\mathrm{kHz}$), where the combination of low current and high frequency produces a field regime not represented in training. The L_2 error approaches approximately 20% when extrapolating beyond the boundaries of the training points. The degradation is gradual and largely confined to the corners of the extrapolation region, suggesting the model has learned the underlying parameter dependence rather than memorising training points.

Thermal surrogate

The thermal surrogate is evaluated on its ability to generalise to unseen electrical material properties μ_r and σ, and to unseen source strengths parameterised by the coil current I and excitation frequency f.

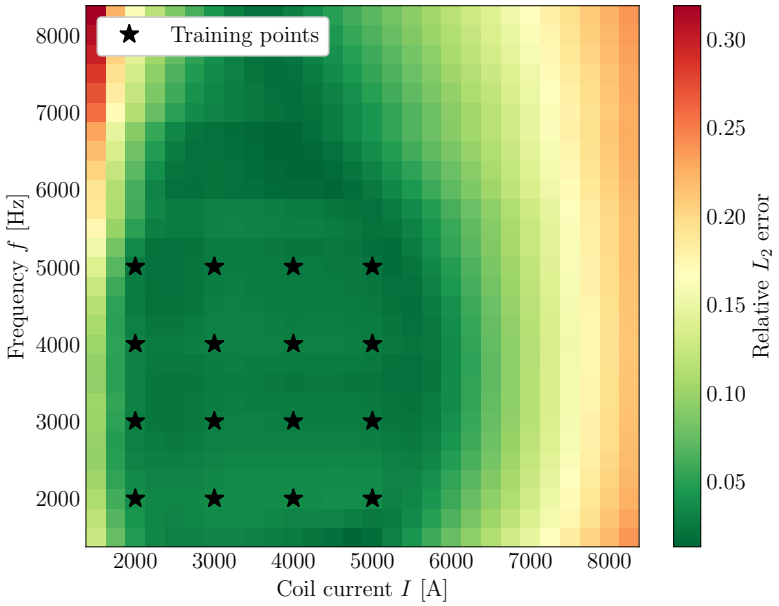


Figure 3.25: Generalisation of the electromagnetic surrogate to unseen process parameters: coil current I and excitation frequency f, with material properties fixed at $\mu_r = 1$ and $\sigma = 4.76\,\mathrm{MS/m}$. The relative L_2 error is shown over a parameter sweep extending beyond the training range; black stars indicate the 16 training points.

Generalisation to material properties μ_r and σ Figure 3.26 shows the relative L_2 error of the thermal surrogate over a two-dimensional sweep of relative permeability μ_r and electrical conductivity σ, mirroring the evaluation performed for the electromagnetic surrogate in Section 3.4.2.

The most noticeable feature of the error map is a sharp band of elevated error ($\gtrsim 30\%$) concentrated at very low relative permeability ($\mu_r \lesssim 10$), visible as a near-vertical stripe along the left edge of the heatmap.

This behaviour is physically analogous to the low-σ artefact observed in the electromagnetic surrogate: at near-unity permeability, the induced eddy currents are weak and penetrate deeply, resulting in a Joule heating source Q that is both spatially diffuse and small in magnitude. A small absolute error in the predicted temperature field, therefore, yields a disproportionately large relative L_2 error. Away from this low-μ_r region, errors decrease and then rise gradually with increasing μ_r, reaching approximately 15–25% over much of the evaluated parameter space.

In contrast to the electromagnetic surrogate, for which the largest errors occurred at low conductivity, the thermal error map shows relatively weak sensitivity to σ across most of the tested range. This is consistent with the coupling structure of the problem: the thermal surrogate receives the Joule heating source Q computed by the electromagnetic surrogate as its input, so variations in σ are partially absorbed into Q and do not directly enter the thermal network as an independent parameter. Linking further to the Jacobi scaling of the loss function (Section 2.4.2), when the heat source Q is small, the source vector f^{n+1} is small relative to the conduction term $\mathbf{K}T^{n+1}$, and the Jacobi-scaled residual R_i^{n+k}/D_i is dominated by the conduction term. The source-term contribution, which is the physical quantity varying across μ_r, has reduced weight in the training signal.

The 1D marginal plots in Figure 3.27 confirm this asymmetry: the error varies more strongly along the μ_r axis than along the σ axis, suggesting that the thermal surrogate

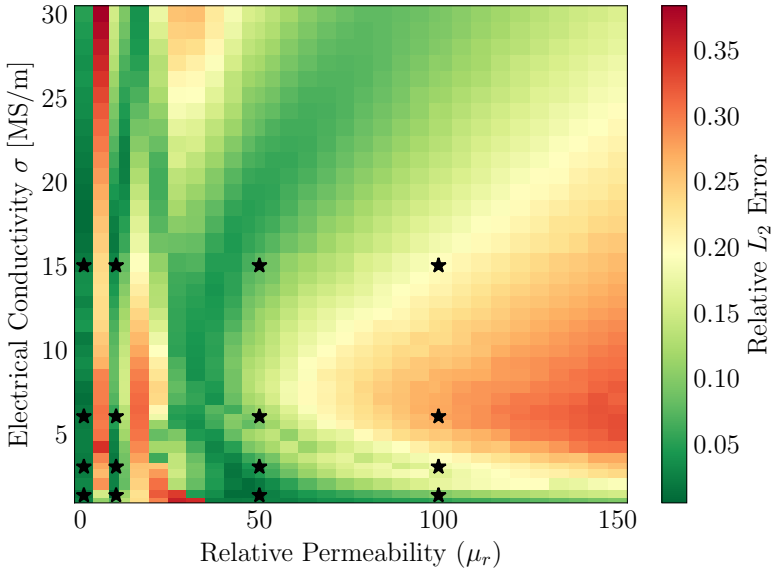
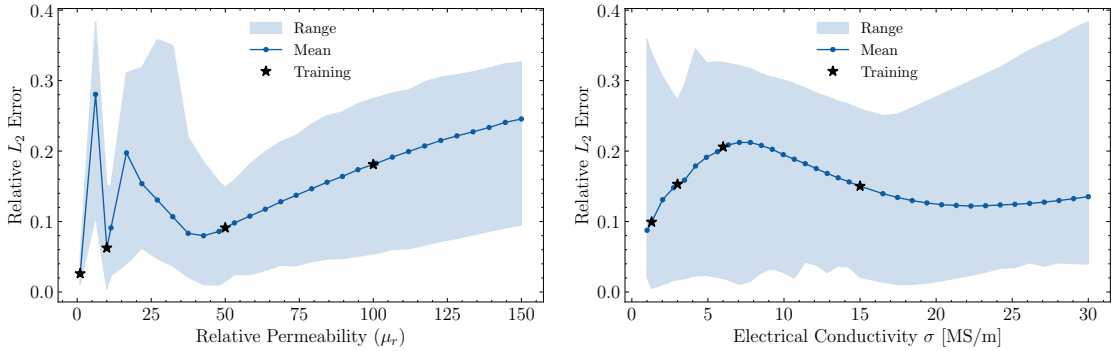


Figure 3.26: Generalisation of the thermal surrogate to unseen material parameters: μ_r and σ. The relative L_2 error is shown over a parameter sweep extending beyond the training range; black stars indicate the 16 training points.



(a) μ_r (b) σ

Figure 3.27: Relative L_2 error of the thermal surrogate as a function of (a) relative permeability μ_r, averaged over all tested σ values, and (b) electrical conductivity σ, averaged over all tested μ_r values. The shaded band shows the min-max range; black stars indicate training values.

generalises more readily to unseen conductivities than to unseen permeabilities within the tested range.

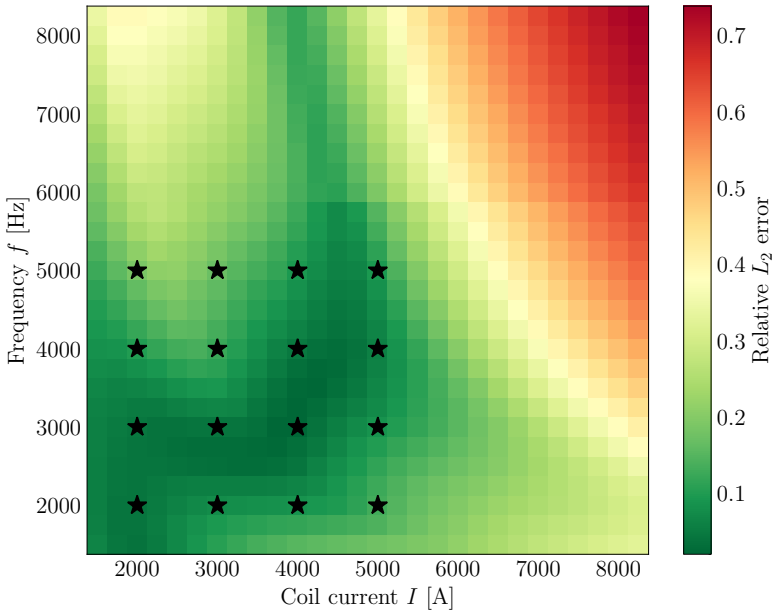


Figure 3.28: Generalisation of the thermal surrogate to unseen source Q strength given by f and I. The relative L_2 error is shown over a parameter sweep extending beyond the training range; black stars indicate the 16 training points.

Generalisation to source strength: coil current I and frequency f Figure 3.28 shows the relative L_2 error of the thermal surrogate over a sweep of coil current I and excitation frequency f, again extending beyond the 4×4 training grid used for both surrogates.

Within the training range, errors remain modest (approximately 5–10%), indicating that the thermal surrogate has learned the dependence of the temperature field on the source strength for conditions well represented in training.

Outside the training range, however, the thermal surrogate degrades more steeply than the electromagnetic surrogate evaluated over the same parameter sweep (see Figure 3.25). Errors reach approximately 70% in the high-I, high-f corner of the extrapolation region, compared with roughly 30% for the electromagnetic surrogate at the same boundary.

This disparity is physically well motivated: the Joule heating source scales approximately as $Q \propto \sigma \mu_r \omega I^2$, so the combined effect of increasing both I and f produces a nonlinear amplification of the heat source that is not represented in training.

Unlike the magnetoquasistatic formulation, in which I and f enter the governing equations linearly through the source term and the $j\omega$ factor, respectively, the thermal problem is driven by $Q \propto I^2 f$, making the temperature field response inherently harder to extrapolate from a linearly spaced training grid.

The elevated errors in the high-I, high-f corner are therefore expected and reflect a genuine limitation of training on a grid that does not cover the full range of relevant source magnitudes.

Discussion and potential improvements. The results above highlight two principal limitations of the thermal surrogate in its current form.

First, the sharp error band at low μ_r suggests that the training grid, while spanning a representative range of material properties, does not provide sufficient coverage near $\mu_r \approx 1$. Adaptive sampling in this region, for instance, by adding training points at $\mu_r \in \{1, 2, 5\}$ for several values of σ, would likely reduce the relative error there substantially.

Second, the rapid degradation of accuracy at high source strengths indicates that the surrogate would benefit from training on a grid that is logarithmically or adaptively spaced in the product $I^2 f$, rather than linearly spaced in f and I separately, so that the full range of physically relevant source magnitudes is represented during training.

3.4.3 Rollout stability

To assess whether the trained thermal surrogates remain stable beyond their training horizon, both the one-way coupled induction-heating surrogate and the TEAM 36 thermal surrogate are rolled out for 2×, 5×, and 10× the training duration, giving normalised times of $t/t_{\text{train}} \in \{1, 2, 5, 10\}$.

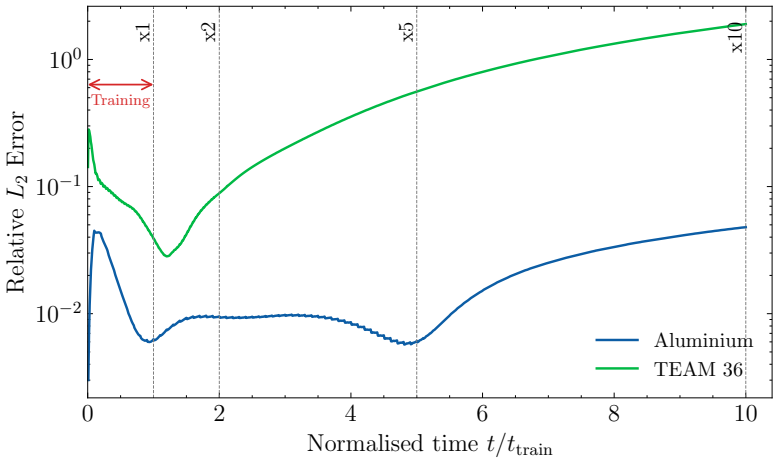


Figure 3.29: Relative L_2 error of the thermal surrogate as a function of normalised rollout time t/t_{train} for the one-way coupled aluminium induction-heating case and the reduced TEAM 36 benchmark. The vertical dashed lines mark the multiplicative length of the temporal rollout.

The aluminium IH case shows a stable error evolution after the training horizon. The error is constant after the training interval in the region from 2× to 5×, the error is $\varepsilon_{L_2} \approx 1 \times 10^{-2}$ and starts to steadily grow in the 5× to 10× region starting from $\approx 1 \times 10^{-2}$ and reaching $\approx 4.7 \times 10^{-2}$ at the 10× boundary.

The TEAM 36 surrogate exhibits higher baseline errors, consistent with the more demanding problem setup (high μ_r, nonlinear $k(T)$, 250 training steps). Its error remains bounded at 2× the training horizon ($\varepsilon_{L_2} \approx 8.9 \times 10^{-2}$), but grows substantially at 5× and 10×, reaching 0.56 and 1.90 respectively. This degradation is consistent with the compounding of two error mechanisms: autoregressive drift in the thermal rollout and the amplification of early-time temperature errors through the temperature-dependent thermal conductivity, as discussed in Section 3.3.2.

These results show how effective the Gaussian noise injection strategy from Section 2.4.4 is at regularising the model by adding noise to input states during training. Both cases used the same noise injection $\mathcal{N}(0, 0.01)$. The aluminium IH surrogate remains stable up to twice the training horizon, then grows gradually thereafter, which matches what we expect from good noise regularisation. On the other hand, the

TEAM 36 surrogate performs much worse beyond twice the training horizon, reaching $\varepsilon_{L_2} = 1.90$ at ten times the training length. This suggests that the noise level or training method was not enough to regularise this more difficult problem, which has nonlinear conductivity and a high-μ_r regime. Overall, the IH surrogate can handle longer rollouts well within moderate time multiples, but the TEAM 36 surrogate works best for rollout horizons close to its training duration. For cases that need long-term predictions of nonlinear, high-μ_r problems, the rollout horizon should be chosen carefully, and the training duration should match the planned simulation length. Also, the amount of injected noise should be treated as a tunable parameter, which should be studied further, and stronger noise may be needed to keep rollouts stable in these nonlinear cases.

3.4.4 Computational speed-up

The computational speed-up of the trained PI-GNN surrogate chain is evaluated on the reduced TEAM 36 benchmark, which involves a single stationary electromagnetic solve followed by a 250-step transient thermal rollout. This problem is representative of the coupled induction-heating setting and provides a concrete basis for comparing surrogate and FEM runtimes across both the electromagnetic and thermal components. The evaluation covers three solver configurations: FEM on CPU, PI-GNN on CPU, and PI-GNN on GPU, to isolate the algorithmic speed-up achieved by the surrogate architecture from the additional acceleration provided by GPU hardware.

Timing protocol

The computational speed-up of the trained PI-GNN surrogates is evaluated. The reported inference times exclude the offline training cost and model loading time, but include graph construction, data transfer, and neural-network inference. This distinction is important because the surrogate is intended for repeated-query settings such as parameter studies, optimisation, or real-time prediction, where the trained model is reused many times.

All timings are repeated 10 times after an initial warm-up run. The reported values correspond to the mean runtime, with the standard deviation used to quantify timing variability. FEM timings are measured on the CPU (AMD Ryzen 9 5900HX). PI-GNN timings are reported on both CPU and GPU (NVIDIA GeForce RTX 3050 Laptop GPU) to distinguish algorithmic speed-up from hardware acceleration.

Problem	Solver	Runtime [s]	Speed-up vs. FEM CPU
Electromagnetic	FEM (CPU)	0.07 ± 1.7e-03	1.00
Electromagnetic	PI-GNN (CPU)	0.58 ± 4.6e-02	0.12
Electromagnetic	PI-GNN (CUDA)	0.16 ± 2.4e-03	0.41
Thermal	FEM (CPU)	10.6 ± 2.9e-01	1.00
Thermal	PI-GNN (CPU)	7.4 ± 3.9e-01	1.42
Thermal	PI-GNN (CUDA)	1.2 ± 6.3e-03	9.10

Table 3.6: Computational runtime of the trained PI-GNN surrogates after training. The reported mean and standard deviation are computed over 10 repeated inference runs after one warm-up run. PI-GNN timings exclude checkpoint loading and include graph construction, data transfer, and neural-network inference. FEM timings are measured on the CPU.

Electromagnetic surrogate runtime

The electromagnetic surrogate does not achieve a wall-clock speed-up over FEM in isolation. The PI-GNN requires 0.58 s on CPU and 0.16 s on GPU, compared with 0.07 s for the FEM reference, corresponding to speed-up factors of 0.12× and 0.41×, respectively. This outcome is expected: the stationary EM problem requires only a single linear solve per query, and the resulting FEM runtime is already small enough that the fixed overhead of graph construction, data transfer, and neural-network inference dominates. The computational benefit of the EM surrogate, therefore, does not lie in per-solve speed but rather in its ability to provide a differentiable, mesh-consistent field prediction that can be embedded in gradient-based optimisation or uncertainty quantification workflows without repeated FEM calls.

Thermal surrogate runtime

The thermal surrogate achieves a meaningful speed-up over FEM. On CPU, the PI-GNN completes the full 250-step transient rollout in 7.4 s, compared with 10.6 s for FEM, yielding a modest 1.42× speed-up that reflects the algorithmic efficiency of the graph-based time-stepping scheme relative to repeated finite-element assembly and solve. On GPU, the surrogate completes the same rollout in 1.2 s, a 9.10× speed-up over the FEM CPU baseline. The low timing variance on GPU (± 6.3 ms) compared with CPU (± 0.39 s) further indicates that GPU execution is more predictable, which is advantageous in latency-sensitive applications.

Coupled surrogate runtime

For the full one-way coupled induction-heating simulation, the EM surrogate is called once and its Joule heat output is passed to the thermal surrogate, which then performs the full transient rollout. The total coupled runtime is therefore the sum of the EM and thermal components. On CPU, the PI-GNN chain requires approximately 8.0 s in total, compared with 10.7 s for the FEM chain, a modest 1.3× overall speed-up. On GPU, the coupled surrogate completes in approximately 1.4 s, achieving an overall speed-up of approximately 7.8× relative to the FEM CPU baseline. The coupled GPU result represents the practically relevant result for many-query applications: each additional simulation of the TEAM 36 reduced benchmark requires only 1.4 s of inference time after the model is trained.

Discussion of offline training cost

The speed-up above reflects only inference time, and the offline training cost must be accounted for when assessing the overall computational trade-off. If the training time is t_{train} and the per-simulation speed-up is $\Delta t = t_{\text{FEM}} - t_{\text{PI-GNN}}$, the surrogate becomes computationally advantageous after approximately

$$N_{\text{break-even}} = \frac{t_{\text{train}}}{\Delta t} \tag{3.10}$$

repeated evaluations. This break-even criterion is only meaningful when $t_{\text{PI-GNN}} < t_{\text{FEM}}$; for the EM surrogate in isolation, where the surrogate is slower than FEM, the criterion does not apply. For the coupled surrogate on GPU, where $\Delta t \approx 9.3$ s per simulation, a training time of several hours translates to a break-even point on the order of 10^3–10^4 evaluations, which is well within the scope of parameter studies, design optimisation

loops, or uncertainty quantification ensembles. The speed-up is therefore most relevant for many-query applications in which the trained model is reused across a large number of operating conditions.

3.5 Ablation/sensitivity study

3.5.1 With/without physics loss

The central question of this ablation is whether incorporating the physics loss reduces the amount of labelled training data required to achieve accurate generalisation over the material-property space (μ_r, σ). Figure 3.30 and Table 3.7 compare a data-only baseline against the PI-GNN across four training-set sizes $N_{\text{train}} \in \{2, 4, 8, 16\}$, using the same interpolation and extrapolation test sets as in Section 3.4.2.

EM surrogate

N_{train}	Data interp. [%]	PI-GNN interp. [%]	Data extrap. [%]	PI-GNN extrap. [%]
2	19.6	24.2	35.6	34.5
4	16.0	7.9	19.9	14.6
8	5.6	5.7	9.4	9.5
16	0.54	0.73	4.0	3.5

Table 3.7: Effect of physics loss on electromagnetic surrogate data efficiency. Errors are reported as relative L_2 errors in per cent of the predicted magnetic vector potential A. The interpolation and extrapolation test sets are identical to those used in the μ_r-σ generalisation study in Section 3.4.2.

Table 3.7 and Figure 3.30 present the relative L_2 error of the predicted magnetic vector potential A for both data-only and PI-GNN surrogates across four training-set sizes. These errors are evaluated at interpolation and extrapolation test points within the parameter space (μ_r, σ).

At $N_{\text{train}} = 2$, neither approach produces a reliable surrogate. The data-only model achieves a lower interpolation error (19.6% vs 24.2%), while the PI-GNN provides a marginal improvement on extrapolation (34.5% vs 35.6%). With only two training points per model, the physics loss does not compensate for the near-absence of data, and neither surrogate generalises meaningfully across the parameter space. The heat maps confirm that both models yield large errors in most evaluated domains at this training set size. Only a relatively narrow band for both models exhibits error below 20%.

At $N_{\text{train}} = 4$, the PI-GNN demonstrates a clear advantage. The interpolation error is reduced by approximately a factor of two relative to the data-only model (7.9% vs 16.0%), and the extrapolation error decreases from 19.9% to 14.6%. The heatmaps indicate that the data-only surrogate retains elevated errors across much of the parameter space, particularly at high μ_r. In contrast, the PI-GNN yields a noticeably smoother and more accurate error distribution. In this regime, the physics loss provides the most practical benefit: it acts as a regulariser that constrains the solution to be consistent with the governing equations, partially compensating for the missing training data.

At $N_{\text{train}} = 8$, the two approaches converge. Interpolation errors are 5.6% and 5.7% for the data-only and PI-GNN models, respectively, and extrapolation errors are 9.3% and 9.5%; these differences are within numerical noise. The heatmaps confirm this

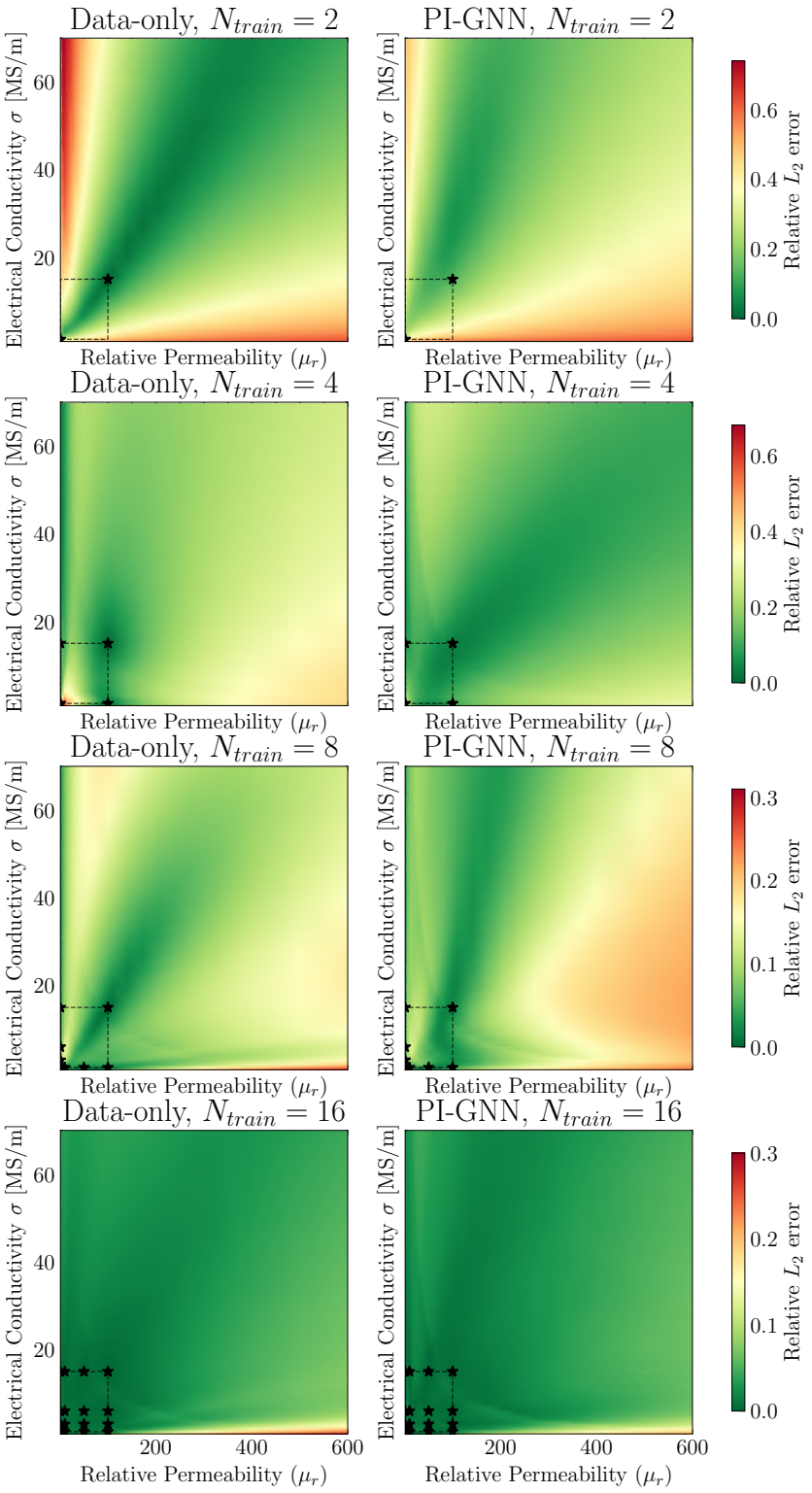


Figure 3.30: Relative L_2 error of the electromagnetic surrogate over the μ_r–σ parameter space for data-only and PI-GNN models trained on $N_{\text{train}} \in \{2, 4, 8, 16\}$ parameter combinations. Black stars mark training points; the dashed rectangle indicates the training range. Left column: data-only baseline. Right column: PI-GNN. Row order corresponds to increasing N_{train}. Note that the colour scale changes between the upper two rows ($N_{\text{train}} \in \{2, 4\}$, max ≈ 0.6) and lower two rows ($N_{\text{train}} \in \{8, 16\}$, max ≈ 0.3), reflecting the overall error reduction with more training data.

similarity: both models yield comparable spatial error distributions, with the remaining elevated errors concentrated in the low-conductivity, high-permeability region of the parameter space, consistent with the challenge of resolving thin skin layers discussed in Section 3.4.2.

At $N_{\text{train}} = 16$, the data-only model achieves a marginally lower interpolation error (0.54% vs 0.73%), while the PI-GNN maintains a slight advantage on extrapolation (3.5% vs 4.0%). Both differences are minimal and not practically significant given the accuracy already achieved. At this training density, the parameter space is sufficiently well sampled for the data alone to constrain the solution effectively, and the additional regularisation from the physics loss yields diminishing returns.

Overall, these results indicate that the benefit of the physics loss is concentrated in the low-data regime, specifically at $N_{\text{train}} = 4$, where it approximately halves the interpolation error relative to a purely data-driven baseline. At $N_{\text{train}} = 2$, the data are too sparse for the physics loss to be effective, and at $N_{\text{train}} \geq 8$ the two approaches are indistinguishable. The physics loss does not yield a consistent extrapolation advantage across all training-set sizes, and its improvement should therefore be interpreted as a data-efficiency gain in the intermediate regime rather than a systematic superiority over data-only training.

Thermal surrogate

Table 3.8 and Figure 3.31 present the analogous data-efficiency comparison for the thermal surrogate, evaluated over the same (μ_r, σ) parameter space and training-set sizes $N_{\text{train}} \in \{2, 4, 8, 16\}$.

N_{train}	Data interp. [%]	PI-GNN interp. [%]	Data extrap. [%]	PI-GNN extrap. [%]
2	22.3	25.7	38.1	36.4
4	45.1	35.9	42.4	39.4
8	35.1	26.0	43.3	47.8
16	32.2	27.0	39.5	28.6

Table 3.8: Effect of physics loss on thermal surrogate data efficiency. Errors are reported as relative L_2 errors in per cent of the predicted temperature field T. The interpolation and extrapolation test sets mirror those used for the electromagnetic surrogate in Table 3.7.

Table 3.8 and Figure 3.31 present the analogous data-efficiency comparison for the thermal surrogate, evaluated across the same (μ_r, σ) parameter space and training-set sizes $N_{\text{train}} \in \{2, 4, 8, 16\}$.

At $N_{\text{train}} = 2$, neither approach produces a reliable surrogate, consistent with the EM results. The data-only model again achieves a marginally lower interpolation error (22.3% vs 25.7%), while the PI-GNN offers a slight improvement on extrapolation (36.4% vs 38.1%). As with the EM surrogate, two training points are insufficient for the physics loss to compensate for the near-absence of labelled data, and both models fail to generalise meaningfully across the parameter space.

With $N_{\text{train}} = 4$, the PI-GNN reduces the interpolation error from 45.1% to 35.9% and the extrapolation error from 42.4% to 39.4%. Both models, however, exhibit higher errors at this training-set size compared to $N_{\text{train}} = 2$. Despite this, the physics loss consistently reduces both interpolation and extrapolation errors in this regime.

For $N_{\text{train}} = 8$, the PI-GNN continues to outperform the data-only baseline in interpolation (26.0% vs 35.1%), but the extrapolation result is reversed: the PI-GNN yields a higher extrapolation error (47.8% vs 43.3%). The heatmaps show that this

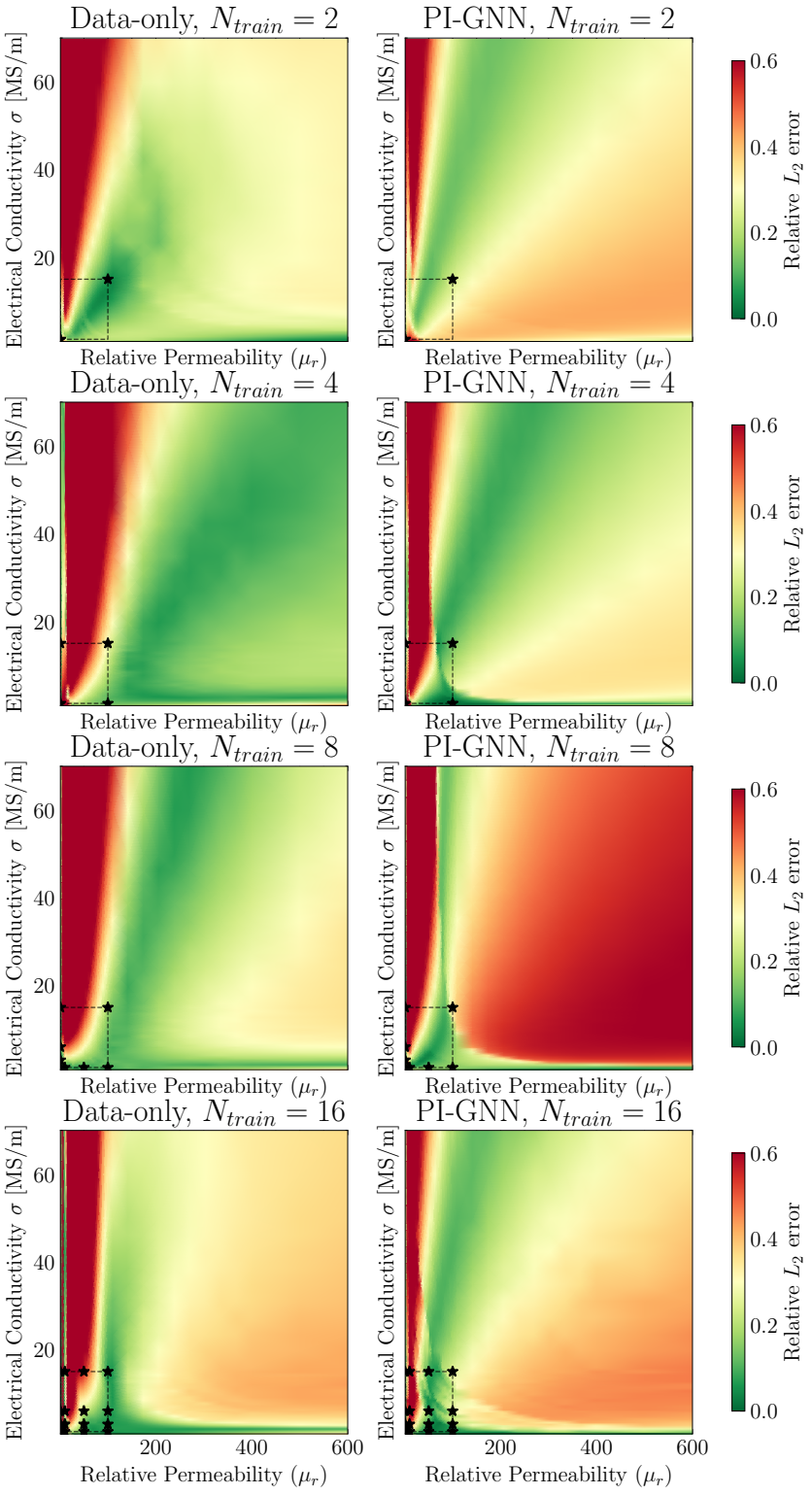


Figure 3.31: Relative L_2 error of the thermal surrogate over the μ_r–σ parameter space for data-only and PI-GNN models trained on $N_{\text{train}} \in \{2, 4, 8, 16\}$ parameter combinations. Black stars mark training points; the dashed rectangle indicates the training range. Left column: data-only baseline. Right column: PI-GNN. Row order corresponds to increasing N_{train}. Note that the colour scale is capped at ≈ 0.6, reflecting the overall error reduction with more training data.

degradation is localised to specific regions of the parameter space. This reversal deserves closer attention because it shows a failure mode unique to coupled physics-informed surrogates. The thermal PDE residual is checked against the models predicted temperature field, not the true one, so it already includes errors from the earlier electromagnetic solution. When the EM predictions are imperfect, which happens at moderate training-set sizes, the thermal physics loss pushes the model to match a flawed forcing term instead of the true governing equation. As a result, gradient updates based on this residual can move the model away from the right generalisation path and make it worse at handling new parameter combinations. This is different from the EM case, where the physics residual depends directly on material parameters, not on a predicted intermediate field. That difference explains why, in this situation, physics-informed regularisation can actually hurt rather than help performance on new data. This outcome contrasts with the EM case, where both models converged at this training-set size.

At $N_{\text{train}} = 16$, the PI-GNN demonstrates a clear advantage in extrapolation (28.6% vs 39.5%) and a modest improvement in interpolation (27.0% vs 32.2%). In contrast to the EM surrogate, which had largely converged by this stage, the thermal surrogate continues to exhibit substantial errors even at the highest training density considered. This outcome reflects the greater difficulty of generalising the temperature field across the (μ_r, σ) space.

In summary, the thermal results are less consistent than those for the EM surrogate. The introduction suggests that PI-GNNs rely less on labelled data, but the ablation results show that this benefit is limited. It is most noticeable at $N_{\text{train}} = 4$ for the EM surrogate, but does not appear or is even reversed for the thermal surrogate at $N_{\text{train}} = 8$. This means the physics loss in the current state is not the best universal regulariser. Its effectiveness depends on how closely the residual relates to the predicted field and on the accuracy of any upstream predictions that affect that residual. The thermal surrogate is driven by Joule heating derived from the electromagnetic solution, so residuals of the thermal PDE are more sensitive to accumulated upstream errors than the EM residuals are to the underlying material parameters. As a result, the physics loss regularises the thermal surrogate less reliably, and its benefit is most apparent only at the highest training density ($N_{\text{train}} = 16$), where it improves extrapolation by approximately 11 percentage points. The overall error levels remain substantially higher than those of the EM surrogate across all training-set sizes, consistent with the findings of Section 3.4.2.

3.6 Summary of results

Overall, the results in this chapter address all the objectives outlined in the introduction. The thermal surrogate was tested on four increasingly challenging benchmarks: analytical diffusion, mixed boundary conditions, source-driven heating, and nonlinear conductivity. These tests showed that it reliably models transient heat transfer, including the effects of temperature-dependent conductivity. The electromagnetic surrogate was tested separately on magnetostatic and harmonic eddy-current cases. Skin-layer tests showed that the graph-based approach can handle steep surface gradients for different relative permeabilities. The coupling interface was checked using the one-way aluminium case and the reduced TEAM 36 benchmark. These tests confirmed that Joule-heat transfer between the two surrogates remains physically accurate, even when high μ_r, a thin skin layer, and temperature-dependent thermal conductivity are present. Generalisation tests showed that both surrogates work well within their training ranges but become less accurate in predictable ways outside those ranges. The main

weaknesses were found with large workpiece diameters and high source strengths ($I^2 f$). The ablation study showed that physics-informed training gives the best data-efficiency in the intermediate-data range, especially at $N_{\text{train}} = 4$ for the electromagnetic surrogate. The thermal surrogate's response to the physics loss was less consistent, likely due to errors building up in the Joule heating term. Finally, the computational performance study showed a 7.8× GPU speed-up for the full coupled surrogate compared to the FEM baseline. The break-even point was about 10^3 to 10^4 evaluations, which matches the needs of the parameter studies and design optimisation tasks that inspired this work.

Chapter 4

Conclusions and Future Work

4.1 Summary and Principal Findings

Induction heating is widely used in industry, and the way temperature changes over time and space affects how materials behave, the quality of products, and how efficient the process is. Predicting these temperature changes is difficult because the heat source comes from Joule heating, which is created by an electromagnetic field. This makes it a complex problem that requires balancing physical accuracy, computational efficiency, and reliability. Many existing surrogate methods only focus on one part of the problem, use overly simple test cases, or are not thoroughly validated. This gap led to the current research.

This thesis developed and tested a physics-informed Graph Neural Network framework in which electromagnetic and thermal behaviours are learned using separate, modular models linked via the Joule heating term. All the goals from the introduction were met: the main electromagnetic and thermal problems were set up for the workpiece, coil, and air geoemtry; both models were built and checked using controlled tests like magnetostatic, harmonic eddy-current, and transient thermal cases; the two models were connected through the Joule heating interface; and the strengths and weaknesses of the approach were examined using generalisation tests, an ablation study, and a computational performance review. The next paragraphs summarise the main results from these evaluations.

Looking at this work from a scientific point of view, its main contribution is not just that the surrogate chain predicts results faster than the standard FEM workflow when used repeatedly. More importantly, it keeps the electromagnetic and thermal subproblems easy to interpret, while making the whole workflow differentiable and reusable for many queries. The graph-based framework we propose sits between traditional reduced-order models and broader operator-learning methods. Projection-based reduced-order models work well for fixed geometries and discretisations, but they need a carefully chosen reduced basis and are less practical when the workpiece, coil layout, or boundary conditions change. General neural operators try to learn solution maps more directly, but using them with irregular industrial meshes and complex multiphysics workflows still means handling geometry, boundary conditions, and local conservation carefully. Our graph approach is appealing because it keeps the mesh connectivity as a physical guide, while still letting us use different discretisations and geometry parameters in the same learning framework.

This helps explain how the method applies to industrial settings. It is not meant to fully replace FEM. Instead, it works best as a way to speed up and add flexibility once a

large enough training database is available. This approach is especially useful for digital twins, design optimisation, parameter studies, and control-focused tasks, where the same trained model might be used thousands of times in similar situations. The verification and performance results suggest this use case is practical, but the generalisation tests show that safe use still depends on knowing the limits of the surrogate's training data.

4.1.1 Surrogate accuracy and physical fidelity

Independent verification of the thermal surrogate was carried out using four increasingly challenging benchmarks: analytical transient diffusion, mixed-boundary-condition heat transfer, source-driven heating, and nonlinear temperature-dependent conductivity. The results showed that the model reliably captures the main physics of induction heating at each level of complexity. For simple cases, relative L_2 errors stayed below 3%, and increased only slightly when temperature-dependent material coefficients were used. This matches the expected challenge of approximating a state-dependent operator with an autoregressive rollout. The model also accurately reproduced Dirichlet, Neumann, and Robin boundary conditions. The close match in integral heat-flux values confirmed that boundary exchanges are handled correctly, which is essential for reliable coupled simulations.

We tested the electromagnetic surrogate on magnetostatic and harmonic eddy-current benchmarks using both single- and dual-coil setups. In the magnetostatic case, the relative errors in the predicted magnetic vector potential stayed below 0.15%, and for the full complex-valued harmonic problem, they were below 0.9%. The skin-layer test showed that separately trained surrogates can capture the exponential current-density profile across a thin surface layer, with relative errors under 2.5% for relative permeabilities of $\mu_r \in \{1, 10, 50, 100\}$. This result confirms that the graph-based representation can handle the steep field gradients found in high-frequency or high-permeability induction problems.

4.1.2 Coupled induction-heating verification

The one-way coupled surrogate chain was first tested on a typical aluminium induction-heating problem. It achieved a final-time temperature-field error of 0.63% and a maximum temperature error of 1.22%. An error-decomposition study showed that the EM and thermal surrogate errors do not simply add together. Instead, the EM-source underprediction and the thermal-rollout drift are separate error mechanisms that partly offset each other in the full chain. This result highlights the need to analyse error sources separately, rather than assuming that the overall accuracy of the coupled surrogate means each part is equally accurate.

The reduced TEAM 36 benchmark tested the framework at the edge of its design limits. It combined high relative permeability ($\mu_r = 100$), a skin depth of about 0.52 mm, a 250-step transient rollout, and a temperature-dependent thermal conductivity $k(T)$. At the end of the simulation, errors were around 3.8–3.9% for both the maximum temperature and the full field L_2 norm. This showed that the surrogate chain stayed physically consistent even under tough conditions. The gradual underprediction of temperature during the rollout came from two main factors: errors building up over time and early temperature deviations getting amplified by the nonlinear $k(T)$ law, which slows heat transport as temperatures rise. Radiative power loss, which scales with temperature as T^4, made boundary-flux errors worse at later times. By the end, the relative error in radiated power reached about 9–10%, even though the temperature-field

error was only 3.8%. This is a clear result of the Stefan–Boltzmann law at temperatures close to 1200 °C.

4.1.3 Generalisation capabilities and limitations

Generalisation tests showed a clear pattern. The surrogates work well within their training envelopes, with errors similar to in-distribution verification. However, their accuracy and predictability drop when tested outside those envelopes.

When testing the electromagnetic surrogate on different geometries, the median relative L_2 error was 1.06×10^{-2} for 42 interpolation cases. For 115 extrapolation cases, the median error increased to 3.74×10^{-2}, and the interquartile range grew by about three times. The largest errors happened when extrapolating to much larger workpiece diameters, where the error increased quickly and unpredictably. This suggests a fundamental change in the electromagnetic coupling geometry, not just a gradual decline in accuracy. In contrast, extrapolating in coil offset and coil inner diameter caused much smaller errors, which indicates the surrogate model has learned smooth patterns for source position and coupling strength.

When we tested different material properties (μ_r, σ), the electromagnetic surrogate model was better at extrapolating along the conductivity axis than the permeability axis. This matches the idea that μ_r affects both skin depth and flux concentration, leading to different field structures as it changes. When extrapolating to $\mu_r = 600$, the error was about 9%, while errors stayed steady across the tested conductivity range.

The thermal surrogate showed a sharp error band when μ_r was low (close to one). This is due to the same relative-error amplification seen in the electromagnetic surrogate at low conductivity. When eddy currents are weak, they create a spread-out Joule heat source with a small value. As a result, even small absolute temperature errors lead to large relative L_2 values. This sensitivity at low μ_r did not appear in the conductivity direction. This matches the coupling structure, since changes in σ are partly reflected in the Joule heating value that the thermal surrogate uses as input. This makes the surrogate less sensitive to σ on its own.

When testing on new process parameters (coil current I and frequency f), a key difference appeared between the EM and thermal surrogates. The magnetoquasistatic model is linear in I and $j\omega$, so the electromagnetic surrogate could predict well even outside its training grid. In contrast, the thermal surrogate depends on a source term that scales as $Q \propto \sigma\mu_r\omega I^2$, which makes the temperature response nonlinear in $I^2 f$. As a result, a training grid that is evenly spaced in I and f does not cover the high-source-strength region well. This led to errors of about 70% for the thermal surrogate and 30% for the electromagnetic surrogate at the grid's high-I, high-f corner.

The preprocessing and graph-generation workflow was also applied to other axisymmetric workpiece types of different geometry and sizes, such as a stepped shaft and a hollow tube. Although some initial tests were done with these shapes, they were not included in this thesis because there was not enough time to verify them as thoroughly as the main benchmark cases. Therefore, these results show that the method can handle several types of industrial workpieces, but they do not yet provide quantitative proof of accuracy for these shapes. A formal study to generalise the geometry across these shapes would be a logical next step.

4.1.4 Physics-informed training and data efficiency

The ablation study compared data-only and PI-GNN (physics-only) surrogates using training set sizes of $N_{\text{train}} = 2, 4, 8$, and 16 in the (μ_r, σ) material-property space. For the electromagnetic surrogate, using the physics loss was most helpful at $N_{\text{train}} = 4$, cutting the interpolation error nearly in half (7.9% compared to 16.0%) and lowering the extrapolation error from 19.9% to 14.6%. When N_{train} was 8 or 16, both methods performed similarly, suggesting that with enough data, the physics constraint is no longer needed as a regulariser. With only $N_{\text{train}} = 2$, the data were too limited for the physics loss to help, and neither method generalised well. To obtain large amounts of data, additional computational resources are needed. A compromise could be to use a hybrid physics and data-driven approach.

The thermal ablation gave less consistent results. The physics loss lowered interpolation error at $N_{\text{train}} \in \{4, 8, 16\}$ but made extrapolation worse at $N_{\text{train}} = 8$. Errors stayed much higher than for the electromagnetic surrogate at all training-set sizes. This inconsistency probably comes from the indirect link between the physics residuals and the thermal field state. The thermal PDE loss is calculated on predicted temperature fields, which collect upstream errors from both the Joule heat source and the autoregressive rollout. As a result, the residual signal is noisier and less reliable as a training guide than in the electromagnetic case, where the governing operator is solved directly from the set boundary conditions. The clearest benefit of the physics loss for the thermal surrogate was at $N_{\text{train}} = 16$, where it improved extrapolation by about 11 percentage points.

4.1.5 Computational efficiency

We compared the surrogate chain to the FEM reference using the reduced TEAM 36 problem, which was the most computationally demanding case in our study. On a CPU, the thermal surrogate finished a 250-step rollout in 7.4 s, compared to 10.6 s for FEM. This gives a modest 1.42× speed-up, mainly because it avoids repeated stiffness-matrix assembly and linear solves. On a GPU, the same rollout took 1.2 s, resulting in a 9.1× speed-up over the FEM CPU baseline and showed much less timing variation. The full coupled surrogate, which runs one EM evaluation followed by the full thermal rollout, finished in about 1.4 s on GPU, giving an overall 7.8× speed-up. The electromagnetic surrogate alone did not run faster than FEM, since the stationary linear solve is already quick. However, its main advantage is that it provides a differentiable, mesh-consistent field prediction suitable for gradient-based optimisation and uncertainty quantification workflows.

The break-even analysis showed that the training investment is recovered after about 10^3 to 10^4 repeated evaluations at GPU inference speed. This situation often happens in parametric studies or iterative design optimisation, which are exactly the many-query settings where surrogate methods are most useful.

4.2 Limitations

This study has some limitations, which also suggest areas for future research, as discussed below.

The surrogate chain works in a one-way coupled mode. Here, the electromagnetic solution drives the thermal field, but changes in temperature do not influence the electromagnetic material properties. Because of this, the model cannot simulate

thermally driven electromagnetic transitions. For example, it cannot capture the ferromagnetic-to-paramagnetic transition at the Curie temperature in steel workpieces, where μ_r drops quickly and the skin depth increases over a small temperature range. Achieving full two-way coupling would require running electromagnetic and thermal calculations in an iterative loop at each time step, which is beyond the scope of this framework.

Each surrogate is trained within a set range of parameters and its performance drops in a predictable way when used outside that range. The ablation study found that this problem is most noticeable for the thermal surrogate at high source strengths and for the electromagnetic surrogate when there are large changes in geometry. Before using these models in situations beyond their tested range, it is necessary to retrain them or use adaptive sampling with a wider data range.

The rollout stability study found that the thermal surrogate stays reliable up to twice its training horizon in the simpler induction-heating case. However, it shows significant errors at five and ten times the training horizon in the more challenging TEAM 36 problem. For longer process simulations, it is important to set the rollout horizon as a design parameter that matches the planned simulation length.

The models used in this study work with two-dimensional axially symmetric shapes. However, real induction heating processes are three-dimensional. Extending these models to 3D will need broader training, greater computational resources, and a fresh look at how graphs are built, how temporal bundling is used, and how physics-based losses are defined.

4.3 Future Work

The modular design of the PI-GNN surrogate chain makes it easy to add new features that can broaden its use and make its results more realistic. Key areas for future work include stronger multiphysics coupling, better extrapolation control, improved geometry handling, and architectures that can handle large industrial meshes more efficiently.

The next step is to implement a two-way coupled surrogate, where the electromagnetic coefficients $\mu_r(T)$ and $\sigma(T)$ are updated at each thermal time step based on the predicted temperature field. This approach allows the simulation to capture the Curie transition and other temperature-driven electromagnetic changes, making the framework suitable for high-carbon and tool steels used in industrial heat treatment. The modular surrogate design supports this by sending the current temperature field as an extra input to the electromagnetic surrogate at each coupling step, without changing the structure of either network.

The generalisation results show that some regions of the parameter space are not well covered by current training data. These include areas near $\mu_r \approx 1$, at large workpiece diameters, and in the high $I^2 f$ regime. To address this, future datasets could use both space-filling and adaptive sampling strategies. For example, Latin Hypercube sampling can help cover the parameter domain at first, while residual-based sampling or Bayesian experimental design can later focus new FEM simulations on areas where the surrogate error, physics residual, or predictive uncertainty is highest [63–66]. These strategies are helpful because they target challenging regions without needing to expand the entire dataset evenly. Before using this method in closed-loop industrial workflows, uncertainty quantification should also be included. Methods like ensembles, Bayesian approximations, Monte Carlo dropout, or conformal calibration can help estimate when

the surrogate is unreliable and when fallback FEM simulations or new training samples are needed.

To extend the framework to 3D, we need to build volumetric graphs using tetrahedral or hexahedral meshes, adjust the message-passing scheme for three-dimensional operators, and update the physics loss for the 3D curl–curl EM operator and the 3D heat equation. The PI-GNN method itself does not depend on a specific geometry, but the main challenges are the high computational cost and finding enough varied 3D training geometries.

Recent mesh-based learning models offer promising ways to extend to 3D and multi-scale problems. X-MeshGraphNet [67] is worth exploring because its partitioned, multi-scale graph structure is built for large graphs and long-range interactions. MeshGraphNet-Transformer [68] is also relevant, as its transformer-based global processing could help solve the under-reaching problem that standard message passing faces on large meshes. The neural-operator element method [69] takes a different approach: instead of replacing the entire FEM workflow with a single global surrogate, it uses reusable neural-operator elements in costly local regions, like skin-depth-resolved zones, while keeping a variational FEM structure elsewhere. Comparing these methods with the current PI-GNN could reveal whether the main challenge is graph scalability, long-range coupling or local multiscale resolution.

The new workpiece shapes developed in this project should be used to create a formal benchmark for geometry generalisation. A plain billet, a stepped shaft, and a tube each have different geometric complexity and thermal boundary-to-volume ratios. These shapes can help test if the surrogate model learns local electromagnetic and thermal rules that apply across different new shapes unseen in this thesis. This approach would directly test a key promise of graph neural networks: handling changing geometries beyond training without the need for retraining.

The surrogate chain is differentiable with respect to its inputs, such as coil geometry, current amplitude, frequency, and material properties. This feature makes it possible to use gradient-based methods for process optimisation. If we treat induction heating design as a constrained optimisation problem using the surrogate's parameters, we can target goals like uniform surface temperature, specific hardening depth, shorter cycle times, or lower energy use. This approach could show the practical benefits of differentiability, which the current architecture has not yet fully used. For industrial applications, the method should include larger geometry libraries, adaptive sampling, uncertainty-aware deployment, multi-coil excitation patterns, and bidirectional coupling, instead of relying on just one trained surrogate chain.

4.4 Concluding Remarks

This thesis set out to create and evaluate a physics-informed surrogate modelling workflow for induction heating. In this approach, electromagnetic and thermal behaviours are learned separately and then linked through Joule heating. The goal was to determine whether this method could maintain sufficient physical accuracy to be useful in multiphysics workflows. The results show that it can, and they also clarify the situations where this accuracy is reliable.

This framework directly addresses the issues found in the literature review. It considers both physical subproblems instead of just one, and it is tested against analytical solutions, manufactured benchmarks, and the TEAM 36 industry standard, not just simple test cases. Its limitations are clearly defined through generalisation and ablation

studies. The framework achieves a 7.8× GPU speed-up over FEM, with errors ranging from less than 1% to low single digits across verification benchmarks. This makes it a practical tool for the many-query applications that inspired this work, such as parameter studies, iterative design optimisation, and digital twin workflows where the trained model is used much more often than it is retrained.

At the same time, the results clearly show the limits of the current approach. Predictions become less accurate when applied to geometries or conditions far outside the training data. The one-way coupling assumption means the method is best for problems without strong electromagnetic and thermal feedback. Also, long simulations on nonlinear problems tend to accumulate errors faster than expected. These challenges suggest a clear direction for future research. The next steps include adding two-way coupling, using adaptive sampling, attaching uncertainty estimates to predictions, improving mesh-transfer strategies, and developing architectures that can handle larger and more varied geometries. These improvements will help make the surrogate more useful in industrial heat-treatment processes.

Overall, the findings in this thesis advance understanding of how physics-informed machine learning can be used in coupled multiphysics simulations. They also lay the groundwork for building more capable, general, and practical surrogate models for industrial use.

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Appendix A

EM additional plots

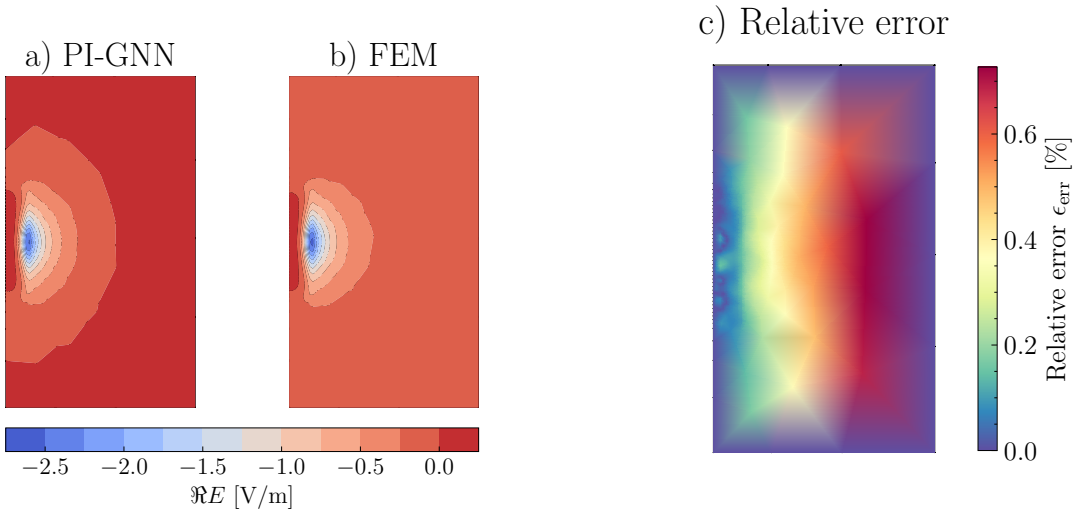


Figure 4.1: One-coil system real part. a) and b) PI-GNN vs FEM, c) Relative error

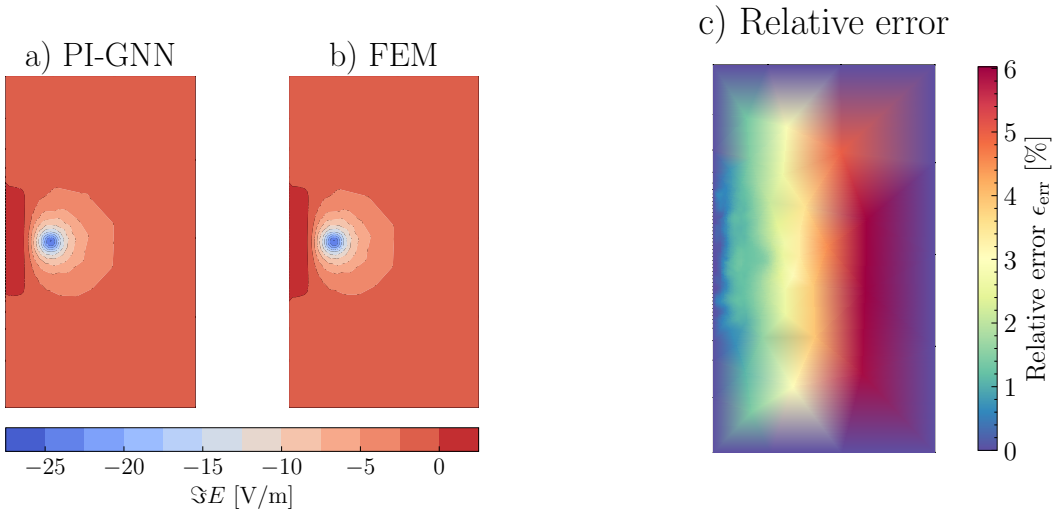


Figure 4.2: One-coil system imaginary part. a) and b) PI-GNN vs FEM, c) Relative error

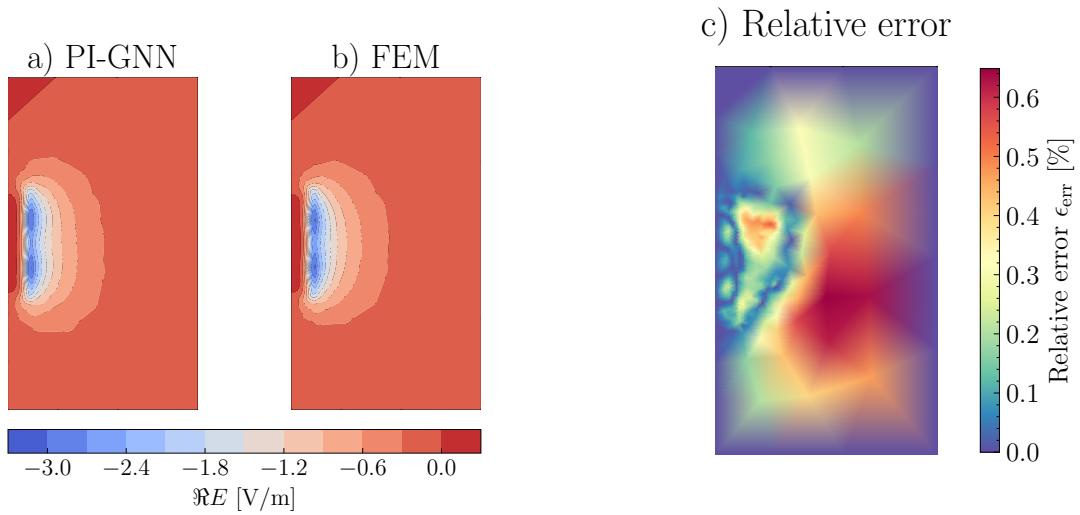


Figure 4.3: Two-coil system real part. a) and b) PI-GNN vs FEM, c) Relative error

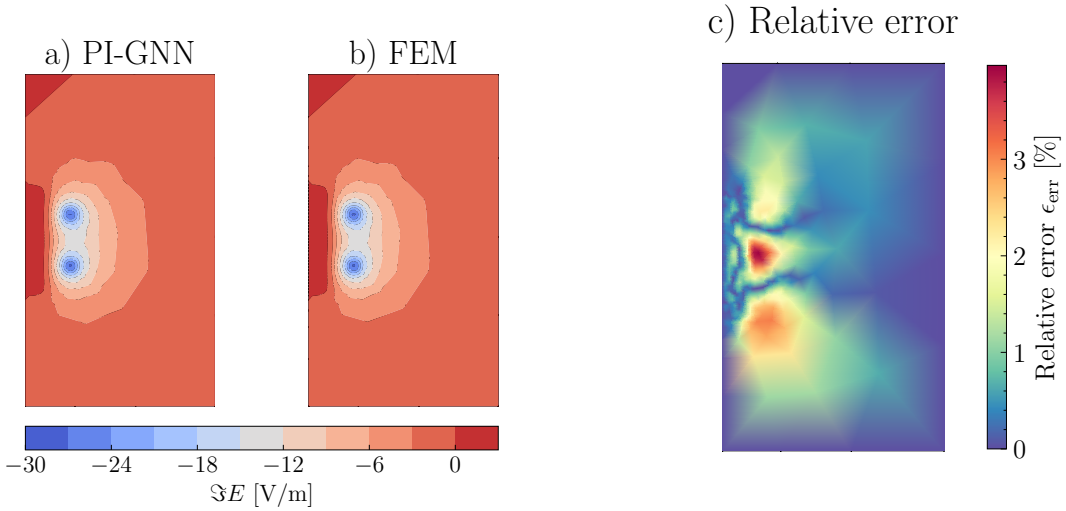
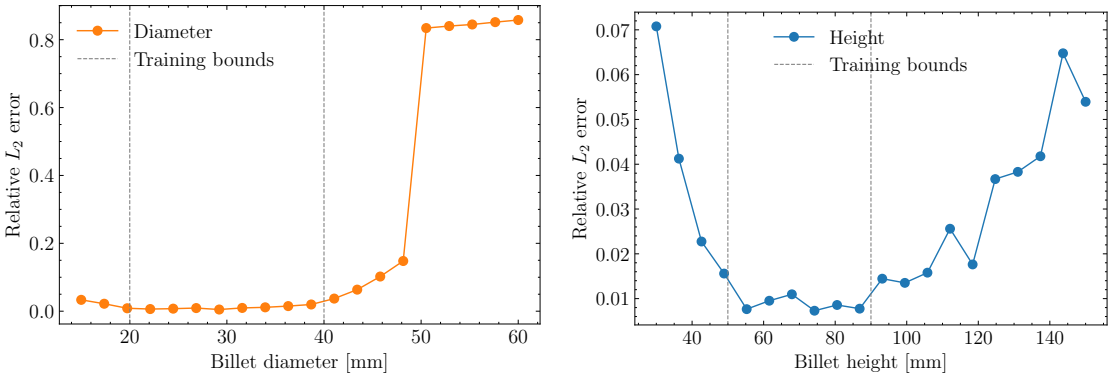


Figure 4.4: Two-coil system imaginary part. a) and b) PI-GNN vs FEM, c) Relative error

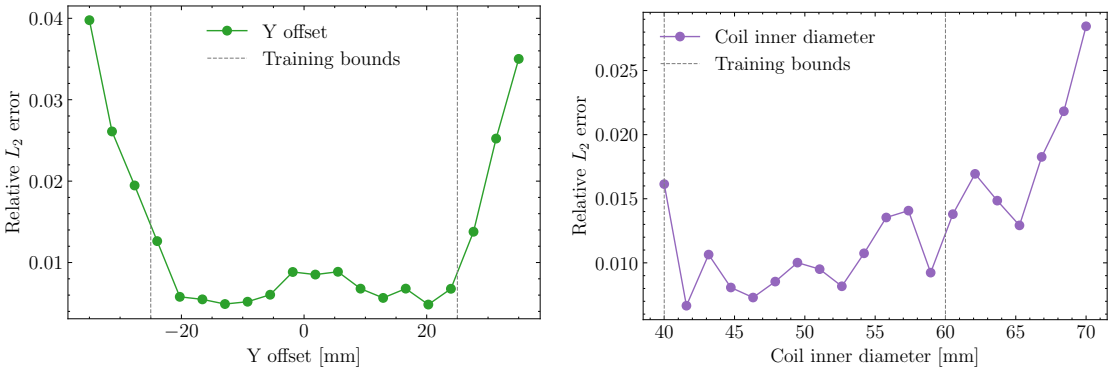
Generalisation additional plots



(a) Billet diameter sweep.

(b) Billet height sweep.

Figure 4.5: The relative L_2 error in $|A|$ is shown for workpiece-size extrapolation. Dashed vertical lines mark the training bounds. When varying height, the error increases gradually outside the training range. In contrast, increasing the diameter leads to a much sharper failure for large values.



(a) Coil offset sweep.

(b) Coil inner diameter sweep.

Figure 4.6: The relative L_2 error in $|A|$ is shown for both source-position and coil to workpiece coupling extrapolation. The dashed vertical lines mark the training limits. When the offset is extrapolated, the error increases smoothly. Increasing the coil's inner diameter results in a moderate drop in performance, due to weaker magnetic coupling.

TEAM 36 $k(T)$ table

Table 4.1: Temperature-dependent thermal conductivity $k(T)$ for TEAM 36.

Temperature T [°C]	Thermal conductivity $k(T)$ [W/(m K)]
0	48.1
100	48.1
200	46.5
300	44.0
400	41.0
500	38.5
600	36.0
700	31.4
750	28.5
800	26.7
900	25.9
1000	26.7
1100	28.0
1200	29.8
1400	35.0
1470	39.0
1800	39.0

Appendix B

International Conference on Heating by Electromagnetic Sources abstract submission

HESUIE 26 - International Conference on Heating by Electromagnetic Sources
September 15-18, 2026 – Padova (Italy)

Physics-Informed Graph Neural Network for Fast Thermal Prediction in Induction Heating

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ABSTRACT. Induction heating needs fast EM–thermal simulations for control and optimisation. The Finite Element Method is accurate but costly. In this paper, we propose a physics-informed GNN for transient heat modelling on unstructured meshes and a physics-informed EM surrogate that predicts A and ϕ in the frequency domain. Each model is validated against FEM on representative geometries. The thermal model predicts multi-step rollouts. We target robust physics-informed graph learning for transient thermal fields and a pathway to coupled EM–thermal prediction suitable for industrial induction heating.

***Keywords*—** Induction heating; Physics-informed graph neural networks; Maxwell's equations; Joule heating; Surrogate modelling

INTRODUCTION

Induction heating is commonly used for heat treatment and process heating; however, accurate multi-physics simulations can be computationally intensive, particularly for iterative design, optimisation, and control [1]. The coupled electromagnetic-thermal problem is generally non-linear, especially when electrical conductivity and magnetic permeability vary with temperature.

Recent advancements in graph neural network models, such as MeshGraphNets, offer a framework for learning on unstructured meshes while maintaining locality and geometric inductive bias [2]. Physics-informed MeshGraphNets have demonstrated that using FEM-style residuals can enable data-efficient learning for transient problems on arbitrary meshes [3].

Building on these advancements, this digest presents a PI-GNN approach for induction heating that utilises separate electromagnetic and thermal surrogates, trained with discretisation-consistent residual losses and validated against FEM simulations using NGSolve [4]. It also outlines a roadmap towards developing a coupled, temperature-dependent surrogate.

PROBLEM DESCRIPTION

We consider a 2D axisymmetric domain: a conductive workpiece, a stranded-coil region, and the surrounding air. The thermal model incorporates a convective boundary condition at the interface between the workpiece and the air. A Neumann symmetry condition is applied along the symmetry axis. The $t_{end} = 1s$ and $\Delta t = 0.1$ s. The current rollout predicts the next n = 5 time steps.

The electromagnetic surrogate focuses on a frequency-domain eddy-current formulation represented in an $A - \phi$ framework [5]. From the predicted fields, the complex electric field is calculated as $E = -j\omega A - \nabla\phi$ and $q(x) = 1/2\ \sigma(T)\ |E(x)|^2$.

The thermal surrogate predicts temperatures $T^{n+1}, T^{n+2}, \dots, T^{n+n_{roll}}$ based on the previous state T^n and the heating source q^n. A discretisation-consistent residual is constructed using FEM assembled mass and stiffness matrices, along with boundary and load contributions. For a single time step, the per-time-step residual $R^{n+1}\left(\tilde{T}^{n+1}, T^n\right)$ is defined as

$$R^{n+1} = (1/\Delta t\ \mathbf{M} + \mathbf{K})\ T^{n+1} - 1/\Delta t\ \mathbf{M}T^n - \mathbf{Q}(T^{n+1}) - g, \quad (1)$$

where **M** and **K** represent the finite element method mass and conductivity matrices, respectively. The $\mathbf{Q}(\cdot)$ denotes the source term, and g is the boundary flux. Given a prediction from the PI-GNN denoted as $\tilde{T}^{n+1}$, the physics-informed loss is calculated for the free DOFs using $\mathcal{L}_{\text{FEM}}^{n} = ||R^{n+1}(\tilde{T}^{n+1}, T^{n})||_2^2$. This loss is normalised by the number of free DOFs and equals zero when $\tilde{T}^{n+1}$ aligns with the FEM solution.

For the electromagnetic surrogate, an objective is formulated based on the complex residual $r = \mathbf{K}a - f$, similar to equation (1), which is separated into real and complex parts and normalised by a number of DOFs.

The induction heating problem is solved by combining the two models, as can be seen in the Figure 1. At time t^n, the EM surrogate is evaluated using the current temperature-dependent properties, starting with $\sigma(T^n)$, to derive q^n. Following this, the thermal surrogate updates the temperature from T^n to T^{n+1} using the method outlined in equation (1).

Both surrogates use a MeshGraphNet [2] architecture (12 message-passing layers, MLP-128).

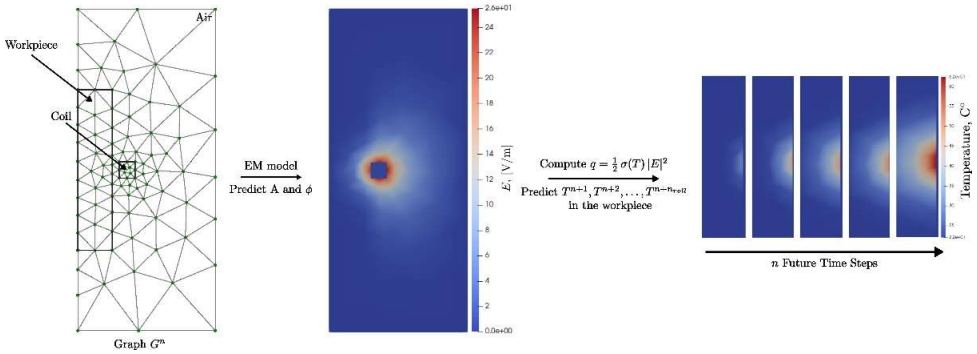


Figure 1. An outlook of the IH surrogate model (EM+Thermal).

CONCLUSIONS

This digest presents a framework for developing induction heating surrogates using physics-informed MeshGraphNets on unstructured meshes.

A harmonic $A - \phi$ electromagnetic Physics-Informed Graph Neural Network is utilised to predict complex potentials and calculate Joule heating. Additionally, a transient thermal PI-GNN performs multi-step rollouts, trained using a FEM—assembled weak residual. The next step is to couple these surrogates via operator splitting, thereby incorporating temperature-dependent properties.

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Declaration on use of generative artificial intelligence

In this scientific work, generative artificial intelligence (AI) is employed. All data and personal information have been processed in accordance with the University of Oslo's regulations. I, as the author of this document, take full responsibility for its content, claims, and references. An overview of the use of generative AI is provided below.

- Grammarly was used to correct English
- OpenAI ChatGPT was utilised to:
  - Brainstorm ideas
  - Find additional resources on the topic